

F2802x Peripheral Driver Library

USER'S GUIDE



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Table of Contents

Copyright	2
Revision Information	2
1 Introduction	5
2 Programming Model	7
3 Analog to Digital Converter (ADC)	9
3.1 ADC	9
4 Capture (CAP)	15
4.1 CAP	15
5 Device Clocking (CLK)	27
5.1 CLK	27
6 Comparater (COMP)	45
6.1 COMP	45
7 Central Processing Unit (CPU)	51
7.1 CPU	51
8 Flash	59
8.1 FLASH	59
9 General Purpose Input/Output (GPIO)	69
9.1 GPIO	69
10 Oscillator (OSC)	81
10.1 OSC	81
11 Peripheral Interrupt Expansion Module (PIE)	87
11.1 PIE	87
12 Phase Locked Loop (PLL)	103
12.1 PLL	103
13 Pulse Width Modulator (PWM)	111
13.1 PWM	111
14 Power Control (PWR)	151
14.1 PWR	151
15 Serial Communications Interface (SCI)	157
15.1 SCI	157
16 Serial Peripheral Interface (SPI)	177
16.1 SPI	177
17 Timer	193
17.1 TIMER	193
18 Watchdog Timer	199
18.1 WDOG	199
IMPORTANT NOTICE	204

1 Introduction

The Texas Instruments® C2000Ware® Peripheral Driver Library is a set of drivers for accessing the peripherals found on the Piccolo Entryline family of C2000 microcontrollers. While they are not drivers in the pure operating system sense (that is, they do not have a common interface and do not connect into a global device driver infrastructure), they do provide a mechanism that makes it easy to use the device's peripherals.

The capabilities and organization of the drivers are governed by the following design goals:

They are written entirely in C except where absolutely not possible.

They demonstrate how to use the peripheral in its common mode of operation.

They are easy to understand.

They are reasonably efficient in terms of memory and processor usage.

They are as self-contained as possible.

Where possible, computations that can be performed at compile time are done there instead of at run time.

Some consequences of these design goals are:

The drivers are not necessarily as efficient as they could be (from a code size and/or execution speed point of view). While the most efficient piece of code for operating a peripheral would be written in assembly and custom tailored to the specific requirements of the application, further size optimizations of the drivers would make them more difficult to understand.

The drivers do not support the full capabilities of the hardware. Some of the peripherals provide complex capabilities which cannot be utilized by the drivers in this library, though the existing code can be used as a reference upon which to add support for the additional capabilities.

For many applications, the drivers can be used as is. But in some cases, the drivers will have to be enhanced or rewritten in order to meet the functionality, memory, or processing requirements of the application. If so, the existing driver can be used as a reference on how to operate the peripheral.

This device support release also contains the traditional peripheral register header files and associated software. Please see chapter 2 of this guide for more information on this software. For future development we recommend the use of this driver infrastructure instead of the traditional header files.

Source Code Overview

The following is an overview of the organization of the peripheral driver library source code.

`common/source/`

This directory contains the source code for the drivers.

`common/include/`

This directory contains the header files for the drivers. These headers define not only values used in the drivers and calls to drivers but also define the register structure of each peripheral.

`common/ccs/`

This directory contains the CCS project for the driver library. The driver library can be rebuilt if modifications are made by importing and building this project.

2 Programming Model

Introduction	??
Direct Register Access Model	??
Software Driver Model	??
Combining The Models	??

The peripheral driver library provides support for two programming models: the direct register access model and the software driver model. Each model can be used independently or combined, based on the needs of the application or the programming environment desired by the developer.

Each programming model has advantages and disadvantages. Use of the direct register access model generally results in smaller and more efficient code than using the software driver model. However, the direct register access model requires detailed knowledge of the operation of each register and bit field, as well as their interactions and any sequencing required for proper operation of the peripheral; the developer is insulated from these details by the software driver model, generally requiring less time to develop applications. In the direct register access model, the peripherals are programmed by the application by writing values directly into the peripheral's registers. Every register is defined in a peripheral's corresponding header file contained in `f2802x/headers/include`. These headers define the locations of each register relative to the other registers in a peripheral as well as the bit fields within each register. All of this is implemented using structures. The header files only define these structures; they do not declare them. To declare the structures a C source file must be included in each project `f2802x/headers/source/F2802x_GlobalVariableDefs.c`. This file declares each structure (and in some cases multiple instances when there are multiple instances of a peripheral) as well as declaring and associating each structure with a code section for the linker. The final piece of the puzzle is a special linker command file that associates each section defined in `F2802x_GlobalVariableDefs.c` with the physical memory of the device. There are two versions of this linker command file, one for use with SYS/BIOS and one for use without, but both can be found in `f2802x/headers/cmd`. One of these linker command files as well as your normal application linker command file should be used to link your project.

Applications that wish to use the direct register access model should define the root of the `f2802x` directory to be an include path and include the file `DSP28x_Project.h` in each source file where register accesses are made.

In practice accessing peripheral registers and bitfields is extremely easy. Below are a few examples: `GpioCtrlRegs.GPAPUD.bit.GPIO0 = 0;` `GpioCtrlRegs.GPAMUX1.bit.GPIO0 = 1;`

In the software driver model, the API provided by the peripheral driver library is used by applications to control the peripherals. Because these drivers provide complete control of the peripherals in their normal mode of operation, it is possible to write an entire application without direct access to the hardware. This method provides for rapid development of the application without requiring detailed knowledge of how to program the peripherals.

Before a driver for a peripheral can be used that peripheral's driver header file should be included and a handle to that peripheral initialized: `GPIO_Handle myGpio; myGpio = GPIO_init((void*)GPIO_BASE_ADDR, sizeof(GPIO_Obj));`

In subsequent calls to that peripheral's driver the initialized handle as well as any parameters are passed to the function: `GPIO_setPullUp(myGpio, GPIO_Number0, GPIO_PullUp_Enable); GPIO_setMode(myGpio, GPIO_Number0, GPIO_`

As you can see from the above sample, using the driver library makes one's code substantially more readable.

In addition to including the appropriate header files for the drivers you are using, the project

should also have `driverlib.lib` linked into it. A CCS project as well as the lib file can be found in `f2802x/common/ccs`. The driver library also contains some basic helper functions that many projects will use as well as the direct register access model source file `F2802x_GlobalVariableDefs.c` which allows you to use the header files without having to directly compile this source file into your project.

The drivers in the peripheral driver library are described in the remaining chapters in this document. They combine to form the software driver model. The direct register access model and software driver model can be used together in a single application, allowing the most appropriate model to be applied as needed to any particular situation within the application. For example, the software driver model can be used to configure the peripherals (because this is not performance critical) and the direct register access model can be used for operation of the peripheral (which may be more performance critical). Or, the software driver model can be used for peripherals that are not performance critical (such as a UART used for data logging) and the direct register access model for performance critical peripherals.

To use both models interchangeably in your application:

Link `driverlib.lib` into your application

Include `DSP28x_Project.h` in files you wish to use the direct register access model

Add `C2000Ware_version/device_support/f2802x/` to your projects include path (Right click on project, Build Properties, Include Path)

Include driver header files from `f2802x/common/include` in any source file that makes calls to that driver.

3 Analog to Digital Converter (ADC)

[Introduction](#)??

[API Functions](#) 9 The Analog to Digital Converter (ADC) APIs provide a set of functions for accessing the Piccolo F2802x ADC modules.

This driver is contained in `f2802x/common/source/adc.c`, with `f2802x/common/include/adc.h` containing the API definitions for use by applications.

3.1 ADC

Data Structures

[_ADC_Obj](#)

Macros

[ADC_ADCCTL1_ADCBGPWD_BITS](#)

[ADC_ADCCTL1_ADCBSY_BITS](#)

[ADC_ADCCTL1_ADCBSYCHAN_BITS](#)

[ADC_ADCCTL1_ADCENABLE_BITS](#)

[ADC_ADCCTL1_ADCPWDN_BITS](#)

[ADC_ADCCTL1_ADCREFPWD_BITS](#)

[ADC_ADCCTL1_ADCREFSEL_BITS](#)

[ADC_ADCCTL1_INTPULSEPOS_BITS](#)

[ADC_ADCCTL1_RESET_BITS](#)

[ADC_ADCCTL1_TEMPCONV_BITS](#)

[ADC_ADCCTL1_VREFLOCONV_BITS](#)

[ADC_ADCSAMPLEMODE_SEPARATE_FLAG](#)

[ADC_ADCSAMPLEMODE_SIMULEN0_BITS](#)

[ADC_ADCSAMPLEMODE_SIMULEN10_BITS](#)

[ADC_ADCSAMPLEMODE_SIMULEN12_BITS](#)

[ADC_ADCSAMPLEMODE_SIMULEN14_BITS](#)

[ADC_ADCSAMPLEMODE_SIMULEN2_BITS](#)

ADC_ADCSAMPLEMODE_SIMULEN4_BITS
ADC_ADCSAMPLEMODE_SIMULEN6_BITS
ADC_ADCSAMPLEMODE_SIMULEN8_BITS
ADC_ADCSOCxCTL_ACQPS_BITS
ADC_ADCSOCxCTL_CHSEL_BITS
ADC_ADCSOCxCTL_TRIGSEL_BITS
ADC_BASE_ADDR
ADC_dataBias
ADC_DELAY_usec
ADC_INTSELxNy_INTCONT_BITS
ADC_INTSELxNy_INTE_BITS
ADC_INTSELxNy_INTSEL_BITS
ADC_INTSELxNy_LOG2_NUMBITS_PER_REG
ADC_INTSELxNy_NUMBITS_PER_REG

Enumerations

ADC_IntMode_e
ADC_IntNumber_e
ADC_IntPulseGenMode_e
ADC_IntSrc_e
ADC_ResultNumber_e
ADC_SampleMode_e
ADC_SocChanNumber_e
ADC_SocNumber_e
ADC_SocSampleWindow_e
ADC_SocTrigSrc_e
ADC_VoltageRefSrc_e

Functions

```
void ADC_clearIntFlag (ADC_Handle adcHandle, const ADC_IntNumber_e intNumber)

void ADC_disable (ADC_Handle adcHandle)

void ADC_disableBandGap (ADC_Handle adcHandle)

void ADC_disableInt (ADC_Handle adcHandle, const ADC_IntNumber_e intNumber)

void ADC_disableRefBuffers (ADC_Handle adcHandle)

void ADC_disableTempSensor (ADC_Handle adcHandle)

void ADC_enable (ADC_Handle adcHandle)

void ADC_enableBandGap (ADC_Handle adcHandle)

void ADC_enableInt (ADC_Handle adcHandle, const ADC_IntNumber_e intNumber)

void ADC_enableRefBuffers (ADC_Handle adcHandle)

void ADC_enableTempSensor (ADC_Handle adcHandle)

void ADC_forceConversion (ADC_Handle adcHandle, const ADC_SocNumber_e socNumber)

bool_t ADC_getIntStatus (ADC_Handle adcHandle, const ADC_IntNumber_e intNumber)

ADC_SocSampleWindow_e ADC_getSocSampleWindow (ADC_Handle adcHandle, const
ADC_SocNumber_e socNumber)

int16_t ADC_getTemperatureC (ADC_Handle adcHandle, int16_t sensorSample)

int16_t ADC_getTemperatureK (ADC_Handle adcHandle, int16_t sensorSample)

ADC_Handle ADC_init (void *pMemory, const size_t numBytes)

void ADC_powerDown (ADC_Handle adcHandle)

void ADC_powerUp (ADC_Handle adcHandle)

uint_least16_t ADC_readResult (ADC_Handle adcHandle, const ADC_ResultNumber_e result-
Number)

void ADC_reset (ADC_Handle adcHandle)

void ADC_setIntMode (ADC_Handle adcHandle, const ADC_IntNumber_e intNumber, const
ADC_IntMode_e intMode)

void ADC_setIntPulseGenMode (ADC_Handle adcHandle, const ADC_IntPulseGenMode_e pulse-
Mode)

void ADC_setIntSrc (ADC_Handle adcHandle, const ADC_IntNumber_e intNumber, const
ADC_IntSrc_e intSrc)

void ADC_setSampleMode (ADC_Handle adcHandle, const ADC_SampleMode_e sampleMode)
```

```
void ADC_setSocChanNumber (ADC_Handle adcHandle, const ADC_SocNumber_e socNumber,
const ADC_SocChanNumber_e chanNumber)
```

```
void ADC_setSocSampleWindow (ADC_Handle adcHandle, const ADC_SocNumber_e socNum-
ber, const ADC_SocSampleWindow_e sampleWindow)
```

```
void ADC_setSocTrigSrc (ADC_Handle adcHandle, const ADC_SocNumber_e socNumber, const
ADC_SocTrigSrc_e trigSrc)
```

```
void ADC_setVoltRefSrc (ADC_Handle adcHandle, const ADC_VoltageRefSrc_e voltRef)
```

3.1.1 Detailed Description

3.1.2 Data Structure Documentation

3.1.2.1 _ADC_Obj_

Definition:

```
typedef struct
{
    uint16_t ADCRESULT[16];
    uint16_t rsvd_1[26096];
    uint16_t ADCCTL1;
    uint16_t rsvd_2[3];
    uint16_t ADCINTFLG;
    uint16_t ADCINTFLGCLR;
    uint16_t ADCINTOVF;
    uint16_t ADCINTOVFCLR;
    uint16_t INTSELxNy[5];
    uint16_t rsvd_3[3];
    uint16_t SOCPRICTRL;
    uint16_t rsvd_4;
    uint16_t ADCSAMPLEMODE;
    uint16_t rsvd_5;
    uint16_t ADCINTSOCSEL1;
    uint16_t ADCINTSOCSEL2;
    uint16_t rsvd_6[2];
    uint16_t ADCSOCFLG1;
    uint16_t rsvd_7;
    uint16_t ADCSOCFRC1;
    uint16_t rsvd_8;
    uint16_t ADCSOCOVF1;
    uint16_t rsvd_9;
    uint16_t ADCSOCOVFCLR1;
    uint16_t rsvd_10;
    uint16_t ADCSOCxCTL[16];
    uint16_t rsvd_11[16];
    uint16_t ADCREFTRIM;
    uint16_t ADCOFFTRIM;
    uint16_t rsvd_12[13];
    uint16_t ADCREV;
```

```
}  
_ADC_Obj_
```

Members:

ADCRESULT ADC result registers.

rsvd_1 Reserved.

ADCCTL1 ADC Control Register 1.

rsvd_2 Reserved.

ADCINTFLG ADC Interrupt Flag Register.

ADCINTFLGCLR ADC Interrupt Flag Clear Register.

ADCINTOVF ADC Interrupt Overflow Register.

ADCINTOVFCLR ADC Interrupt Overflow Clear Register.

INTSELxNy ADC Interrupt Select x and y Register.

rsvd_3 Reserved.

SOCPRICTRL ADC Start Of Conversion Priority Control Register.

rsvd_4 Reserved.

ADCSAMPLEMODE ADC Sample Mode Register.

rsvd_5 Reserved.

ADCINTSOCSEL1 ADC Interrupt Trigger SOC Select 1 Register.

ADCINTSOCSEL2 ADC Interrupt Trigger SOC Select 2 Register.

rsvd_6 Reserved.

ADCSOCFLG1 ADC SOC Flag 1 Register.

rsvd_7 Reserved.

ADCSOCFRC1 ADC SOC Force 1 Register.

rsvd_8 Reserved.

ADCSOCOVF1 ADC SOC Overflow 1 Register.

rsvd_9 Reserved.

ADCSOCOVFCLR1 ADC SOC Overflow Clear 1 Register.

rsvd_10 Reserved.

ADCSOCxCTL ADC SOCx Control Registers.

rsvd_11 Reserved.

ADCREFTRIM ADC Reference/Gain Trim Register.

ADCOFFTRIM ADC Offset Trim Register.

rsvd_12 Reserved.

ADCREV ADC Revision Register.

Description:

Defines the analog-to-digital converter (ADC) object.

4 Capture (CAP)

[Introduction](#) ??

[API Functions](#) 15 The Enhanced Capture (CAP) API provides a set of functions and data structs for configuring and capturing data as well as generating PWMs with the capture peripheral.

This driver is contained in `f2802x/common/source/cap.c`, with `f2802x/common/include/cap.h` containing the API definitions and data structs for use by applications.

4.1 CAP

Data Structures

struct `_CAP_Obj`

Macros

```
#define CAP_ECCTL1_CAP1POL_BITS
#define CAP_ECCTL1_CAP2POL_BITS
#define CAP_ECCTL1_CAP3POL_BITS
#define CAP_ECCTL1_CAP4POL_BITS
#define CAP_ECCTL1_CAPLDEN_BITS
#define CAP_ECCTL1_CTRRST1_BITS
#define CAP_ECCTL1_CTRRST2_BITS
#define CAP_ECCTL1_CTRRST3_BITS
#define CAP_ECCTL1_CTRRST4_BITS
#define CAP_ECCTL1_FREESOFT_BITS
#define CAP_ECCTL1_PRESCALE_BITS
#define CAP_ECCTL2_APWMPOL_BITS
#define CAP_ECCTL2_CAPAPWM_BITS
#define CAP_ECCTL2_CONTONESHOT_BITS
#define CAP_ECCTL2_REARM_BITS
#define CAP_ECCTL2_STOP_WRAP_BITS
#define CAP_ECCTL2_SWSYNC_BITS
```

```
#define CAP_ECCTL2_SYNCIEN_BITS
#define CAP_ECCTL2_SYNCOSSEL_BITS
#define CAP_ECCTL2_TSCTRSTOP_BITS
#define CAP_ECCxxx_C EVT1_BITS
#define CAP_ECCxxx_C EVT2_BITS
#define CAP_ECCxxx_C EVT3_BITS
#define CAP_ECCxxx_C EVT4_BITS
#define CAP_ECCxxx_C TRCOMP_BITS
#define CAP_ECCxxx_C TROVF_BITS
#define CAP_ECCxxx_C TRPRD_BITS
#define CAP_ECCxxx_INT_BITS
#define CAPA_BASE_ADDR
```

Typedefs

```
typedef struct _CAP_Obj * CAP_Handle
typedef struct _CAP_Obj CAP_Obj
```

Enumerations

```
enum CAP_Event_e { CAP_Event_1, CAP_Event_2, CAP_Event_3, CAP_Event_4 }

enum CAP_Int_Type_e {
    CAP_Int_Type_CTR_CMP,          CAP_Int_Type_CTR_PRD,          CAP_Int_Type_CTR_OVF,
    CAP_Int_Type_C EVT4,
    CAP_Int_Type_C EVT3, CAP_Int_Type_C EVT2, CAP_Int_Type_C EVT1, CAP_Int_Type_Global,
    CAP_Int_Type_All }

enum CAP_Polarity_e { CAP_Polarity_Rising, CAP_Polarity_Falling }

enum CAP_Prescale_e {
    CAP_Prescale_By_1, CAP_Prescale_By_2, CAP_Prescale_By_4, CAP_Prescale_By_6,
    CAP_Prescale_By_8, CAP_Prescale_By_10, CAP_Prescale_By_12, CAP_Prescale_By_14,
    CAP_Prescale_By_16, CAP_Prescale_By_18, CAP_Prescale_By_20, CAP_Prescale_By_22,
    CAP_Prescale_By_24, CAP_Prescale_By_26, CAP_Prescale_By_28, CAP_Prescale_By_30,
    CAP_Prescale_By_32, CAP_Prescale_By_34, CAP_Prescale_By_36, CAP_Prescale_By_38,
    CAP_Prescale_By_40, CAP_Prescale_By_42, CAP_Prescale_By_44, CAP_Prescale_By_46,
    CAP_Prescale_By_48, CAP_Prescale_By_50, CAP_Prescale_By_52, CAP_Prescale_By_54,
    CAP_Prescale_By_56, CAP_Prescale_By_58, CAP_Prescale_By_60, CAP_Prescale_By_62 }
```

```
enum CAP_Reset_e { CAP_Reset_Disable, CAP_Reset_Enable }

enum CAP_RunMode_e

enum CAP_Stop_Wrap_e { CAP_Stop_Wrap_C EVT1, CAP_Stop_Wrap_C EVT2,
CAP_Stop_Wrap_C EVT3, CAP_Stop_Wrap_C EVT4 }

enum CAP_SyncOut_e { CAP_SyncOut_SyncIn, CAP_SyncOut_CTRPRD,
CAP_SyncOut_Disable }
```

Functions

```
void CAP_clearInt (CAP_Handle capHandle, const CAP_Int_Type_e intType)

void CAP_disableCaptureLoad (CAP_Handle capHandle)

void CAP_disableInt (CAP_Handle capHandle, const CAP_Int_Type_e intType)

void CAP_disableSyncIn (CAP_Handle capHandle)

void CAP_disableTimestampCounter (CAP_Handle capHandle)

void CAP_enableCaptureLoad (CAP_Handle capHandle)

void CAP_enableInt (CAP_Handle capHandle, const CAP_Int_Type_e intType)

void CAP_enableSyncIn (CAP_Handle capHandle)

void CAP_enableTimestampCounter (CAP_Handle capHandle)

uint32_t CAP_getCap1 (CAP_Handle capHandle)

uint32_t CAP_getCap2 (CAP_Handle capHandle)

uint32_t CAP_getCap3 (CAP_Handle capHandle)

uint32_t CAP_getCap4 (CAP_Handle capHandle)

CAP_Handle CAP_init (void *pMemory, const size_t numBytes)

void CAP_rearm (CAP_Handle capHandle)

void CAP_setApwmCompare (CAP_Handle capHandle, const uint32_t compare)

void CAP_setApwmPeriod (CAP_Handle capHandle, const uint32_t period)

void CAP_setApwmShadowPeriod (CAP_Handle capHandle, const uint32_t shadowPeriod)

void CAP_setCapContinuous (CAP_Handle capHandle)

void CAP_setCapEvtPolarity (CAP_Handle capHandle, const CAP_Event_e event, const
CAP_Polarity_e polarity)

void CAP_setCapEvtReset (CAP_Handle capHandle, const CAP_Event_e event, const
CAP_Reset_e reset)
```

```
void CAP_setCapOneShot (CAP_Handle capHandle)
void CAP_setModeApwm (CAP_Handle capHandle)
void CAP_setModeCap (CAP_Handle capHandle)
void CAP_setStopWrap (CAP_Handle capHandle, const CAP_Stop_Wrap_e stopWrap)
void CAP_setSyncOut (CAP_Handle capHandle, const CAP_SyncOut_e syncOut)
```

4.1.1 Detailed Description

4.1.2 Enumeration Type Documentation

4.1.2.1 enum **CAP_Event_e**

Enumeration to define the capture (CAP) events.

Enumerator

CAP_Event_1 Capture Event 1.
CAP_Event_2 Capture Event 2.
CAP_Event_3 Capture Event 3.
CAP_Event_4 Capture Event 4.

4.1.2.2 enum **CAP_Int_Type_e**

Enumeration to define the capture (CAP) interrupts.

Enumerator

CAP_Int_Type_CTR_CMP Denotes CTR = CMP interrupt.
CAP_Int_Type_CTR_PRD Denotes CTR = PRD interrupt.
CAP_Int_Type_CTR_OVF Denotes CTROVF interrupt.
CAP_Int_Type_C EVT4 Denotes CEVT4 interrupt.
CAP_Int_Type_C EVT3 Denotes CEVT3 interrupt.
CAP_Int_Type_C EVT2 Denotes CEVT2 interrupt.
CAP_Int_Type_C EVT1 Denotes CEVT1 interrupt.
CAP_Int_Type_Global Denotes Capture global interrupt.
CAP_Int_Type_All Denotes All interrupts.

4.1.2.3 enum **CAP_Polarity_e**

Enumeration to define the capture (CAP) event polarities.

Enumerator

CAP_Polarity_Rising Rising Edge Triggered.
CAP_Polarity_Falling Falling Edge Triggered.

4.1.2.4 enum **CAP_Prescale_e**

Enumeration to define the capture (CAP) prescaler values.

Enumerator

CAP_Prescale_By_1 Divide by 1.
CAP_Prescale_By_2 Divide by 2.
CAP_Prescale_By_4 Divide by 4.
CAP_Prescale_By_6 Divide by 6.
CAP_Prescale_By_8 Divide by 8.
CAP_Prescale_By_10 Divide by 10.
CAP_Prescale_By_12 Divide by 12.
CAP_Prescale_By_14 Divide by 14.
CAP_Prescale_By_16 Divide by 16.
CAP_Prescale_By_18 Divide by 18.
CAP_Prescale_By_20 Divide by 20.
CAP_Prescale_By_22 Divide by 22.
CAP_Prescale_By_24 Divide by 24.
CAP_Prescale_By_26 Divide by 26.
CAP_Prescale_By_28 Divide by 28.
CAP_Prescale_By_30 Divide by 30.
CAP_Prescale_By_32 Divide by 32.
CAP_Prescale_By_34 Divide by 34.
CAP_Prescale_By_36 Divide by 36.
CAP_Prescale_By_38 Divide by 38.
CAP_Prescale_By_40 Divide by 40.
CAP_Prescale_By_42 Divide by 42.
CAP_Prescale_By_44 Divide by 44.
CAP_Prescale_By_46 Divide by 46.
CAP_Prescale_By_48 Divide by 48.
CAP_Prescale_By_50 Divide by 50.
CAP_Prescale_By_52 Divide by 52.
CAP_Prescale_By_54 Divide by 54.
CAP_Prescale_By_56 Divide by 56.
CAP_Prescale_By_58 Divide by 58.
CAP_Prescale_By_60 Divide by 60.
CAP_Prescale_By_62 Divide by 62.

4.1.2.5 enum **CAP_Reset_e**

Enumeration to define the capture (CAP) event resets.

Enumerator

CAP_Reset_Disable Disable counter reset on capture event.
CAP_Reset_Enable Enable counter reset on capture event.

4.1.2.6 enum **CAP_Stop_Wrap_e**

Enumeration to define the capture (CAP) Stop/Wrap modes.

Enumerator

CAP_Stop_Wrap_C EVT1 Stop/Wrap after Capture Event 1.
CAP_Stop_Wrap_C EVT2 Stop/Wrap after Capture Event 2.
CAP_Stop_Wrap_C EVT3 Stop/Wrap after Capture Event 3.
CAP_Stop_Wrap_C EVT4 Stop/Wrap after Capture Event 4.

4.1.2.7 enum **CAP_SyncOut_e**

Enumeration to define the Sync Out options.

Enumerator

CAP_SyncOut_SyncIn Sync In used for Sync Out.
CAP_SyncOut_CTRPRD CTR = PRD used for Sync Out.
CAP_SyncOut_Disable Disables Sync Out.

4.1.3 Function Documentation

4.1.3.1 void CAP_clearInt (

CAP_Handle capHandle,

const **CAP_Int_Type_e** intType) [inline]

Clears capture (CAP) interrupt flag.

[1]Parameters in *capHandle* The capture (CAP) object handle

in *intType* The capture interrupt to be cleared

References **_CAP_Obj_::ECECLR**.

4.1.3.2 void CAP_disableCaptureLoad (

CAP_Handle capHandle)

Disables loading of CAP1-4 on capture event.

[1]Parameters in *capHandle* The capture (CAP) object handle

4.1.3.3 void CAP_disableInt (

CAP_Handle capHandle,

const **CAP_Int_Type_e** intType)

Disables capture (CAP) interrupt source.

[1]Parameters in *capHandle* The capture (CAP) object handle

in *intType* The capture interrupt type to be disabled

4.1.3.4 void CAP_disableSyncIn (

CAP_Handle capHandle)

Disables counter synchronization.

[1]Parameters in *capHandle* The capture (CAP) object handle

4.1.3.5 void CAP_disableTimestampCounter (

CAP_Handle capHandle)

Disables Time Stamp counter from running.

[1]Parameters in *capHandle* The capture (CAP) object handle

4.1.3.6 void CAP_enableCaptureLoad (

CAP_Handle capHandle)

Enables loading of CAP1-4 on capture event.

[1]Parameters in *capHandle* The capture (CAP) object handle

4.1.3.7 void CAP_enableInt (

CAP_Handle capHandle,

const **CAP_Int_Type_e** intType)

Enables capture (CAP) interrupt source.

[1]Parameters in *capHandle* The capture (CAP) object handle

in *intType* The capture interrupt type to be enabled

4.1.3.8 void CAP_enableSyncln (
CAP_Handle capHandle)

Enables counter synchronization.

[1]Parameters in *capHandle* The capture (CAP) object handle

4.1.3.9 void CAP_enableTimestampCounter (
CAP_Handle capHandle)

Enables Time Stamp counter to running.

[1]Parameters in *capHandle* The capture (CAP) object handle

4.1.3.10 uint32_t CAP_getCap1 (
CAP_Handle capHandle) [inline]

Gets the CAP1 register value.

[1]Parameters in *capHandle* The capture (CAP) object handle

References _CAP_Obj_::CAP1.

4.1.3.11 uint32_t CAP_getCap2 (
CAP_Handle capHandle) [inline]

Gets the CAP2 register value.

[1]Parameters in *capHandle* The capture (CAP) object handle

References _CAP_Obj_::CAP2.

4.1.3.12 uint32_t CAP_getCap3 (
CAP_Handle capHandle) [inline]

Gets the CAP3 register value.

[1]Parameters in *capHandle* The capture (CAP) object handle

References _CAP_Obj_::CAP3.

4.1.3.13 `uint32_t CAP_getCap4 (``CAP_Handle capHandle) [inline]`

Gets the CAP4 register value.

[1]Parameters in *capHandle* The capture (CAP) object handle

References `_CAP_Obj_::CAP4`.

4.1.3.14 `CAP_Handle CAP_init (``void * pMemory,``const size_t numBytes)`

Initializes the capture (CAP) object handle.

[1]Parameters in *pMemory* A pointer to the base address of the CAP registers

in *numBytes* The number of bytes allocated for the CAP object, bytes

Returns The capture (CAP) object handle

4.1.3.15 `void CAP_rearm (``CAP_Handle capHandle) [inline]`

(Re-)Arm the capture module

[1]Parameters in *capHandle* The capture (CAP) object handle

References `CAP_ECCTL2_REARM_BITS`, and `_CAP_Obj_::ECCTL2`.

4.1.3.16 `void CAP_setApwmCompare (``CAP_Handle capHandle,``const uint32_t compare) [inline]`

Sets the APWM compare value.

[1]Parameters in *capHandle* The capture (CAP) object handle

in *compare* The APWM compare value

References `_CAP_Obj_::CAP2`.

722.7

in polarity The polarity to configure the event for

4.1.3.21 void CAP_setCapEvtReset (
 CAP_Handle capHandle,
 const **CAP_Event_e** event,
 const **CAP_Reset_e** reset)

Sets the capture event counter reset configuration.

[1]Parameters *in capHandle* The capture (CAP) object handle

in event The event to configure

in reset Whether the event should reset the counter or not

4.1.3.22 void CAP_setCapOneShot (
 CAP_Handle capHandle)

Sets up for one-shot Capture.

[1]Parameters *in capHandle* The capture (CAP) object handle

4.1.3.23 void CAP_setModeApwm (
 CAP_Handle capHandle)

Sets capture peripheral up for APWM mode.

[1]Parameters *in capHandle* The capture (CAP) object handle

4.1.3.24 void CAP_setModeCap (
 CAP_Handle capHandle)

Sets capture peripheral up for capture mode.

[1]Parameters *in capHandle* The capture (CAP) object handle

4.1.3.25 void CAP_setStopWrap (
 CAP_Handle capHandle,
 const **CAP_Stop_Wrap_e** stopWrap)

Set the stop/wrap mode.

[1]Parameters in *capHandle* The capture (CAP) object handle

in *stopWrap* The stop/wrap mode to set

4.1.3.26 void CAP_setSyncOut (
 CAP_Handle capHandle,
 const **CAP_SyncOut_e** syncOut)

Set the sync out mode.

[1]Parameters in *capHandle* The capture (CAP) object handle

in *syncOut* The sync out mode to set

5 Device Clocking (CLK)

Introduction	??
API Functions	27

The CLK API provides functions to control the clocking subsystem of the device. Clock dividers and prescalers as well as peripheral clocks can all be set or enabled via this API.

This driver is contained in `f2802x/common/source/clk.c`, with `f2802x/common/include/clk.h` containing the API definitions for use by applications.

5.1 CLK

Data Structures

```
struct _CLK_Obj_
```

Macros

```
#define CLK_BASE_ADDR
#define CLK_CLKCTL_INTOSC1HALTI_BITS
#define CLK_CLKCTL_INTOSC1OFF_BITS
#define CLK_CLKCTL_INTOSC2HALTI_BITS
#define CLK_CLKCTL_INTOSC2OFF_BITS
#define CLK_CLKCTL_NMIRESETSEL_BITS
#define CLK_CLKCTL_OSCCLKSRC2SEL_BITS
#define CLK_CLKCTL_OSCCLKSRCSEL_BITS
#define CLK_CLKCTL_TMR2CLKPRESCALE_BITS
#define CLK_CLKCTL_TMR2CLKSRCSEL_BITS
#define CLK_CLKCTL_WDCLKSRCSEL_BITS
#define CLK_CLKCTL_WDHALTI_BITS
#define CLK_CLKCTL_XCLKINOFF_BITS
#define CLK_CLKCTL_XTALOSCOFF_BITS
#define CLK_LOSPCP_LSPCLK_BITS
#define CLK_PCLKCR0_ADCENCLK_BITS
```

```
#define CLK_PCLKCR0_ECANAENCLK_BITS
#define CLK_PCLKCR0_HRPWMENCLK_BITS
#define CLK_PCLKCR0_I2CAENCLK_BITS
#define CLK_PCLKCR0_LINAENCLK_BITS
#define CLK_PCLKCR0_SCIAENCLK_BITS
#define CLK_PCLKCR0_SPIAENCLK_BITS
#define CLK_PCLKCR0_SPIBENCLK_BITS
#define CLK_PCLKCR0_TBCLKSYNC_BITS
#define CLK_PCLKCR1_ECAP1ENCLK_BITS
#define CLK_PCLKCR1_EPWM1ENCLK_BITS
#define CLK_PCLKCR1_EPWM2ENCLK_BITS
#define CLK_PCLKCR1_EPWM3ENCLK_BITS
#define CLK_PCLKCR1_EPWM4ENCLK_BITS
#define CLK_PCLKCR1_EPWM5ENCLK_BITS
#define CLK_PCLKCR1_EPWM6ENCLK_BITS
#define CLK_PCLKCR1_EPWM7ENCLK_BITS
#define CLK_PCLKCR1_EQEP1ENCLK_BITS
#define CLK_PCLKCR3_CLA1ENCLK_BITS
#define CLK_PCLKCR3_COMP1ENCLK_BITS
#define CLK_PCLKCR3_COMP2ENCLK_BITS
#define CLK_PCLKCR3_COMP3ENCLK_BITS
#define CLK_PCLKCR3_CPUTIMER0ENCLK_BITS
#define CLK_PCLKCR3_CPUTIMER1ENCLK_BITS
#define CLK_PCLKCR3_CPUTIMER2ENCLK_BITS
#define CLK_PCLKCR3_GPIOINENCLK_BITS
#define CLK_XCLK_XCLKINSEL_BITS
#define CLK_XCLK_XCLKOUTDIV_BITS
```

Typedefs

```
typedef struct _CLK_Obj * CLK_Handle
```

```
typedef struct _CLK_Obj CLK_Obj
```

Enumerations

```
enum CLK_ClkInSrc_e
```

```
enum CLK_ClkOutPreScaler_e { CLK_ClkOutPreScaler_SysClkOut_by_4,  
CLK_ClkOutPreScaler_SysClkOut_by_2, CLK_ClkOutPreScaler_SysClkOut_by_1,  
CLK_ClkOutPreScaler_Off }
```

```
enum CLK_CompNumber_e { CLK_CompNumber_1, CLK_CompNumber_2,  
CLK_CompNumber_3 }
```

```
enum CLK_CpuTimerNumber_e { CLK_CpuTimerNumber_0, CLK_CpuTimerNumber_1,  
CLK_CpuTimerNumber_2 }
```

```
enum CLK_LowSpdPreScaler_e {  
CLK_LowSpdPreScaler_SysClkOut_by_1, CLK_LowSpdPreScaler_SysClkOut_by_2,  
CLK_LowSpdPreScaler_SysClkOut_by_4, CLK_LowSpdPreScaler_SysClkOut_by_6,  
CLK_LowSpdPreScaler_SysClkOut_by_8, CLK_LowSpdPreScaler_SysClkOut_by_10,  
CLK_LowSpdPreScaler_SysClkOut_by_12, CLK_LowSpdPreScaler_SysClkOut_by_14 }
```

```
enum CLK_Osc2Src_e { CLK_Osc2Src_Internal, CLK_Osc2Src_External }
```

```
enum CLK_OscSrc_e { CLK_OscSrc_Internal, CLK_OscSrc_External }
```

```
enum CLK_Timer2PreScaler_e {  
CLK_Timer2PreScaler_by_1, CLK_Timer2PreScaler_by_2, CLK_Timer2PreScaler_by_4,  
CLK_Timer2PreScaler_by_8,  
CLK_Timer2PreScaler_by_16 }
```

```
enum CLK_Timer2Src_e { CLK_Timer2Src_SysClk, CLK_Timer2Src_ExtOsc,  
CLK_Timer2Src_IntOsc1, CLK_Timer2Src_IntOsc2 }
```

```
enum CLK_WdClkSrc_e { CLK_WdClkSrc_IntOsc1, CLK_WdClkSrc_ExtOscOrIntOsc2 }
```

Functions

```
void CLK_disableAdcClock (CLK_Handle clkHandle)
```

```
void CLK_disableClaClock (CLK_Handle clkHandle)
```

```
void CLK_disableCkIn (CLK_Handle clkHandle)
```

```
void CLK_disableCompClock (CLK_Handle clkHandle, const CLK_CompNumber_e compNumber)
```

```
void CLK_disableCpuTimerClock (CLK_Handle clkHandle, const CLK_CpuTimerNumber_e  
cpuTimerNumber)
```

```
void CLK_disableCrystalOsc (CLK_Handle clkHandle)
void CLK_disableEcanaClock (CLK_Handle clkHandle)
void CLK_disableEcap1Clock (CLK_Handle clkHandle)
void CLK_disableEqep1Clock (CLK_Handle clkHandle)
void CLK_disableGpioInputClock (CLK_Handle clkHandle)
void CLK_disableHrPwmClock (CLK_Handle clkHandle)
void CLK_disableI2cClock (CLK_Handle clkHandle)
void CLK_disableLinAClock (CLK_Handle clkHandle)
void CLK_disableOsc1 (CLK_Handle clkHandle)
void CLK_disableOsc1HaltMode (CLK_Handle clkHandle)
void CLK_disableOsc2 (CLK_Handle clkHandle)
void CLK_disableOsc2HaltMode (CLK_Handle clkHandle)
void CLK_disablePwmClock (CLK_Handle clkHandle, const PWM_Number_e pwmNumber)
void CLK_disableSciaClock (CLK_Handle clkHandle)
void CLK_disableSpiaClock (CLK_Handle clkHandle)
void CLK_disableSpibClock (CLK_Handle clkHandle)
void CLK_disableTbClockSync (CLK_Handle clkHandle)
void CLK_disableWatchDogHaltMode (CLK_Handle clkHandle)
void CLK_enableAdcClock (CLK_Handle clkHandle)
void CLK_enableCiaClock (CLK_Handle clkHandle)
void CLK_enableClkIn (CLK_Handle clkHandle)
void CLK_enableCompClock (CLK_Handle clkHandle, const CLK_CompNumber_e compNumber)
void CLK_enableCpuTimerClock (CLK_Handle clkHandle, const CLK_CpuTimerNumber_e
cpuTimerNumber)
void CLK_enableCrystalOsc (CLK_Handle clkHandle)
void CLK_enableEcanaClock (CLK_Handle clkHandle)
void CLK_enableEcap1Clock (CLK_Handle clkHandle)
void CLK_enableEqep1Clock (CLK_Handle clkHandle)
void CLK_enableGpioInputClock (CLK_Handle clkHandle)
void CLK_enableHrPwmClock (CLK_Handle clkHandle)
```

```
void CLK_enableI2cClock (CLK_Handle clkHandle)
void CLK_enableLinAClock (CLK_Handle clkHandle)
void CLK_enableOsc1 (CLK_Handle clkHandle)
void CLK_enableOsc1HaltMode (CLK_Handle clkHandle)
void CLK_enableOsc2 (CLK_Handle clkHandle)
void CLK_enableOsc2HaltMode (CLK_Handle clkHandle)
void CLK_enablePwmClock (CLK_Handle clkHandle, const PWM_Number_e pwmNumber)
void CLK_enableSciaClock (CLK_Handle clkHandle)
void CLK_enableSpiaClock (CLK_Handle clkHandle)
void CLK_enableSpibClock (CLK_Handle clkHandle)
void CLK_enableTbClockSync (CLK_Handle clkHandle)
void CLK_enableWatchDogHaltMode (CLK_Handle clkHandle)
CLK_Handle CLK_init (void *pMemory, const size_t numBytes)
void CLK_setClkOutPreScaler (CLK_Handle clkHandle, const CLK_ClkOutPreScaler_e preScaler)
void CLK_setLowSpdPreScaler (CLK_Handle clkHandle, const CLK_LowSpdPreScaler_e preScaler)
void CLK_setOsc2Src (CLK_Handle clkHandle, const CLK_Osc2Src_e src)
void CLK_setOscSrc (CLK_Handle clkHandle, const CLK_OscSrc_e src)
void CLK_setTimer2PreScaler (CLK_Handle clkHandle, const CLK_Timer2PreScaler_e preScaler)
void CLK_setTimer2Src (CLK_Handle clkHandle, const CLK_Timer2Src_e src)
void CLK_setWatchDogSrc (CLK_Handle clkHandle, const CLK_WdClkSrc_e src)
```

5.1.1 Detailed Description

5.1.2 Enumeration Type Documentation

5.1.2.1 enum **CLK_ClkOutPreScaler_e**

Enumeration to define the external clock output frequency.

Enumerator

CLK_ClkOutPreScaler_SysClkOut_by_4 Denotes XCLKOUT = SYSCLKOUT/4.

CLK_ClkOutPreScaler_SysClkOut_by_2 Denotes XCLKOUT = SYSCLKOUT/2.

CLK_ClkOutPreScaler_SysClkOut_by_1 Denotes XCLKOUT = SYSCLKOUT/1.

CLK_ClkOutPreScaler_Off Denotes XCLKOUT = Off.

5.1.2.2 enum **CLK_CompNumber_e**

Enumeration to define the comparator numbers.

Enumerator

CLK_CompNumber_1 Denotes comparator number 1.

CLK_CompNumber_2 Denotes comparator number 2.

CLK_CompNumber_3 Denotes comparator number 3.

5.1.2.3 enum **CLK_CpuTimerNumber_e**

Enumeration to define the CPU timer numbers.

Enumerator

CLK_CpuTimerNumber_0 Denotes CPU timer number 0.

CLK_CpuTimerNumber_1 Denotes CPU timer number 1.

CLK_CpuTimerNumber_2 Denotes CPU timer number 2.

5.1.2.4 enum **CLK_LowSpdPreScaler_e**

Enumeration to define the low speed clock prescaler, which sets the clock frequency.

Enumerator

CLK_LowSpdPreScaler_SysClkOut_by_1 Denotes Low Speed Clock = SYSCLKOUT/1.

CLK_LowSpdPreScaler_SysClkOut_by_2 Denotes Low Speed Clock = SYSCLKOUT/2.

CLK_LowSpdPreScaler_SysClkOut_by_4 Denotes Low Speed Clock = SYSCLKOUT/4.

CLK_LowSpdPreScaler_SysClkOut_by_6 Denotes Low Speed Clock = SYSCLKOUT/6.

CLK_LowSpdPreScaler_SysClkOut_by_8 Denotes Low Speed Clock = SYSCLKOUT/8.

CLK_LowSpdPreScaler_SysClkOut_by_10 Denotes Low Speed Clock = SYSCLKOUT/10.

CLK_LowSpdPreScaler_SysClkOut_by_12 Denotes Low Speed Clock = SYSCLKOUT/12.

CLK_LowSpdPreScaler_SysClkOut_by_14 Denotes Low Speed Clock = SYSCLKOUT/14.

5.1.2.5 enum **CLK_Osc2Src_e**

Enumeration to define the clock oscillator 2 source.

Enumerator

CLK_Osc2Src_Internal Denotes an internal oscillator 2 source.

CLK_Osc2Src_External Denotes an external oscillator 2 source.

5.1.2.6 enum **CLK_OscSrc_e**

Enumeration to define the clock oscillator source.

Enumerator

CLK_OscSrc_Internal Denotes an internal oscillator source.

CLK_OscSrc_External Denotes an external oscillator source.

5.1.2.7 enum **CLK_Timer2PreScaler_e**

Enumeration to define the timer 2 prescaler, which sets the timer 2 frequency.

Enumerator

CLK_Timer2PreScaler_by_1 Denotes a CPU timer 2 clock pre-scaler value of divide by 1.

CLK_Timer2PreScaler_by_2 Denotes a CPU timer 2 clock pre-scaler value of divide by 2.

CLK_Timer2PreScaler_by_4 Denotes a CPU timer 2 clock pre-scaler value of divide by 4.

CLK_Timer2PreScaler_by_8 Denotes a CPU timer 2 clock pre-scaler value of divide by 8.

CLK_Timer2PreScaler_by_16 Denotes a CPU timer 2 clock pre-scaler value of divide by 16.

5.1.2.8 enum **CLK_Timer2Src_e**

Enumeration to define the timer 2 source.

Enumerator

CLK_Timer2Src_SysClk Denotes the CPU timer 2 clock source is SYSCLKOUT.

CLK_Timer2Src_ExtOsc Denotes the CPU timer 2 clock source is external oscillator.

CLK_Timer2Src_IntOsc1 Denotes the CPU timer 2 clock source is internal oscillator 1.

CLK_Timer2Src_IntOsc2 Denotes the CPU timer 2 clock source is internal oscillator 2.

5.1.2.9 enum **CLK_WdClkSrc_e**

Enumeration to define the watchdog clock source.

Enumerator

CLK_WdClkSrc_IntOsc1 Denotes the watchdog clock source is internal oscillator 1.

CLK_WdClkSrc_ExtOscOrIntOsc2 Denotes the watchdog clock source is external oscillator or internal oscillator 2.

5.1.3 Function Documentation

5.1.3.1 void CLK_disableAdcClock (

CLK_Handle clkHandle)

Disables the ADC clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.2 void CLK_disableClaClock (

CLK_Handle clkHandle)

Disables the CLA clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.3 void CLK_disableClkIn (

CLK_Handle clkHandle)

Disables the XCLKIN oscillator input.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.4 void CLK_disableCompClock (

CLK_Handle clkHandle,

const **CLK_CompNumber_e** compNumber)

Disables the comparator clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

in *compNumber* The comparator number

5.1.3.5 void CLK_disableCpuTimerClock (

CLK_Handle clkHandle,

const **CLK_CpuTimerNumber_e** cpuTimerNumber)

Disables the CPU timer clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

in *cpuTimerNumber* The CPU timer number

5.1.3.6 void CLK_disableCrystalOsc (

CLK_Handle clkHandle)

Disables the crystal oscillator.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.7 void CLK_disableEcanaClock (

CLK_Handle clkHandle)

Disables the ECANA clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.8 void CLK_disableEcap1Clock (

CLK_Handle clkHandle)

Disables the ECAP1 clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.9 void CLK_disableEqep1Clock (

CLK_Handle clkHandle)

Disables the EQEP1 clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.10 void CLK_disableGpioInputClock (

CLK_Handle clkHandle)

Disables the GPIO input clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.11 void CLK_disableHrPwmClock (

CLK_Handle clkHandle)

Disables the HRPWM clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.12 void CLK_disableI2cClock (

CLK_Handle clkHandle)

Disables the I2C clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.13 void CLK_disableLinAClock (

CLK_Handle clkHandle)

Disables the LIN-A clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.14 void CLK_disableOsc1 (

CLK_Handle clkHandle)

Disables internal oscillator 1.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.15 void CLK_disableOsc1HaltMode (

CLK_Handle clkHandle)

Disables internal oscillator 1 halt mode ignore.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.16 void CLK_disableOsc2 (

CLK_Handle clkHandle)

Disables internal oscillator 2.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.17 void CLK_disableOsc2HaltMode (

CLK_Handle clkHandle)

Disables internal oscillator 2 halt mode ignore.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.18 void CLK_disablePwmClock (
 CLK_Handle clkHandle,
 const **PWM_Number_e** pwmNumber)

Disables the pwm clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

in *pwmNumber* The PWM number

5.1.3.19 void CLK_disableSciaClock (
 CLK_Handle clkHandle)

Disables the SCI-A clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.20 void CLK_disableSpiaClock (
 CLK_Handle clkHandle)

Disables the SPI-A clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.21 void CLK_disableSpibClock (
 CLK_Handle clkHandle)

Disables the SPI-B clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.22 void CLK_disableTbClockSync (
 CLK_Handle clkHandle)

Disables the ePWM module time base clock sync signal.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.23 void CLK_disableWatchDogHaltMode (

CLK_Handle clkHandle)

Disables the watchdog halt mode ignore.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.24 void CLK_enableAdcClock (

CLK_Handle clkHandle)

Enables the ADC clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.25 void CLK_enableClaClock (

CLK_Handle clkHandle)

Enables the CLA clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.26 void CLK_enableClkIn (

CLK_Handle clkHandle)

Enables the XCLKIN oscillator input.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.27 void CLK_enableCompClock (

CLK_Handle clkHandle,

const **CLK_CompNumber_e** compNumber)

Enables the comparator clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

in *compNumber* The comparator number

5.1.3.28 void CLK_enableCpuTimerClock (

CLK_Handle clkHandle,

const **CLK_CpuTimerNumber_e** cpuTimerNumber)

Enables the CPU timer clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

in *cpuTimerNumber* The CPU timer number

5.1.3.29 void CLK_enableCrystalOsc (

CLK_Handle clkHandle)

Enables the crystal oscillator.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.30 void CLK_enableEcanaClock (

CLK_Handle clkHandle)

Enables the ECANA clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.31 void CLK_enableEcap1Clock (

CLK_Handle clkHandle)

Enables the ECAP1 clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.32 void CLK_enableEqep1Clock (

CLK_Handle clkHandle)

Enables the EQEP1 clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.33 void CLK_enableGpioInputClock (

CLK_Handle clkHandle)

Enables the GPIO input clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.34 void CLK_enableHrPwmClock (

CLK_Handle clkHandle)

Enables the HRPWM clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.35 void CLK_enableI2cClock (

CLK_Handle clkHandle)

Enables the I2C clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.36 void CLK_enableLinAClock (

CLK_Handle clkHandle)

Enables the LIN-A clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.37 void CLK_enableOsc1 (

CLK_Handle clkHandle)

Enables internal oscillator 1.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.38 void CLK_enableOsc1HaltMode (

CLK_Handle clkHandle)

Enables internal oscillator 1 halt mode ignore.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.39 void CLK_enableOsc2 (

CLK_Handle clkHandle)

Enables internal oscillator 2.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.40 void CLK_enableOsc2HaltMode (

CLK_Handle clkHandle)

Enables internal oscillator 2 halt mode ignore.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.41 void CLK_enablePwmClock (

CLK_Handle clkHandle,

const **PWM_Number_e** pwmNumber)

Enables the pwm clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

in *pwmNumber* The PWM number

5.1.3.42 void CLK_enableSciaClock (

CLK_Handle clkHandle)

Enables the SCI-A clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

5.1.3.43 void CLK_enableSpiaClock (

CLK_Handle clkHandle)

Enables the SPI-A clock.

[1]Parameters in *clkHandle* The clock (CLK) object handle

722.7

in preScaler The prescaler value

5.1.3.49 void CLK_setLowSpdPreScaler (
 CLK_Handle clkHandle,
 const **CLK_LowSpdPreScaler_e** preScaler)

Sets the low speed peripheral clock prescaler.

[1]Parameters *in clkHandle* The clock (CLK) object handle

in preScaler The prescaler value

5.1.3.50 void CLK_setOsc2Src (
 CLK_Handle clkHandle,
 const **CLK_Osc2Src_e** src)

Sets the oscillator 2 clock source.

[1]Parameters *in clkHandle* The clock (CLK) object handle

in src The oscillator 2 clock source

5.1.3.51 void CLK_setOscSrc (
 CLK_Handle clkHandle,
 const **CLK_OscSrc_e** src)

Sets the oscillator clock source.

[1]Parameters *in clkHandle* The clock (CLK) object handle

in src The oscillator clock source

5.1.3.52 void CLK_setTimer2PreScaler (
 CLK_Handle clkHandle,
 const **CLK_Timer2PreScaler_e** preScaler)

Sets the timer 2 clock prescaler.

[1]Parameters in *clkHandle* The clock (CLK) object handle

in *preScaler* The prescaler value

5.1.3.53 void CLK_setTimer2Src (

CLK_Handle clkHandle,

const **CLK_Timer2Src_e** src)

Sets the timer 2 clock source.

[1]Parameters in *clkHandle* The clock (CLK) object handle

in *src* The timer 2 clock source

5.1.3.54 void CLK_setWatchDogSrc (

CLK_Handle clkHandle,

const **CLK_WdClkSrc_e** src)

Sets the watchdog clock source.

[1]Parameters in *clkHandle* The clock (CLK) object handle

in *src* The watchdog clock source

6 Comparater (COMP)

[Introduction](#)??

[API Functions](#) 45 The Comparator (COMP) API provides the set of functions to configure the analog comparators present on this device.

This driver is contained in `f2802x/common/source/comp.c`, with `f2802x/common/include/comp.h` containing the API definitions for use by applications.

6.1 COMP

Data Structures

```
struct _COMP_Obj_
```

Macros

```
#define COMP1_BASE_ADDR
#define COMP2_BASE_ADDR
#define COMP_COMPCTL_CMPINV_BITS
#define COMP_COMPCTL_COMPDACE_BITS
#define COMP_COMPCTL_COMPSOURCE_BITS
#define COMP_COMPCTL_QUALSEL_BITS
#define COMP_COMPCTL_SYNCSEL_BITS
#define COMP_COMPSTS_COMPSTS_BITS
#define COMP_DACCTL_DACSOURCE_BITS
#define COMP_DACCTL_FREESOFT_BITS
#define COMP_DACCTL_RAMPSOURCE_BITS
```

Typedefs

```
typedef struct _COMP_Obj_ * COMP_Handle
typedef struct _COMP_Obj_ COMP_Obj
```

Enumerations

```
enum COMP_QualSel_e {  
    COMP_QualSel_Sync,           COMP_QualSel_Qual_2,           COMP_QualSel_Qual_3,  
    COMP_QualSel_Qual_4,         COMP_QualSel_Qual_6,           COMP_QualSel_Qual_7,  
    COMP_QualSel_Qual_8,         COMP_QualSel_Qual_10,         COMP_QualSel_Qual_11,  
    COMP_QualSel_Qual_9,         COMP_QualSel_Qual_12,         COMP_QualSel_Qual_13,  
    COMP_QualSel_Qual_10,        COMP_QualSel_Qual_14,         COMP_QualSel_Qual_15,  
    COMP_QualSel_Qual_11,        COMP_QualSel_Qual_16,         COMP_QualSel_Qual_17,  
    COMP_QualSel_Qual_12,        COMP_QualSel_Qual_18,         COMP_QualSel_Qual_19,  
    COMP_QualSel_Qual_13,        COMP_QualSel_Qual_20,         COMP_QualSel_Qual_21,  
    COMP_QualSel_Qual_14,        COMP_QualSel_Qual_22,         COMP_QualSel_Qual_23,  
    COMP_QualSel_Qual_15,        COMP_QualSel_Qual_24,         COMP_QualSel_Qual_25,  
    COMP_QualSel_Qual_16,        COMP_QualSel_Qual_26,         COMP_QualSel_Qual_27,  
    COMP_QualSel_Qual_17,        COMP_QualSel_Qual_28,         COMP_QualSel_Qual_29,  
    COMP_QualSel_Qual_18,        COMP_QualSel_Qual_30,         COMP_QualSel_Qual_31,  
    COMP_QualSel_Qual_19,        COMP_QualSel_Qual_32,         COMP_QualSel_Qual_33 }  
  
enum COMP_RampSyncSrc_e { COMP_RampSyncSrc_PWMSYNC1,  
    COMP_RampSyncSrc_PWMSYNC2, COMP_RampSyncSrc_PWMSYNC3,  
    COMP_RampSyncSrc_PWMSYNC4 }
```

Functions

```
void COMP_disable (COMP_Handle compHandle)  
void COMP_disableDac (COMP_Handle compHandle)  
void COMP_enable (COMP_Handle compHandle)  
void COMP_enableDac (COMP_Handle compHandle)  
COMP_Handle COMP_init (void *pMemory, const size_t numBytes)  
void COMP_setDacValue (COMP_Handle compHandle, uint16_t dacValue)
```

6.1.1 Detailed Description

6.1.2 Enumeration Type Documentation

6.1.2.1 enum **COMP_QualSel_e**

Enumeration to define the comparator (COMP) output qualification.

Enumerator

COMP_QualSel_Sync Synchronize comparator output.

COMP_QualSel_Qual_2 Qualify comparator output with 2 cycles.
COMP_QualSel_Qual_3 Qualify comparator output with 3 cycles.
COMP_QualSel_Qual_4 Qualify comparator output with 4 cycles.
COMP_QualSel_Qual_5 Qualify comparator output with 5 cycles.
COMP_QualSel_Qual_6 Qualify comparator output with 6 cycles.
COMP_QualSel_Qual_7 Qualify comparator output with 7 cycles.
COMP_QualSel_Qual_8 Qualify comparator output with 8 cycles.
COMP_QualSel_Qual_9 Qualify comparator output with 9 cycles.
COMP_QualSel_Qual_10 Qualify comparator output with 10 cycles.
COMP_QualSel_Qual_11 Qualify comparator output with 11 cycles.
COMP_QualSel_Qual_12 Qualify comparator output with 12 cycles.
COMP_QualSel_Qual_13 Qualify comparator output with 13 cycles.
COMP_QualSel_Qual_14 Qualify comparator output with 14 cycles.
COMP_QualSel_Qual_15 Qualify comparator output with 15 cycles.
COMP_QualSel_Qual_16 Qualify comparator output with 16 cycles.
COMP_QualSel_Qual_17 Qualify comparator output with 17 cycles.
COMP_QualSel_Qual_18 Qualify comparator output with 18 cycles.
COMP_QualSel_Qual_19 Qualify comparator output with 19 cycles.
COMP_QualSel_Qual_20 Qualify comparator output with 20 cycles.
COMP_QualSel_Qual_21 Qualify comparator output with 21 cycles.
COMP_QualSel_Qual_22 Qualify comparator output with 22 cycles.
COMP_QualSel_Qual_23 Qualify comparator output with 23 cycles.
COMP_QualSel_Qual_24 Qualify comparator output with 24 cycles.
COMP_QualSel_Qual_25 Qualify comparator output with 25 cycles.
COMP_QualSel_Qual_26 Qualify comparator output with 26 cycles.
COMP_QualSel_Qual_27 Qualify comparator output with 27 cycles.
COMP_QualSel_Qual_28 Qualify comparator output with 28 cycles.
COMP_QualSel_Qual_29 Qualify comparator output with 29 cycles.
COMP_QualSel_Qual_30 Qualify comparator output with 30 cycles.
COMP_QualSel_Qual_31 Qualify comparator output with 31 cycles.
COMP_QualSel_Qual_32 Qualify comparator output with 32 cycles.
COMP_QualSel_Qual_33 Qualify comparator output with 33 cycles.

6.1.2.2 enum **COMP_RampSyncSrc_e**

Enumeration to define the comparator (COMP) ramp generator sync source.

Enumerator

COMP_RampSyncSrc_PWMSYNC1 PWMSync1 used as Ramp Sync.
COMP_RampSyncSrc_PWMSYNC2 PWMSync2 used as Ramp Sync.
COMP_RampSyncSrc_PWMSYNC3 PWMSync3 used as Ramp Sync.
COMP_RampSyncSrc_PWMSYNC4 PWMSync4 used as Ramp Sync.

6.1.3 Function Documentation

6.1.3.1 void COMP_disable (

COMP_Handle compHandle)

Disables the comparator (COMP)

[1]Parameters in *compHandle* The comparator (COMP) object handle

6.1.3.2 void COMP_disableDac (

COMP_Handle compHandle)

Disables the DAC.

[1]Parameters in *compHandle* The comparator (COMP) object handle

6.1.3.3 void COMP_enable (

COMP_Handle compHandle)

Enables the comparator (COMP)

[1]Parameters in *compHandle* The comparator (COMP) object handle

6.1.3.4 void COMP_enableDac (

COMP_Handle compHandle)

Enables the DAC.

[1]Parameters in *compHandle* The comparator (COMP) object handle

6.1.3.5 **COMP_Handle** COMP_init (

void * pMemory,

const size_t numBytes)

Initializes the comparator (COMP) object handle.

[1]Parameters in *pMemory* A pointer to the base address of the COMP registers

in *numBytes* The number of bytes allocated for the COMP object, bytes

Returns The comparator (COMP) object handle

6.1.3.6 void COMP_setDacValue (
 COMP_Handle compHandle,
 uint16_t dacValue) [inline]

Sets the DAC's value.

[1]Parameters in *compHandle* The comparator (COMP) object handle

in *dacValue* The DAC's value

References _COMP_Obj_:DACVAL.

7 Central Processing Unit (CPU)

Introduction??
 API Functions51 The CPU API provides a set of functions for controlling the central processing unit of this device. This driver is unique in that when initialized `NULL` should be passed as the peripheral base address.

This driver is contained in `f2802x/common/source/cpu.c`, with `f2802x/common/include/cpu.h` containing the API definitions for use by applications.

7.1 CPU

Data Structures

struct `_CPU_Obj_`

Macros

```
#define CPU_DBGIER_DLOGINT_BITS
#define CPU_DBGIER_INT10_BITS
#define CPU_DBGIER_INT11_BITS
#define CPU_DBGIER_INT12_BITS
#define CPU_DBGIER_INT13_BITS
#define CPU_DBGIER_INT14_BITS
#define CPU_DBGIER_INT1_BITS
#define CPU_DBGIER_INT2_BITS
#define CPU_DBGIER_INT3_BITS
#define CPU_DBGIER_INT4_BITS
#define CPU_DBGIER_INT5_BITS
#define CPU_DBGIER_INT6_BITS
#define CPU_DBGIER_INT7_BITS
#define CPU_DBGIER_INT8_BITS
#define CPU_DBGIER_INT9_BITS
#define CPU_DBGIER_RTOSINT_BITS
```

```
#define CPU_IER_DLOGINT_BITS
#define CPU_IER_INT10_BITS
#define CPU_IER_INT11_BITS
#define CPU_IER_INT12_BITS
#define CPU_IER_INT13_BITS
#define CPU_IER_INT14_BITS
#define CPU_IER_INT1_BITS
#define CPU_IER_INT2_BITS
#define CPU_IER_INT3_BITS
#define CPU_IER_INT4_BITS
#define CPU_IER_INT5_BITS
#define CPU_IER_INT6_BITS
#define CPU_IER_INT7_BITS
#define CPU_IER_INT8_BITS
#define CPU_IER_INT9_BITS
#define CPU_IER_RTOSINT_BITS
#define CPU_IFR_DLOGINT_BITS
#define CPU_IFR_INT10_BITS
#define CPU_IFR_INT11_BITS
#define CPU_IFR_INT12_BITS
#define CPU_IFR_INT13_BITS
#define CPU_IFR_INT14_BITS
#define CPU_IFR_INT1_BITS
#define CPU_IFR_INT2_BITS
#define CPU_IFR_INT3_BITS
#define CPU_IFR_INT4_BITS
#define CPU_IFR_INT5_BITS
#define CPU_IFR_INT6_BITS
#define CPU_IFR_INT7_BITS
```

```
#define CPU_IFR_INT8_BITS
#define CPU_IFR_INT9_BITS
#define CPU_IFR_RTOSINT_BITS
#define CPU_ST0_C_BITS
#define CPU_ST0_N_BITS
#define CPU_ST0_OVCOVCU_BITS
#define CPU_ST0_OVM_BITS
#define CPU_ST0_PW_BITS
#define CPU_ST0_SXM_BITS
#define CPU_ST0_TC_BITS
#define CPU_ST0_V_BITS
#define CPU_ST0_Z_BITS
#define CPU_ST1_AMODE_BITS
#define CPU_ST1_ARP_BITS
#define CPU_ST1_DBGM_BITS
#define CPU_ST1_EALLOW_BITS
#define CPU_ST1_IDLESTAT_BITS
#define CPU_ST1_INTM_BITS
#define CPU_ST1_LOOP_BITS
#define CPU_ST1_MOM1MAP_BITS
#define CPU_ST1_OBJMODE_BITS
#define CPU_ST1_PAGE0_BITS
#define CPU_ST1_SPA_BITS
#define CPU_ST1_VMAP_BITS
#define CPU_ST1_XF_BITS
#define DINT
#define DISABLE_INTERRUPTS
#define DISABLE_PROTECTED_REGISTER_WRITE_MODE
#define DRTM
```

```
#define EALLOW
#define EDIS
#define EINT
#define ENABLE_INTERRUPTS
#define ENABLE_PROTECTED_REGISTER_WRITE_MODE
#define ERTM
#define ESTOP0
#define IDLE
```

Typedefs

```
typedef struct _CPU_Obj * CPU_Handle
typedef struct _CPU_Obj CPU_Obj
```

Enumerations

```
enum CPU_ExtIntNumber_e { CPU_ExtIntNumber_1, CPU_ExtIntNumber_2,
CPU_ExtIntNumber_3 }

enum CPU_IntNumber_e {
CPU_IntNumber_1, CPU_IntNumber_2, CPU_IntNumber_3, CPU_IntNumber_4,
CPU_IntNumber_5, CPU_IntNumber_6, CPU_IntNumber_7, CPU_IntNumber_8,
CPU_IntNumber_9, CPU_IntNumber_10, CPU_IntNumber_11, CPU_IntNumber_12,
CPU_IntNumber_13, CPU_IntNumber_14 }
```

Functions

```
void CPU_clearIntFlags (CPU_Handle cpuHandle)
void CPU_disableDebugInt (CPU_Handle cpuHandle)
void CPU_disableGlobalInts (CPU_Handle cpuHandle)
void CPU_disableInt (CPU_Handle cpuHandle, const CPU_IntNumber_e intNumber)
void CPU_disableInts (CPU_Handle cpuHandle)
void CPU_disableProtectedRegisterWrite (CPU_Handle cpuHandle)
void CPU_enableDebugInt (CPU_Handle cpuHandle)
void CPU_enableGlobalInts (CPU_Handle cpuHandle)
```

```
void CPU_enableInt (CPU_Handle cpuHandle, const CPU_IntNumber_e intNumber)
```

```
void CPU_enableProtectedRegisterWrite (CPU_Handle cpuHandle)
```

```
CPU_Handle CPU_init (void *pMemory, const size_t numBytes)
```

Variables

`CPU_Obj` `cpu`

register volatile unsigned int `IER`

register volatile unsigned int `IFR`

7.1.1 Detailed Description

7.1.2 Enumeration Type Documentation

7.1.2.1 enum `CPU_ExtIntNumber_e`

Enumeration to define the external interrupt numbers.

Enumerator

`CPU_ExtIntNumber_1` Denotes external interrupt number 1.

`CPU_ExtIntNumber_2` Denotes external interrupt number 2.

`CPU_ExtIntNumber_3` Denotes external interrupt number 3.

7.1.2.2 enum `CPU_IntNumber_e`

Enumeration to define the interrupt numbers.

Enumerator

`CPU_IntNumber_1` Denotes interrupt number 1.

`CPU_IntNumber_2` Denotes interrupt number 2.

`CPU_IntNumber_3` Denotes interrupt number 3.

`CPU_IntNumber_4` Denotes interrupt number 4.

`CPU_IntNumber_5` Denotes interrupt number 5.

`CPU_IntNumber_6` Denotes interrupt number 6.

`CPU_IntNumber_7` Denotes interrupt number 7.

`CPU_IntNumber_8` Denotes interrupt number 8.

`CPU_IntNumber_9` Denotes interrupt number 9.

`CPU_IntNumber_10` Denotes interrupt number 10.

`CPU_IntNumber_11` Denotes interrupt number 11.

`CPU_IntNumber_12` Denotes interrupt number 12.

`CPU_IntNumber_13` Denotes interrupt number 13.

`CPU_IntNumber_14` Denotes interrupt number 14.

7.1.3 Function Documentation

7.1.3.1 void CPU_clearIntFlags (

CPU_Handle cpuHandle)

Clears all interrupt flags.

[1]Parameters in *cpuHandle* The central processing unit (CPU) object handle

7.1.3.2 void CPU_disableDebugInt (

CPU_Handle cpuHandle)

Disables the debug interrupt.

[1]Parameters in *cpuHandle* The central processing unit (CPU) object handle

7.1.3.3 void CPU_disableGlobalInts (

CPU_Handle cpuHandle)

Disables global interrupts.

[1]Parameters in *cpuHandle* The CPU handle

7.1.3.4 void CPU_disableInt (

CPU_Handle cpuHandle,

const **CPU_IntNumber_e** intNumber)

Disables a specified interrupt number.

[1]Parameters in *cpuHandle* The central processing unit (CPU) object handle

in *intNumber* The interrupt number

7.1.3.5 void CPU_disableInts (

CPU_Handle cpuHandle)

Disables all interrupts.

[1]Parameters in *cpuHandle* The central processing unit (CPU) object handle

7.1.3.6 void CPU_disableProtectedRegisterWrite (
 CPU_Handle cpuHandle)

Disables protected register writes.

[1]Parameters in *cpuHandle* The central processing unit (CPU) object handle

7.1.3.7 void CPU_enableDebugInt (
 CPU_Handle cpuHandle)

Enables the debug interrupt.

[1]Parameters in *cpuHandle* The CPU handle

7.1.3.8 void CPU_enableGlobalInts (
 CPU_Handle cpuHandle)

Enables global interrupts.

[1]Parameters in *cpuHandle* The CPU handle

7.1.3.9 void CPU_enableInt (
 CPU_Handle cpuHandle,
 const **CPU_IntNumber_e** intNumber)

Enables a specified interrupt number.

[1]Parameters in *cpuHandle* The central processing unit (CPU) object handle

in *intNumber* The interrupt number

7.1.3.10 void CPU_enableProtectedRegisterWrite (
 CPU_Handle cpuHandle)

Enables protected register writes.

[1]Parameters in *cpuHandle* The central processing unit (CPU) object handle

7.1.3.11 **CPU_Handle** CPU_init (

void * pMemory,

const size_t numBytes)

Initializes the central processing unit (CPU) object handle.

[1]Parameters in *pMemory* A pointer to the memory for the CPU object

in *numBytes* The number of bytes allocated for the CPU object, bytes

Returns The central processing unit (CPU) object handle

8 Flash

Introduction	??
API Functions	59

The Flash API contains functions for configuring the wait states as well as run and sleep modes of flash in the device.

CAUTION: The flash function(s) should only be run from RAM. Please copy the function to RAM using the memcpy function found in the run time support library before calling any of them.

This driver is contained in `f2802x/common/source/flash.c`, with `f2802x/common/include/flash.h` containing the API definitions for use by applications.

8.1 FLASH

Data Structures

struct `_FLASH_Obj_`

Macros

```
#define FLASH_ACTIVE_WAIT_COUNT_DEFAULT
#define FLASH_BASE_ADDR
#define FLASH_FACTIVEWAIT_ACTIVWAIT_BITS
#define FLASH_FBANKWAIT_PAGEWAIT_BITS
#define FLASH_FBANKWAIT_RANDWAIT_BITS
#define FLASH_FOPT_ENPIPE_BITS
#define FLASH_FOTPWAIT_OTPWAIT_BITS
#define FLASH_FPWR_PWR_BITS
#define FLASH_FSTATUS_3VSTAT_BITS
#define FLASH_FSTATUS_ACTIVWAITS_BITS
#define FLASH_FSTATUS_PWRS_BITS
#define FLASH_FSTATUS_STDBYWAITS_BITS
#define FLASH_FSTDBYWAIT_STDBYWAIT_BITS
#define FLASH_STANDBY_WAIT_COUNT_DEFAULT
```

Typedefs

```
typedef struct _FLASH_Obj_ * FLASH_Handle
```

```
typedef struct _FLASH_Obj_ FLASH_Obj
```

Enumerations

```
enum FLASH_3VStatus_e { FLASH_3VStatus_InRange, FLASH_3VStatus_OutOfRange }
```

```
enum FLASH_CounterStatus_e { FLASH_CounterStatus_NotCounting, FLASH_CounterStatus_Counting }
```

```
enum FLASH_NumOtpWaitStates_e {  
    FLASH_NumOtpWaitStates_1, FLASH_NumOtpWaitStates_2, FLASH_NumOtpWaitStates_3,  
    FLASH_NumOtpWaitStates_4,  
    FLASH_NumOtpWaitStates_5, FLASH_NumOtpWaitStates_6, FLASH_NumOtpWaitStates_7,  
    FLASH_NumOtpWaitStates_8,  
    FLASH_NumOtpWaitStates_9, FLASH_NumOtpWaitStates_10, FLASH_NumOtpWaitStates_11,  
    FLASH_NumOtpWaitStates_12,  
    FLASH_NumOtpWaitStates_13, FLASH_NumOtpWaitStates_14, FLASH_NumOtpWaitStates_15  
}
```

```
enum FLASH_NumPagedWaitStates_e {  
    FLASH_NumPagedWaitStates_0, FLASH_NumPagedWaitStates_1,  
    FLASH_NumPagedWaitStates_2, FLASH_NumPagedWaitStates_3,  
    FLASH_NumPagedWaitStates_4, FLASH_NumPagedWaitStates_5,  
    FLASH_NumPagedWaitStates_6, FLASH_NumPagedWaitStates_7,  
    FLASH_NumPagedWaitStates_8, FLASH_NumPagedWaitStates_9,  
    FLASH_NumPagedWaitStates_10, FLASH_NumPagedWaitStates_11,  
    FLASH_NumPagedWaitStates_12, FLASH_NumPagedWaitStates_13,  
    FLASH_NumPagedWaitStates_14, FLASH_NumPagedWaitStates_15 }
```

```
enum FLASH_NumRandomWaitStates_e {  
    FLASH_NumRandomWaitStates_1, FLASH_NumRandomWaitStates_2,  
    FLASH_NumRandomWaitStates_3, FLASH_NumRandomWaitStates_4,  
    FLASH_NumRandomWaitStates_5, FLASH_NumRandomWaitStates_6,  
    FLASH_NumRandomWaitStates_7, FLASH_NumRandomWaitStates_8,  
    FLASH_NumRandomWaitStates_9, FLASH_NumRandomWaitStates_10,  
    FLASH_NumRandomWaitStates_11, FLASH_NumRandomWaitStates_12,  
    FLASH_NumRandomWaitStates_13, FLASH_NumRandomWaitStates_14,  
    FLASH_NumRandomWaitStates_15 }
```

```
enum FLASH_PowerMode_e { FLASH_PowerMode_PumpAndBankSleep, FLASH_PowerMode_PumpAndBankStandby, FLASH_PowerMode_PumpAndBankActive }
```

Functions

```
void FLASH_clear3VStatus (FLASH_Handle flashHandle)
```

```
void FLASH_disablePipelineMode (FLASH_Handle flashHandle)
```

```

void FLASH_enablePipelineMode (FLASH_Handle flashHandle)

FLASH_3VStatus_e FLASH_get3VStatus (FLASH_Handle flashHandle)

uint16_t FLASH_getActiveWaitCount (FLASH_Handle flashHandle)

FLASH_CounterStatus_e FLASH_getActiveWaitStatus (FLASH_Handle flashHandle)

FLASH_PowerMode_e FLASH_getPowerMode (FLASH_Handle flashHandle)

uint16_t FLASH_getStandbyWaitCount (FLASH_Handle flashHandle)

FLASH_CounterStatus_e FLASH_getStandbyWaitStatus (FLASH_Handle flashHandle)

FLASH_Handle FLASH_init (void *pMemory, const size_t numBytes)

void FLASH_setActiveWaitCount (FLASH_Handle flashHandle, const uint16_t count)

void FLASH_setNumPagedReadWaitStates (FLASH_Handle flashHandle, const
FLASH_NumPagedWaitStates_e numStates)

void FLASH_setNumRandomReadWaitStates (FLASH_Handle flashHandle, const
FLASH_NumRandomWaitStates_e numStates)

void FLASH_setOtpWaitStates (FLASH_Handle flashHandle, const FLASH_NumOtpWaitStates_e
numStates)

void FLASH_setPowerMode (FLASH_Handle flashHandle, const FLASH_PowerMode_e mode)

void FLASH_setStandbyWaitCount (FLASH_Handle flashHandle, const uint16_t count)

void FLASH_setup (FLASH_Handle flashHandle)

```

8.1.1 Detailed Description

8.1.2 Enumeration Type Documentation

8.1.2.1 enum **FLASH_3VStatus_e**

Enumeration to define the 3V status.

Enumerator

FLASH_3VStatus_InRange Denotes the 3V flash voltage is in range.

FLASH_3VStatus_OutOfRange Denotes the 3V flash voltage went out of range.

8.1.2.2 enum **FLASH_CounterStatus_e**

Enumeration to define the counter status.

Enumerator

FLASH_CounterStatus_NotCounting Denotes the flash counter is not counting.

FLASH_CounterStatus_Counting Denotes the flash counter is counting.

8.1.2.3 enum **FLASH_NumOtpWaitStates_e**

Enumeration to define the number of one-time programmable wait states.

Enumerator

- FLASH_NumOtpWaitStates_1** Denotes the number of one-time programmable (OTP) wait states is 1.
- FLASH_NumOtpWaitStates_2** Denotes the number of one-time programmable (OTP) wait states is 2.
- FLASH_NumOtpWaitStates_3** Denotes the number of one-time programmable (OTP) wait states is 3.
- FLASH_NumOtpWaitStates_4** Denotes the number of one-time programmable (OTP) wait states is 4.
- FLASH_NumOtpWaitStates_5** Denotes the number of one-time programmable (OTP) wait states is 5.
- FLASH_NumOtpWaitStates_6** Denotes the number of one-time programmable (OTP) wait states is 6.
- FLASH_NumOtpWaitStates_7** Denotes the number of one-time programmable (OTP) wait states is 7.
- FLASH_NumOtpWaitStates_8** Denotes the number of one-time programmable (OTP) wait states is 8.
- FLASH_NumOtpWaitStates_9** Denotes the number of one-time programmable (OTP) wait states is 9.
- FLASH_NumOtpWaitStates_10** Denotes the number of one-time programmable (OTP) wait states is 10.
- FLASH_NumOtpWaitStates_11** Denotes the number of one-time programmable (OTP) wait states is 11.
- FLASH_NumOtpWaitStates_12** Denotes the number of one-time programmable (OTP) wait states is 12.
- FLASH_NumOtpWaitStates_13** Denotes the number of one-time programmable (OTP) wait states is 13.
- FLASH_NumOtpWaitStates_14** Denotes the number of one-time programmable (OTP) wait states is 14.
- FLASH_NumOtpWaitStates_15** Denotes the number of one-time programmable (OTP) wait states is 15.

8.1.2.4 enum **FLASH_NumPagedWaitStates_e**

Enumeration to define the number of paged wait states.

Enumerator

- FLASH_NumPagedWaitStates_0** Denotes the number of paged read wait states is 0.
- FLASH_NumPagedWaitStates_1** Denotes the number of paged read wait states is 1.
- FLASH_NumPagedWaitStates_2** Denotes the number of paged read wait states is 2.
- FLASH_NumPagedWaitStates_3** Denotes the number of paged read wait states is 3.
- FLASH_NumPagedWaitStates_4** Denotes the number of paged read wait states is 4.
- FLASH_NumPagedWaitStates_5** Denotes the number of paged read wait states is 5.
- FLASH_NumPagedWaitStates_6** Denotes the number of paged read wait states is 6.

FLASH_NumPagedWaitStates_7 Denotes the number of paged read wait states is 7.
FLASH_NumPagedWaitStates_8 Denotes the number of paged read wait states is 8.
FLASH_NumPagedWaitStates_9 Denotes the number of paged read wait states is 9.
FLASH_NumPagedWaitStates_10 Denotes the number of paged read wait states is 10.
FLASH_NumPagedWaitStates_11 Denotes the number of paged read wait states is 11.
FLASH_NumPagedWaitStates_12 Denotes the number of paged read wait states is 12.
FLASH_NumPagedWaitStates_13 Denotes the number of paged read wait states is 13.
FLASH_NumPagedWaitStates_14 Denotes the number of paged read wait states is 14.
FLASH_NumPagedWaitStates_15 Denotes the number of paged read wait states is 15.

8.1.2.5 enum **FLASH_NumRandomWaitStates_e**

Enumeration to define the number of random wait states.

Enumerator

FLASH_NumRandomWaitStates_1 Denotes the number of random read wait states is 1.
FLASH_NumRandomWaitStates_2 Denotes the number of random read wait states is 2.
FLASH_NumRandomWaitStates_3 Denotes the number of random read wait states is 3.
FLASH_NumRandomWaitStates_4 Denotes the number of random read wait states is 4.
FLASH_NumRandomWaitStates_5 Denotes the number of random read wait states is 5.
FLASH_NumRandomWaitStates_6 Denotes the number of random read wait states is 6.
FLASH_NumRandomWaitStates_7 Denotes the number of random read wait states is 7.
FLASH_NumRandomWaitStates_8 Denotes the number of random read wait states is 8.
FLASH_NumRandomWaitStates_9 Denotes the number of random read wait states is 9.
FLASH_NumRandomWaitStates_10 Denotes the number of random read wait states is 10.
FLASH_NumRandomWaitStates_11 Denotes the number of random read wait states is 11.
FLASH_NumRandomWaitStates_12 Denotes the number of random read wait states is 12.
FLASH_NumRandomWaitStates_13 Denotes the number of random read wait states is 13.
FLASH_NumRandomWaitStates_14 Denotes the number of random read wait states is 14.
FLASH_NumRandomWaitStates_15 Denotes the number of random read wait states is 15.

8.1.2.6 enum **FLASH_PowerMode_e**

Enumeration to define the power modes.

Enumerator

FLASH_PowerMode_PumpAndBankSleep Denotes a pump and bank sleep power mode.
FLASH_PowerMode_PumpAndBankStandby Denotes a pump and bank standby power mode.
FLASH_PowerMode_PumpAndBankActive Denotes a pump and bank active power mode.

8.1.3 Function Documentation

8.1.3.1 void FLASH_clear3VStatus (

FLASH_Handle flashHandle)

Clears the 3V status.

[1]Parameters in *flashHandle* The flash (FLASH) object handle

8.1.3.2 void FLASH_disablePipelineMode (

FLASH_Handle flashHandle)

Disables the pipeline mode.

[1]Parameters in *flashHandle* The flash (FLASH) object handle

8.1.3.3 void FLASH_enablePipelineMode (

FLASH_Handle flashHandle)

Enables the pipeline mode.

[1]Parameters in *flashHandle* The flash (FLASH) object handle

8.1.3.4 **FLASH_3VStatus_e** FLASH_get3VStatus (

FLASH_Handle flashHandle)

Gets the 3V status.

[1]Parameters in *flashHandle* The flash (FLASH) object handle

Returns The 3V status

8.1.3.5 uint16_t FLASH_getActiveWaitCount (

FLASH_Handle flashHandle)

Gets the active wait count.

[1]Parameters in *flashHandle* The flash (FLASH) object handle

Returns The active wait count

722.7

in numBytes The number of bytes allocated for the FLASH object, bytes

Returns The flash (FLASH) object handle

8.1.3.11 void FLASH_setActiveWaitCount (

FLASH_Handle flashHandle,

const uint16_t count)

Sets the active wait count.

[1]Parameters *in flashHandle* The flash (FLASH) object handle

in count The active wait count

8.1.3.12 void FLASH_setNumPagedReadWaitStates (

FLASH_Handle flashHandle,

const **FLASH_NumPagedWaitStates_e** numStates)

Sets the number of paged read wait states.

[1]Parameters *in flashHandle* The flash (FLASH) object handle

in numStates The number of paged read wait states

8.1.3.13 void FLASH_setNumRandomReadWaitStates (

FLASH_Handle flashHandle,

const **FLASH_NumRandomWaitStates_e** numStates)

Sets the number of random read wait states.

[1]Parameters *in flashHandle* The flash (FLASH) object handle

in numStates The number of random read wait states

8.1.3.14 void FLASH_setOtpWaitStates (

FLASH_Handle flashHandle,

const **FLASH_NumOtpWaitStates_e** numStates)

Sets the number of one-time programmable (OTP) wait states.

[1]Parameters in *flashHandle* The flash (FLASH) object handle

in *numStates* The number of one-time programmable (OTP) wait states

8.1.3.15 void FLASH_setPowerMode (

FLASH_Handle flashHandle,

const **FLASH_PowerMode_e** mode)

Sets the power mode.

[1]Parameters in *flashHandle* The flash (FLASH) object handle

in *mode* The power mode

8.1.3.16 void FLASH_setStandbyWaitCount (

FLASH_Handle flashHandle,

const uint16_t count)

Sets the standby wait count.

[1]Parameters in *flashHandle* The flash (FLASH) object handle

in *count* The standby wait count

8.1.3.17 void FLASH_setup (

FLASH_Handle flashHandle)

Setup flash for optimal performance.

[1]Parameters in *flashHandle* The flash (FLASH) object handle

9 General Purpose Input/Output (GPIO)

[Introduction](#)??

[API Functions](#) 69 The GPIO API provides functions to configure and control the GPIO of this device. Pins can be set as inputs, outputs, or a peripheral function and their value set or read via this API.

This driver is contained in `f2802x/common/source/gpio.c`, with `f2802x/common/include/gpio.h` containing the API definitions for use by applications.

9.1 GPIO

Data Structures

`struct _GPIO_Obj_`

Macros

```
#define GPIO_BASE_ADDR
#define GPIO_GPMUX_CONFIG_BITS
#define GPIO_GPMUX_NUMGPIOs_PER_REG
#define GPIO_GPxCtrl_QUALPRDx_BITS
#define GPIO_GPxCtrl_QUALPRDx_NUMBITS_PER_REG
#define GPIO_GPxQSELx_NUMGPIOs_PER_REG
#define GPIO_GPxQSELy_GPIOx_BITS
```

Typedefs

```
typedef struct _GPIO_Obj_ * GPIO_Handle
typedef struct _GPIO_Obj_ GPIO_Obj
```

Enumerations

```
enum GPIO_Direction_e { GPIO_Direction_Input, GPIO_Direction_Output }

enum GPIO_Mode_e {
    GPIO_0_Mode_GeneralPurpose,    GPIO_0_Mode_EPWM1A,    GPIO_0_Mode_Rsvd_2,
    GPIO_0_Mode_Rsvd_3,            GPIO_1_Mode_EPWM1B,    GPIO_1_Mode_Rsvd_2,
    GPIO_1_Mode_GeneralPurpose,
```

```
GPIO_1_Mode_COMP1OUT,
GPIO_2_Mode_GeneralPurpose, GPIO_2_Mode_EPWM2A, GPIO_2_Mode_Rsvd_2,
GPIO_2_Mode_Rsvd_3,
GPIO_3_Mode_GeneralPurpose, GPIO_3_Mode_EPWM2B, GPIO_3_Mode_Rsvd_2,
GPIO_3_Mode_COMP2OUT,
GPIO_4_Mode_GeneralPurpose, GPIO_4_Mode_EPWM3A, GPIO_4_Mode_Rsvd_2,
GPIO_4_Mode_Rsvd_3,
GPIO_5_Mode_GeneralPurpose, GPIO_5_Mode_EPWM3B, GPIO_5_Mode_Rsvd_2,
GPIO_5_Mode_ECAP1,
GPIO_6_Mode_GeneralPurpose, GPIO_6_Mode_EPWM4A, GPIO_6_Mode_EPWMSYNCI,
GPIO_6_Mode_EPWMSYNCO,
GPIO_7_Mode_GeneralPurpose, GPIO_7_Mode_EPWM4B, GPIO_7_Mode_SCIRXDA,
GPIO_7_Mode_Rsvd_3,
GPIO_12_Mode_GeneralPurpose, GPIO_12_Mode_TZ1_NOT, GPIO_12_Mode_SCITXDA,
GPIO_12_Mode_Rsvd_3,
GPIO_16_Mode_GeneralPurpose, GPIO_16_Mode_SPISIMOA, GPIO_16_Mode_Rsvd_2,
GPIO_16_Mode_TZ2_NOT,
GPIO_17_Mode_GeneralPurpose, GPIO_17_Mode_SPISOMIA, GPIO_17_Mode_Rsvd_2,
GPIO_17_Mode_TZ3_NOT,
GPIO_18_Mode_GeneralPurpose, GPIO_18_Mode_SPICLKA, GPIO_18_Mode_SCITXDA,
GPIO_18_Mode_XCLKOUT,
GPIO_19_Mode_GeneralPurpose, GPIO_19_Mode_SPISTEA_NOT, GPIO_19_Mode_SCIRXDA,
GPIO_19_Mode_ECAP1,
GPIO_28_Mode_GeneralPurpose, GPIO_28_Mode_SCIRXDA, GPIO_28_Mode_SDDA,
GPIO_28_Mode_TZ2_NOT,
GPIO_29_Mode_GeneralPurpose, GPIO_29_Mode_SCITXDA, GPIO_29_Mode_SCLA,
GPIO_29_Mode_TZ3_NOT,
GPIO_32_Mode_GeneralPurpose, GPIO_32_Mode_SDAA, GPIO_32_Mode_EPWMSYNCI,
GPIO_32_Mode_ADCSOCOA_NOT,
GPIO_33_Mode_GeneralPurpose, GPIO_33_Mode_SCLA, GPIO_33_Mode_EPWMSYNCO,
GPIO_33_Mode_ADCSOCBO_NOT,
GPIO_34_Mode_GeneralPurpose, GPIO_34_Mode_COMP2OUT, GPIO_34_Mode_Rsvd_2,
GPIO_34_Mode_Rsvd_3,
GPIO_35_Mode_JTAG_TDI, GPIO_35_Mode_Rsvd_1, GPIO_35_Mode_Rsvd_2,
GPIO_35_Mode_Rsvd_3,
GPIO_36_Mode_JTAG_TMS, GPIO_36_Mode_Rsvd_1, GPIO_36_Mode_Rsvd_2,
GPIO_36_Mode_Rsvd_3,
GPIO_37_Mode_JTAG_TDO, GPIO_37_Mode_Rsvd_1, GPIO_37_Mode_Rsvd_2,
GPIO_37_Mode_Rsvd_3,
GPIO_38_Mode_JTAG_TCK, GPIO_38_Mode_Rsvd_1, GPIO_38_Mode_Rsvd_2,
GPIO_38_Mode_Rsvd_3 }

enum GPIO_Number_e {
GPIO_Number_0, GPIO_Number_1, GPIO_Number_2, GPIO_Number_3,
GPIO_Number_4, GPIO_Number_5, GPIO_Number_6, GPIO_Number_7,
GPIO_Rsvd_8, GPIO_Rsvd_9, GPIO_Rsvd_10, GPIO_Rsvd_11,
GPIO_Number_12, GPIO_Rsvd_13, GPIO_Rsvd_14, GPIO_Rsvd_15,
GPIO_Number_16, GPIO_Number_17, GPIO_Number_18, GPIO_Number_19,
GPIO_Rsvd_20, GPIO_Rsvd_21, GPIO_Rsvd_22, GPIO_Rsvd_23,
GPIO_Rsvd_24, GPIO_Rsvd_25, GPIO_Rsvd_26, GPIO_Rsvd_27,
GPIO_Number_28, GPIO_Number_29, GPIO_Rsvd_30, GPIO_Rsvd_31,
GPIO_Number_32, GPIO_Number_33, GPIO_Number_34, GPIO_Number_35,
GPIO_Number_36, GPIO_Number_37, GPIO_Number_38 }
```

```
enum GPIO_Port_e { GPIO_Port_A, GPIO_Port_B }

enum GPIO_PullUp_e { GPIO_PullUp_Enable, GPIO_PullUp_Disable }

enum GPIO_Qual_e { GPIO_Qual_Sync, GPIO_Qual_Sample_3, GPIO_Qual_Sample_6,
GPIO_Qual_ASynC }
```

Functions

```
uint16_t GPIO_getData (GPIO_Handle gpioHandle, const GPIO_Number_e gpioNumber)

uint32_t GPIO_getPortData (GPIO_Handle gpioHandle, const GPIO_Port_e gpioPort)

GPIO_Handle GPIO_init (void *pMemory, const size_t numBytes)

void GPIO_lpmSelect (GPIO_Handle gpioHandle, const GPIO_Number_e gpioNumber)

void GPIO_setDirection (GPIO_Handle gpioHandle, const GPIO_Number_e gpioNumber, const
GPIO_Direction_e direction)

void GPIO_setExtInt (GPIO_Handle gpioHandle, const GPIO_Number_e gpioNumber, const
CPU_ExtIntNumber_e intNumber)

void GPIO_setHigh (GPIO_Handle gpioHandle, const GPIO_Number_e gpioNumber)

void GPIO_setLow (GPIO_Handle gpioHandle, const GPIO_Number_e gpioNumber)

void GPIO_setMode (GPIO_Handle gpioHandle, const GPIO_Number_e gpioNumber, const
GPIO_Mode_e mode)

void GPIO_setPortData (GPIO_Handle gpioHandle, const GPIO_Port_e gpioPort, const uint32_t
data)

void GPIO_setPullUp (GPIO_Handle gpioHandle, const GPIO_Number_e gpioNumber, const
GPIO_PullUp_e pullUp)

void GPIO_setQualification (GPIO_Handle gpioHandle, const GPIO_Number_e gpioNumber, const
GPIO_Qual_e qualification)

void GPIO_setQualificationPeriod (GPIO_Handle gpioHandle, const GPIO_Number_e gpioNum-
ber, const uint8_t period)

void GPIO_toggle (GPIO_Handle gpioHandle, const GPIO_Number_e gpioNumber)
```

9.1.1 Detailed Description

9.1.2 Enumeration Type Documentation

9.1.2.1 enum **GPIO_Direction_e**

Enumeration to define the general purpose I/O (GPIO) directions.

Enumerator

GPIO_Direction_Input Denotes an input direction.
GPIO_Direction_Output Denotes an output direction.

9.1.2.2 enum **GPIO_Mode_e**

Enumeration to define the general purpose I/O (GPIO) modes for each pin.

Enumerator

GPIO_0_Mode_GeneralPurpose Denotes a general purpose function.
GPIO_0_Mode_EPWM1A Denotes a EPWM1A function.
GPIO_0_Mode_Rsvd_2 Denotes a reserved function.
GPIO_0_Mode_Rsvd_3 Denotes a reserved function.
GPIO_1_Mode_GeneralPurpose Denotes a general purpose function.
GPIO_1_Mode_EPWM1B Denotes a EPWM1B function.
GPIO_1_Mode_Rsvd_2 Denotes a reserved function.
GPIO_1_Mode_COMP1OUT Denotes a COMP1OUT function.
GPIO_2_Mode_GeneralPurpose Denotes a general purpose function.
GPIO_2_Mode_EPWM2A Denotes a EPWM2A function.
GPIO_2_Mode_Rsvd_2 Denotes a reserved function.
GPIO_2_Mode_Rsvd_3 Denotes a reserved function.
GPIO_3_Mode_GeneralPurpose Denotes a general purpose function.
GPIO_3_Mode_EPWM2B Denotes a EPWM2B function.
GPIO_3_Mode_Rsvd_2 Denotes a reserved function.
GPIO_3_Mode_COMP2OUT Denotes a COMP2OUT function.
GPIO_4_Mode_GeneralPurpose Denotes a general purpose function.
GPIO_4_Mode_EPWM3A Denotes a EPWM3A function.
GPIO_4_Mode_Rsvd_2 Denotes a reserved function.
GPIO_4_Mode_Rsvd_3 Denotes a reserved function.
GPIO_5_Mode_GeneralPurpose Denotes a general purpose function.
GPIO_5_Mode_EPWM3B Denotes a EPWM3B function.
GPIO_5_Mode_Rsvd_2 Denotes a reserved function.
GPIO_5_Mode_ECAP1 Denotes a ECAP1 function.
GPIO_6_Mode_GeneralPurpose Denotes a general purpose function.
GPIO_6_Mode_EPWM4A Denotes a EPWM4A function.
GPIO_6_Mode_EPWMSYNCl Denotes a EPWMSYNCl function.
GPIO_6_Mode_EPWMSYNCO Denotes a EPWMSYNCO function.
GPIO_7_Mode_GeneralPurpose Denotes a general purpose function.
GPIO_7_Mode_EPWM4B Denotes a EPWM4B function.
GPIO_7_Mode_SCIRXDA Denotes a SCIRXDA function.
GPIO_7_Mode_Rsvd_3 Denotes a reserved function.
GPIO_12_Mode_GeneralPurpose Denotes a general purpose function.
GPIO_12_Mode_TZ1_NOT Denotes a TZ1_NOT function.
GPIO_12_Mode_SCITXDA Denotes a SCITXDA function.
GPIO_12_Mode_Rsvd_3 Denotes a reserved function.

GPIO_16_Mode_GeneralPurpose Denotes a general purpose function.
GPIO_16_Mode_SPISIMOA Denotes a SPISIMOA function.
GPIO_16_Mode_Rsvd_2 Denotes a reserved function.
GPIO_16_Mode_TZ2_NOT Denotes a TZ2_NOT function.
GPIO_17_Mode_GeneralPurpose Denotes a general purpose function.
GPIO_17_Mode_SPISOMIA Denotes a SPISOMIA function.
GPIO_17_Mode_Rsvd_2 Denotes a reserved function.
GPIO_17_Mode_TZ3_NOT Denotes a TZ3_NOT function.
GPIO_18_Mode_GeneralPurpose Denotes a general purpose function.
GPIO_18_Mode_SPICLKA Denotes a SPICLKA function.
GPIO_18_Mode_SCITXDA Denotes a SCITXDA function.
GPIO_18_Mode_XCLKOUT Denotes a XCLKOUT function.
GPIO_19_Mode_GeneralPurpose Denotes a general purpose function.
GPIO_19_Mode_SPISTEA_NOT Denotes a SPISTEA_NOT function.
GPIO_19_Mode_SCIRXDA Denotes a SCIRXDA function.
GPIO_19_Mode_ECAP1 Denotes a ECAP1 function.
GPIO_28_Mode_GeneralPurpose Denotes a general purpose function.
GPIO_28_Mode_SCIRXDA Denotes a SCIRXDA function.
GPIO_28_Mode_SDDA Denotes a SDDA function.
GPIO_28_Mode_TZ2_NOT Denotes a TZ2_NOT function.
GPIO_29_Mode_GeneralPurpose Denotes a general purpose function.
GPIO_29_Mode_SCITXDA Denotes a SCITXDA function.
GPIO_29_Mode_SCLA Denotes a SCLA function.
GPIO_29_Mode_TZ3_NOT Denotes a TZ2_NOT function.
GPIO_32_Mode_GeneralPurpose Denotes a general purpose function.
GPIO_32_Mode_SDAA Denotes a SDDA function.
GPIO_32_Mode_EPWMSYNCI Denotes a EPWMSYNCI function.
GPIO_32_Mode_ADCSOCOA_NOT Denotes a ADCSOCOA_NOT function.
GPIO_33_Mode_GeneralPurpose Denotes a general purpose function.
GPIO_33_Mode_SCLA Denotes a SCLA function.
GPIO_33_Mode_EPWMSYNCO Denotes a EPWMSYNCO function.
GPIO_33_Mode_ADCSOCBO_NOT Denotes a ADCSOCBO_NOT function.
GPIO_34_Mode_GeneralPurpose Denotes a general purpose function.
GPIO_34_Mode_COMP2OUT Denotes a COMP2OUT function.
GPIO_34_Mode_Rsvd_2 Denotes a reserved function.
GPIO_34_Mode_Rsvd_3 Denotes a reserved function.
GPIO_35_Mode_JTAG_TDI Denotes a JTAG_TDI function.
GPIO_35_Mode_Rsvd_1 Denotes a reserved function.
GPIO_35_Mode_Rsvd_2 Denotes a reserved function.
GPIO_35_Mode_Rsvd_3 Denotes a reserved function.
GPIO_36_Mode_JTAG_TMS Denotes a JTAG_TMS function.
GPIO_36_Mode_Rsvd_1 Denotes a reserved function.
GPIO_36_Mode_Rsvd_2 Denotes a reserved function.
GPIO_36_Mode_Rsvd_3 Denotes a reserved function.
GPIO_37_Mode_JTAG_TDO Denotes a JTAG_TDO function.
GPIO_37_Mode_Rsvd_1 Denotes a reserved function.

GPIO_37_Mode_Rsvd_2 Denotes a reserved function.
GPIO_37_Mode_Rsvd_3 Denotes a reserved function.
GPIO_38_Mode_JTAG_TCK Denotes a JTAG_TCK function.
GPIO_38_Mode_Rsvd_1 Denotes a reserved function.
GPIO_38_Mode_Rsvd_2 Denotes a reserved function.
GPIO_38_Mode_Rsvd_3 Denotes a reserved function.

9.1.2.3 enum **GPIO_Number_e**

Enumeration to define the general purpose I/O (GPIO) numbers.

Enumerator

GPIO_Number_0 Denotes GPIO number 0.
GPIO_Number_1 Denotes GPIO number 1.
GPIO_Number_2 Denotes GPIO number 2.
GPIO_Number_3 Denotes GPIO number 3.
GPIO_Number_4 Denotes GPIO number 4.
GPIO_Number_5 Denotes GPIO number 5.
GPIO_Number_6 Denotes GPIO number 6.
GPIO_Number_7 Denotes GPIO number 7.
GPIO_Rsvd_8 This GPIO not present.
GPIO_Rsvd_9 This GPIO not present.
GPIO_Rsvd_10 This GPIO not present.
GPIO_Rsvd_11 This GPIO not present.
GPIO_Number_12 Denotes GPIO number 12.
GPIO_Rsvd_13 This GPIO not present.
GPIO_Rsvd_14 This GPIO not present.
GPIO_Rsvd_15 This GPIO not present.
GPIO_Number_16 Denotes GPIO number 16.
GPIO_Number_17 Denotes GPIO number 17.
GPIO_Number_18 Denotes GPIO number 18.
GPIO_Number_19 Denotes GPIO number 19.
GPIO_Rsvd_20 This GPIO not present.
GPIO_Rsvd_21 This GPIO not present.
GPIO_Rsvd_22 This GPIO not present.
GPIO_Rsvd_23 This GPIO not present.
GPIO_Rsvd_24 This GPIO not present.
GPIO_Rsvd_25 This GPIO not present.
GPIO_Rsvd_26 This GPIO not present.
GPIO_Rsvd_27 This GPIO not present.
GPIO_Number_28 Denotes GPIO number 28.
GPIO_Number_29 Denotes GPIO number 29.
GPIO_Rsvd_30 This GPIO not present.
GPIO_Rsvd_31 This GPIO not present.
GPIO_Number_32 Denotes GPIO number 32.

GPIO_Number_33 Denotes GPIO number 33.
GPIO_Number_34 Denotes GPIO number 34.
GPIO_Number_35 Denotes GPIO number 35.
GPIO_Number_36 Denotes GPIO number 36.
GPIO_Number_37 Denotes GPIO number 37.
GPIO_Number_38 Denotes GPIO number 38.

9.1.2.4 enum **GPIO_Port_e**

Enumeration to define the general purpose I/O (GPIO) ports.

Enumerator

GPIO_Port_A GPIO Port A.
GPIO_Port_B GPIO Port B.

9.1.2.5 enum **GPIO_PullUp_e**

Enumeration to define the general purpose I/O (GPIO) pull ups.

Enumerator

GPIO_PullUp_Enable Denotes pull up will be enabled.
GPIO_PullUp_Disable Denotes pull up will be disabled.

9.1.2.6 enum **GPIO_Qual_e**

Enumeration to define the general purpose I/O (GPIO) qualification.

Enumerator

GPIO_Qual_Sync Denotes input will be synchronized to SYSCLK.
GPIO_Qual_Sample_3 Denotes input is qualified with 3 samples.
GPIO_Qual_Sample_6 Denotes input is qualified with 6 samples.
GPIO_Qual_ASync Denotes input is asynchronous.

9.1.3 Function Documentation

9.1.3.1 uint16_t GPIO_getData (

GPIO_Handle gpioHandle,
const **GPIO_Number_e** gpioNumber)

Returns the data value present on a pin (either input or output)

[1]Parameters in *gpioHandle* The general purpose I/O (GPIO) object handle

in gpioNumber The GPIO number

Returns The boolean state of a pin (high/low)

9.1.3.2 uint32_t GPIO_getPortData (

GPIO_Handle gpioHandle,

const **GPIO_Port_e** gpioPort)

Returns the data value present on a GPIO port.

[1]Parameters *in gpioHandle* The general purpose I/O (GPIO) object handle

in gpioPort The GPIO port

Returns The data values for the specified port

9.1.3.3 **GPIO_Handle** GPIO_init (

void * pMemory,

const size_t numBytes)

Initializes the general purpose I/O (GPIO) object handle.

[1]Parameters *in pMemory* A pointer to the base address of the GPIO registers

in numBytes The number of bytes allocated for the GPIO object, bytes

Returns The general purpose I/O (GPIO) object handle

9.1.3.4 void GPIO_lpmSelect (

GPIO_Handle gpioHandle,

const **GPIO_Number_e** gpioNumber)

Selects a gpio pin to wake up device from STANDBY and HALT LPM.

[1]Parameters *in gpioHandle* The general purpose I/O (GPIO) object handle

in gpioNumber The GPIO number

9.1.3.5 void GPIO_setDirection (

GPIO_Handle gpioHandle,

```
const GPIO_Number_e gpioNumber,  
const GPIO_Direction_e direction )
```

Sets the general purpose I/O (GPIO) signal direction.

[1]Parameters in *gpioHandle* The general purpose I/O (GPIO) object handle

in *gpioNumber* The GPIO number

in *direction* The signal direction

```
9.1.3.6 void GPIO_setExtInt (  
    GPIO_Handle gpioHandle,  
    const GPIO_Number_e gpioNumber,  
    const CPU_ExtIntNumber_e intNumber )
```

Sets the general purpose I/O (GPIO) external interrupt number.

[1]Parameters in *gpioHandle* The general purpose I/O (GPIO) object handle

in *gpioNumber* The GPIO number

in *intNumber* The interrupt number

```
9.1.3.7 void GPIO_setHigh (  
    GPIO_Handle gpioHandle,  
    const GPIO_Number_e gpioNumber )
```

Sets the specified general purpose I/O (GPIO) signal high.

[1]Parameters in *gpioHandle* The general purpose I/O (GPIO) object handle

in *gpioNumber* The GPIO number

```
9.1.3.8 void GPIO_setLow (  
    GPIO_Handle gpioHandle,  
    const GPIO_Number_e gpioNumber )
```

Sets the specified general purpose I/O (GPIO) signal low.

[1]Parameters in *gpioHandle* The general purpose I/O (GPIO) object handle

in *gpioNumber* The GPIO number

9.1.3.9 void GPIO_setMode (
 GPIO_Handle gpioHandle,
 const **GPIO_Number_e** gpioNumber,
 const **GPIO_Mode_e** mode)

Sets the mode for the specified general purpose I/O (GPIO) signal.

[1]Parameters in *gpioHandle* The general purpose I/O (GPIO) object handle

in *gpioNumber* The GPIO number

in *mode* The mode

9.1.3.10 void GPIO_setPortData (
 GPIO_Handle gpioHandle,
 const **GPIO_Port_e** gpioPort,
 const uint32_t data)

Sets data output on a given GPIO port.

[1]Parameters in *gpioHandle* The general purpose I/O (GPIO) object handle

in *gpioPort* The GPIO number

in *data* The data to write to the port

9.1.3.11 void GPIO_setPullUp (
 GPIO_Handle gpioHandle,
 const **GPIO_Number_e** gpioNumber,
 const **GPIO_PullUp_e** pullUp)

Sets the general purpose I/O (GPIO) signal pullups.

[1]Parameters in *gpioHandle* The general purpose I/O (GPIO) object handle

in gpioNumber The GPIO number

in pullUp The pull up enable or disable

9.1.3.12 void GPIO_setQualification (

GPIO_Handle gpioHandle,
const **GPIO_Number_e** gpioNumber,
const **GPIO_Qual_e** qualification)

Sets the qualification for the specified general purpose I/O (GPIO)

[1]Parameters *in gpioHandle* The general purpose I/O (GPIO) object handle

in gpioNumber The GPIO number

in qualification The desired input qualification

9.1.3.13 void GPIO_setQualificationPeriod (

GPIO_Handle gpioHandle,
const **GPIO_Number_e** gpioNumber,
const uint8_t period)

Sets the qualification period for the specified general purpose I/O block (8 I/O's per block)

[1]Parameters *in gpioHandle* The general purpose I/O (GPIO) object handle

in gpioNumber The GPIO number

in period The desired input qualification period

9.1.3.14 void GPIO_toggle (

GPIO_Handle gpioHandle,
const **GPIO_Number_e** gpioNumber)

Toggles the specified general purpose I/O (GPIO) signal.

[1]Parameters *in gpioHandle* The general purpose I/O (GPIO) object handle

in gpioNumber The GPIO number

10 Oscillator (OSC)

[Introduction](#) ??

[OSC API Drivers](#) 81

The oscillator (OSC) API provides functions for configuring an external or internal oscillator as well as compensating the internal oscillator for temperature drift.

This driver is contained in `f2802x/common/source/osc.c`, with `f2802x/common/include/osc.h` containing the API definitions for use by applications.

10.1 OSC

Data Structures

`struct _OSC_Obj_`

Macros

`#define OSC_BASE_ADDR`

`#define OSC_INTOSCnTRIM_COARSE_BITS`

`#define OSC_INTOSCnTRIM_FINE_BITS`

`#define OSC_OTP_COURSE_TRIM1`

`#define OSC_OTP_COURSE_TRIM2`

`#define OSC_OTP_FINE_TRIM_OFFSET1`

`#define OSC_OTP_FINE_TRIM_OFFSET2`

`#define OSC_OTP_FINE_TRIM_SLOPE1`

`#define OSC_OTP_FINE_TRIM_SLOPE2`

`#define OSC_OTP_REF_TEMP_OFFSET`

Typedefs

`typedef struct _OSC_Obj_ * OSC_Handle`

`typedef struct _OSC_Obj_ OSC_Obj`

Enumerations

enum [OSC_Number_e](#) { [OSC_Number_1](#), [OSC_Number_2](#) }

enum [OSC_Osc2Src_e](#) { [OSC_Osc2Src_Internal](#), [OSC_Osc2Src_External](#) }

enum [OSC_Src_e](#) { [OSC_Src_Internal](#), [OSC_Src_External](#) }

Functions

int16_t [OSC_getCourseTrim1](#) ([OSC_Handle](#) oscHandle)

int16_t [OSC_getCourseTrim2](#) ([OSC_Handle](#) oscHandle)

int16_t [OSC_getFineTrimOffset1](#) ([OSC_Handle](#) oscHandle)

int16_t [OSC_getFineTrimOffset2](#) ([OSC_Handle](#) oscHandle)

int16_t [OSC_getFineTrimSlope1](#) ([OSC_Handle](#) oscHandle)

int16_t [OSC_getFineTrimSlope2](#) ([OSC_Handle](#) oscHandle)

int16_t [OSC_getRefTempOffset](#) ([OSC_Handle](#) oscHandle)

[OSC_Handle](#) [OSC_init](#) (void *pMemory, const size_t numBytes)

void [OSC_setCoarseTrim](#) ([OSC_Handle](#) oscHandle, const [OSC_Number_e](#) oscNumber, const uint8_t trimValue)

void [OSC_setFineTrim](#) ([OSC_Handle](#) oscHandle, const [OSC_Number_e](#) oscNumber, const uint8_t trimValue)

10.1.1 Detailed Description

10.1.2 Enumeration Type Documentation

10.1.2.1 enum [OSC_Number_e](#)

Enumeration to define the oscillator (OSC) number.

Enumerator

[OSC_Number_1](#) Denotes oscillator number 1.

[OSC_Number_2](#) Denotes oscillator number 2.

10.1.2.2 enum [OSC_Osc2Src_e](#)

Enumeration to define the oscillator (OSC) 2 source.

Enumerator

OSC_Osc2Src_Internal Denotes an internal oscillator source for oscillator 2.

OSC_Osc2Src_External Denotes an external oscillator source for oscillator 2.

10.1.2.3 enum **OSC_Src_e**

Enumeration to define the oscillator (OSC) source.

Enumerator

OSC_Src_Internal Denotes an internal oscillator.

OSC_Src_External Denotes an external oscillator.

10.1.3 Function Documentation

10.1.3.1 int16_t OSC_getCourseTrim1 (

OSC_Handle oscHandle) [inline]

Gets the fine trim offset for oscillator 1.

[1]Parameters in *oscHandle* The oscillator (OSC) object handle

Returns The fine trim offset for oscillator 1

References OSC_OTP_COURSE_TRIM1.

10.1.3.2 int16_t OSC_getCourseTrim2 (

OSC_Handle oscHandle) [inline]

Gets the fine trim offset for oscillator 2.

[1]Parameters in *oscHandle* The oscillator (OSC) object handle

Returns The fine trim offset for oscillator 2

References OSC_OTP_COURSE_TRIM2.

10.1.3.3 int16_t OSC_getFineTrimOffset1 (

OSC_Handle oscHandle) [inline]

Gets the fine trim offset for oscillator 1.

[1]Parameters in *oscHandle* The oscillator (OSC) object handle

Returns The fine trim offset for oscillator 1

References OSC_OTP_FINE_TRIM_OFFSET1.

10.1.3.4 int16_t OSC_getFineTrimOffset2 (

OSC_Handle oscHandle) [inline]

Gets the fine trim offset for oscillator 2.

[1]Parameters in *oscHandle* The oscillator (OSC) object handle

Returns The fine trim offset for oscillator 2

References OSC_OTP_FINE_TRIM_OFFSET2.

10.1.3.5 int16_t OSC_getFineTrimSlope1 (

OSC_Handle oscHandle) [inline]

Gets the fine trim offset for oscillator 1.

[1]Parameters in *oscHandle* The oscillator (OSC) object handle

Returns The fine trim offset for oscillator 1

References OSC_OTP_FINE_TRIM_SLOPE1.

10.1.3.6 int16_t OSC_getFineTrimSlope2 (

OSC_Handle oscHandle) [inline]

Gets the fine trim offset for oscillator 2.

[1]Parameters in *oscHandle* The oscillator (OSC) object handle

Returns The fine trim offset for oscillator 2

References OSC_OTP_FINE_TRIM_SLOPE2.

10.1.3.7 int16_t OSC_getRefTempOffset (

OSC_Handle oscHandle) [inline]

Gets the reference temperature offset.

[1]Parameters in *oscHandle* The oscillator (OSC) object handle

Returns The reference temperature offset

References OSC_OTP_REF_TEMP_OFFSET.

10.1.3.8 **OSC_Handle** OSC_init (

void * pMemory,

```
const size_t numBytes )
```

Initializes the oscillator (OSC) handle.

[1]Parameters in *pMemory* A pointer to the base address of the FLASH registers

in *numBytes* The number of bytes allocated for the FLASH object, bytes

Returns The flash (FLASH) object handle

10.1.3.9 void OSC_setCoarseTrim (

```
    OSC_Handle oscHandle,
```

```
    const OSC_Number_e oscNumber,
```

```
    const uint8_t trimValue )
```

Sets the coarse trim value for a specified oscillator.

[1]Parameters in *oscHandle* The oscillator (OSC) object handle

in *oscNumber* The oscillator number

in *trimValue* The coarse trim value

10.1.3.10 void OSC_setFineTrim (

```
    OSC_Handle oscHandle,
```

```
    const OSC_Number_e oscNumber,
```

```
    const uint8_t trimValue )
```

Sets the fine trim value for a specified oscillator.

[1]Parameters in *oscHandle* The oscillator (OSC) object handle

in *oscNumber* The oscillator number

in *trimValue* The fine trim value

11 Peripheral Interrupt Expansion Module (PIE)

[Introduction](#)??

[API Functions](#)87 The Peripheral Interrupt Expansion controller (PIE) API provides functions for configuring and using the PIE on Piccolo devices. This API provides functions for enabling and disabling individual interrupt sources as well as acknowledging interrupts that have occurred.

This driver is contained in `f2802x/common/source/pie.c`, with `f2802x/common/include/pie.h` containing the API definitions for use by applications.

11.1 PIE

Data Structures

`struct _PIE_IERIFR_t`

`struct _PIE_Obj_`

Macros

`#define PIE_BASE_ADDR`

`#define PIE_DBGIER_DLOGINT_BITS`

`#define PIE_DBGIER_INT10_BITS`

`#define PIE_DBGIER_INT11_BITS`

`#define PIE_DBGIER_INT12_BITS`

`#define PIE_DBGIER_INT13_BITS`

`#define PIE_DBGIER_INT14_BITS`

`#define PIE_DBGIER_INT1_BITS`

`#define PIE_DBGIER_INT2_BITS`

`#define PIE_DBGIER_INT3_BITS`

`#define PIE_DBGIER_INT4_BITS`

`#define PIE_DBGIER_INT5_BITS`

`#define PIE_DBGIER_INT6_BITS`

`#define PIE_DBGIER_INT7_BITS`

`#define PIE_DBGIER_INT8_BITS`

```
#define PIE_DBGIER_INT9_BITS
#define PIE_DBGIER_RTOSINT_BITS
#define PIE_IER_DLOGINT_BITS
#define PIE_IER_INT10_BITS
#define PIE_IER_INT11_BITS
#define PIE_IER_INT12_BITS
#define PIE_IER_INT13_BITS
#define PIE_IER_INT14_BITS
#define PIE_IER_INT1_BITS
#define PIE_IER_INT2_BITS
#define PIE_IER_INT3_BITS
#define PIE_IER_INT4_BITS
#define PIE_IER_INT5_BITS
#define PIE_IER_INT6_BITS
#define PIE_IER_INT7_BITS
#define PIE_IER_INT8_BITS
#define PIE_IER_INT9_BITS
#define PIE_IER_RTOSINT_BITS
#define PIE_IERx_INTx1_BITS
#define PIE_IERx_INTx2_BITS
#define PIE_IERx_INTx3_BITS
#define PIE_IERx_INTx4_BITS
#define PIE_IERx_INTx5_BITS
#define PIE_IERx_INTx6_BITS
#define PIE_IERx_INTx7_BITS
#define PIE_IERx_INTx8_BITS
#define PIE_IFR_DLOGINT_BITS
#define PIE_IFR_INT10_BITS
#define PIE_IFR_INT11_BITS
```

```
#define PIE_IFR_INT12_BITS
#define PIE_IFR_INT13_BITS
#define PIE_IFR_INT14_BITS
#define PIE_IFR_INT1_BITS
#define PIE_IFR_INT2_BITS
#define PIE_IFR_INT3_BITS
#define PIE_IFR_INT4_BITS
#define PIE_IFR_INT5_BITS
#define PIE_IFR_INT6_BITS
#define PIE_IFR_INT7_BITS
#define PIE_IFR_INT8_BITS
#define PIE_IFR_INT9_BITS
#define PIE_IFR_RTOSINT_BITS
#define PIE_IFRx_INTx1_BITS
#define PIE_IFRx_INTx2_BITS
#define PIE_IFRx_INTx3_BITS
#define PIE_IFRx_INTx4_BITS
#define PIE_IFRx_INTx5_BITS
#define PIE_IFRx_INTx6_BITS
#define PIE_IFRx_INTx7_BITS
#define PIE_IFRx_INTx8_BITS
#define PIE_PIEACK_GROUP10_BITS
#define PIE_PIEACK_GROUP11_BITS
#define PIE_PIEACK_GROUP12_BITS
#define PIE_PIEACK_GROUP1_BITS
#define PIE_PIEACK_GROUP2_BITS
#define PIE_PIEACK_GROUP3_BITS
#define PIE_PIEACK_GROUP4_BITS
#define PIE_PIEACK_GROUP5_BITS
```

```
#define PIE_PIEACK_GROUP6_BITS
#define PIE_PIEACK_GROUP7_BITS
#define PIE_PIEACK_GROUP8_BITS
#define PIE_PIEACK_GROUP9_BITS
#define PIE_PIECTRL_ENPIE_BITS
#define PIE_PIECTRL_PIEVECT_BITS
```

Typedefs

```
typedef interrupt void(* intVec_t)(void)
typedef struct _PIE_Obj * PIE_Handle
typedef struct _PIE_IERIFR_t PIE_IERIFR_t
typedef struct _PIE_Obj PIE_Obj
```

Enumerations

```
enum PIE_ExtIntPolarity_e { PIE_ExtIntPolarity_FallingEdge, PIE_ExtIntPolarity_RisingEdge,
PIE_ExtIntPolarity_RisingAndFallingEdge }

enum PIE_GroupNumber_e {
PIE_GroupNumber_1, PIE_GroupNumber_2, PIE_GroupNumber_3, PIE_GroupNumber_4,
PIE_GroupNumber_5, PIE_GroupNumber_6, PIE_GroupNumber_7, PIE_GroupNumber_8,
PIE_GroupNumber_9, PIE_GroupNumber_10, PIE_GroupNumber_11, PIE_GroupNumber_12 }

enum PIE_InterruptSource_e {
PIE_InterruptSource_ADCINT_1_1, PIE_InterruptSource_ADCINT_1_2,
PIE_InterruptSource_XINT_1, PIE_InterruptSource_XINT_2,
PIE_InterruptSource_ADCINT_9, PIE_InterruptSource_TIMER_0, PIE_InterruptSource_WAKE,
PIE_InterruptSource_TZ1,
PIE_InterruptSource_TZ2, PIE_InterruptSource_TZ3, PIE_InterruptSource_TZ4,
PIE_InterruptSource_EPWM1,
PIE_InterruptSource_EPWM2, PIE_InterruptSource_EPWM3, PIE_InterruptSource_EPWM4,
PIE_InterruptSource_ECAP1,
PIE_InterruptSource_SPIARX, PIE_InterruptSource_SPIATX, PIE_InterruptSource_I2CA1,
PIE_InterruptSource_I2CA2,
PIE_InterruptSource_SCIARX, PIE_InterruptSource_SCIATX, PIE_InterruptSource_ADCINT_10_1,
PIE_InterruptSource_ADCINT_10_2,
PIE_InterruptSource_ADCINT_3, PIE_InterruptSource_ADCINT_4,
PIE_InterruptSource_ADCINT_5, PIE_InterruptSource_ADCINT_6,
PIE_InterruptSource_ADCINT_7, PIE_InterruptSource_ADCINT_8, PIE_InterruptSource_XINT_3
}

enum PIE_SubGroupNumber_e {
PIE_SubGroupNumber_1, PIE_SubGroupNumber_2, PIE_SubGroupNumber_3,
```

```
PIE_SubGroupNumber_4,  
PIE_SubGroupNumber_5,    PIE_SubGroupNumber_6,    PIE_SubGroupNumber_7,  
PIE_SubGroupNumber_8 }  
  
enum PIE_SystemInterrupts_e {  
    PIE_SystemInterrupts_Reset,    PIE_SystemInterrupts_INT1,    PIE_SystemInterrupts_INT2,  
    PIE_SystemInterrupts_INT3,    PIE_SystemInterrupts_INT5,    PIE_SystemInterrupts_INT6,  
    PIE_SystemInterrupts_INT4,    PIE_SystemInterrupts_INT7,    PIE_SystemInterrupts_INT8,  
    PIE_SystemInterrupts_INT9,    PIE_SystemInterrupts_INT10,  
    PIE_SystemInterrupts_INT11,    PIE_SystemInterrupts_TINT1,    PIE_SystemInterrupts_TINT2,  
    PIE_SystemInterrupts_DATALOG,  
    PIE_SystemInterrupts_RTOSINT,    PIE_SystemInterrupts_EMUINT,    PIE_SystemInterrupts_NMI,  
    PIE_SystemInterrupts_ILLEGAL,  
    PIE_SystemInterrupts_USER1,    PIE_SystemInterrupts_USER2,    PIE_SystemInterrupts_USER3,  
    PIE_SystemInterrupts_USER4,    PIE_SystemInterrupts_USER5,    PIE_SystemInterrupts_USER6,  
    PIE_SystemInterrupts_USER7,    PIE_SystemInterrupts_USER8,  
    PIE_SystemInterrupts_USER9,    PIE_SystemInterrupts_USER10,    PIE_SystemInterrupts_USER11,  
    PIE_SystemInterrupts_USER12 }  
}
```

Functions

```
void PIE_clearAllFlags (PIE_Handle pieHandle)  
  
void PIE_clearAllInts (PIE_Handle pieHandle)  
  
void PIE_clearInt (PIE_Handle pieHandle, const PIE_GroupNumber_e groupNumber)  
  
void PIE_disable (PIE_Handle pieHandle)  
  
void PIE_disableAllInts (PIE_Handle pieHandle)  
  
void PIE_disableCaptureInt (PIE_Handle pieHandle)  
  
void PIE_disableInt (PIE_Handle pieHandle, const PIE_GroupNumber_e group, const  
PIE_InterruptSource_e intSource)  
  
void PIE_enable (PIE_Handle pieHandle)  
  
void PIE_enableAdcInt (PIE_Handle pieHandle, const ADC_IntNumber_e intNumber)  
  
void PIE_enableCaptureInt (PIE_Handle pieHandle)  
  
void PIE_enableExtInt (PIE_Handle pieHandle, const CPU_ExtIntNumber_e intNumber)  
  
void PIE_enableInt (PIE_Handle pieHandle, const PIE_GroupNumber_e group, const  
PIE_InterruptSource_e intSource)  
  
void PIE_enablePwmInt (PIE_Handle pieHandle, const PWM_Number_e pwmNumber)  
  
void PIE_enablePwmTzInt (PIE_Handle pieHandle, const PWM_Number_e pwmNumber)
```

```
void PIE_enableTimer0Int (PIE_Handle pieHandle)

void PIE_forceInt (PIE_Handle pieHandle, const PIE_GroupNumber_e group, const
PIE_InterruptSource_e intSource)

uint16_t PIE_getExtIntCount (PIE_Handle pieHandle, const CPU_ExtIntNumber_e intNumber)

uint16_t PIE_getIntEnables (PIE_Handle pieHandle, const PIE_GroupNumber_e group)

uint16_t PIE_getIntFlags (PIE_Handle pieHandle, const PIE_GroupNumber_e group)

interrupt void PIE_illegalSr (void)

PIE_Handle PIE_init (void *pMemory, const size_t numBytes)

void PIE_registerPieIntHandler (PIE_Handle pieHandle, const PIE_GroupNumber_e groupNum-
ber, const PIE_SubGroupNumber_e subGroupNumber, const intVec_t vector)

void PIE_registerSystemIntHandler (PIE_Handle pieHandle, const PIE_SystemInterrupts_e sys-
temInt, const intVec_t vector)

void PIE_setDefaultIntVectorTable (PIE_Handle pieHandle)

void PIE_setExtIntPolarity (PIE_Handle pieHandle, const CPU_ExtIntNumber_e intNumber, const
PIE_ExtIntPolarity_e polarity)

void PIE_unregisterPieIntHandler (PIE_Handle pieHandle, const PIE_GroupNumber_e groupNum-
ber, const PIE_SubGroupNumber_e subGroupNumber)

void PIE_unregisterSystemIntHandler (PIE_Handle pieHandle, const PIE_SystemInterrupts_e sys-
temInt)
```

11.1.1 Detailed Description

11.1.2 Enumeration Type Documentation

11.1.2.1 enum **PIE_ExtIntPolarity_e**

Enumeration to define the external interrupt polarity.

Enumerator

PIE_ExtIntPolarity_FallingEdge Denotes an interrupt is generated on the falling edge.

PIE_ExtIntPolarity_RisingEdge Denotes an interrupt is generated on the rising edge.

PIE_ExtIntPolarity_RisingAndFallingEdge Denotes an interrupt is generated on the falling and rising edges.

11.1.2.2 enum **PIE_GroupNumber_e**

Enumeration to define the peripheral interrupt expansion (PIE) group numbers.

Enumerator

- PIE_GroupNumber_1*** Denotes PIE group number 1.
- PIE_GroupNumber_2*** Denotes PIE group number 2.
- PIE_GroupNumber_3*** Denotes PIE group number 3.
- PIE_GroupNumber_4*** Denotes PIE group number 4.
- PIE_GroupNumber_5*** Denotes PIE group number 5.
- PIE_GroupNumber_6*** Denotes PIE group number 6.
- PIE_GroupNumber_7*** Denotes PIE group number 7.
- PIE_GroupNumber_8*** Denotes PIE group number 8.
- PIE_GroupNumber_9*** Denotes PIE group number 9.
- PIE_GroupNumber_10*** Denotes PIE group number 10.
- PIE_GroupNumber_11*** Denotes PIE group number 11.
- PIE_GroupNumber_12*** Denotes PIE group number 12.

11.1.2.3 enum `PIE_InterruptSource_e`

Enumeration to define the peripheral interrupt expansion (PIE) individual interrupt sources.

Enumerator

- PIE_InterruptSource_ADCINT_1_1*** Group 1 ADC Interrupt 1.
- PIE_InterruptSource_ADCINT_1_2*** Group 1 ADC Interrupt 2.
- PIE_InterruptSource_XINT_1*** External Interrupt 1.
- PIE_InterruptSource_XINT_2*** External Interrupt 2.
- PIE_InterruptSource_ADCINT_9*** ADC Interrupt 9.
- PIE_InterruptSource_TIMER_0*** Timer Interrupt 0.
- PIE_InterruptSource_WAKE*** Wake Up Interrupt.
- PIE_InterruptSource_TZ1*** EPWM TZ1 Interrupt.
- PIE_InterruptSource_TZ2*** EPWM TZ2 Interrupt.
- PIE_InterruptSource_TZ3*** EPWM TZ3 Interrupt.
- PIE_InterruptSource_TZ4*** EPWM TZ4 Interrupt.
- PIE_InterruptSource_EPWM1*** EPWM 1 Interrupt.
- PIE_InterruptSource_EPWM2*** EPWM 2 Interrupt.
- PIE_InterruptSource_EPWM3*** EPWM 3 Interrupt.
- PIE_InterruptSource_EPWM4*** EPWM 4 Interrupt.
- PIE_InterruptSource_ECAP1*** ECAP 1 Interrupt.
- PIE_InterruptSource_SPIARX*** SPI A RX Interrupt.
- PIE_InterruptSource_SPIATX*** SPI A TX Interrupt.
- PIE_InterruptSource_I2CA1*** I2C A Interrupt 1.
- PIE_InterruptSource_I2CA2*** I2C A Interrupt 2.
- PIE_InterruptSource_SCIARX*** SCI A RX Interrupt.
- PIE_InterruptSource_SCIATX*** SCI A TX Interrupt.
- PIE_InterruptSource_ADCINT_10_1*** Group 10 ADC Interrupt 1.
- PIE_InterruptSource_ADCINT_10_2*** Group 10 ADC Interrupt 2.
- PIE_InterruptSource_ADCINT_3*** ADC Interrupt 3.
- PIE_InterruptSource_ADCINT_4*** ADC Interrupt 4.

PIE_InterruptSource_ADCINT_5 ADC Interrupt 5.
PIE_InterruptSource_ADCINT_6 ADC Interrupt 6.
PIE_InterruptSource_ADCINT_7 ADC Interrupt 7.
PIE_InterruptSource_ADCINT_8 ADC Interrupt 8.
PIE_InterruptSource_XINT_3 External Interrupt 3.

11.1.2.4 enum **PIE_SubGroupNumber_e**

Enumeration to define the peripheral interrupt expansion (PIE) sub-group numbers.

Enumerator

PIE_SubGroupNumber_1 Denotes PIE group number 1.
PIE_SubGroupNumber_2 Denotes PIE group number 2.
PIE_SubGroupNumber_3 Denotes PIE group number 3.
PIE_SubGroupNumber_4 Denotes PIE group number 4.
PIE_SubGroupNumber_5 Denotes PIE group number 5.
PIE_SubGroupNumber_6 Denotes PIE group number 6.
PIE_SubGroupNumber_7 Denotes PIE group number 7.
PIE_SubGroupNumber_8 Denotes PIE group number 8.

11.1.2.5 enum **PIE_SystemInterrupts_e**

Enumeration to define the system interrupts.

Enumerator

PIE_SystemInterrupts_Reset Reset interrupt vector.
PIE_SystemInterrupts_INT1 INT1 interrupt vector.
PIE_SystemInterrupts_INT2 INT2 interrupt vector.
PIE_SystemInterrupts_INT3 INT3 interrupt vector.
PIE_SystemInterrupts_INT4 INT4 interrupt vector.
PIE_SystemInterrupts_INT5 INT5 interrupt vector.
PIE_SystemInterrupts_INT6 INT6 interrupt vector.
PIE_SystemInterrupts_INT7 INT7 interrupt vector.
PIE_SystemInterrupts_INT8 INT8 interrupt vector.
PIE_SystemInterrupts_INT9 INT9 interrupt vector.
PIE_SystemInterrupts_INT10 INT10 interrupt vector.
PIE_SystemInterrupts_INT11 INT11 interrupt vector.
PIE_SystemInterrupts_INT12 INT12 interrupt vector.
PIE_SystemInterrupts_TINT1 INT13 interrupt vector.
PIE_SystemInterrupts_TINT2 INT14 interrupt vector.
PIE_SystemInterrupts_DATALOG DATALOG interrupt vector.
PIE_SystemInterrupts_RTOSINT RTOSINT interrupt vector.
PIE_SystemInterrupts_EMUINT EMUINT interrupt vector.
PIE_SystemInterrupts_NMI NMI interrupt vector.
PIE_SystemInterrupts_ILLEGAL ILLEGAL interrupt vector.

PIE_SystemInterrupts_USER1 USER1 interrupt vector.
PIE_SystemInterrupts_USER2 USER2 interrupt vector.
PIE_SystemInterrupts_USER3 USER3 interrupt vector.
PIE_SystemInterrupts_USER4 USER4 interrupt vector.
PIE_SystemInterrupts_USER5 USER5 interrupt vector.
PIE_SystemInterrupts_USER6 USER6 interrupt vector.
PIE_SystemInterrupts_USER7 USER7 interrupt vector.
PIE_SystemInterrupts_USER8 USER8 interrupt vector.
PIE_SystemInterrupts_USER9 USER9 interrupt vector.
PIE_SystemInterrupts_USER10 USER10 interrupt vector.
PIE_SystemInterrupts_USER11 USER11 interrupt vector.
PIE_SystemInterrupts_USER12 USER12 interrupt vector.

11.1.3 Function Documentation

11.1.3.1 void PIE_clearAllFlags (

PIE_Handle pieHandle)

Clears all the interrupt flags.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) object handle

11.1.3.2 void PIE_clearAllInts (

PIE_Handle pieHandle)

Clears all the interrupts.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) object handle

11.1.3.3 void PIE_clearInt (

PIE_Handle pieHandle,

const **PIE_GroupNumber_e** groupNumber) [inline]

Clears an interrupt defined by group number.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) object handle

in *groupNumber* The group number

References `_PIE_Obj_::PIEACK`.

11.1.3.4 void PIE_disable (
 PIE_Handle pieHandle)

Disables the peripheral interrupt expansion (PIE)

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) object handle

11.1.3.5 void PIE_disableAllInts (
 PIE_Handle pieHandle)

Disables all of the interrupts.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) object handle

11.1.3.6 void PIE_disableCaptureInt (
 PIE_Handle pieHandle)

Disables the capture interrupt.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) object handle

11.1.3.7 void PIE_disableInt (
 PIE_Handle pieHandle,
 const **PIE_GroupNumber_e** group,
 const **PIE_InterruptSource_e** intSource)

Disable a specific PIE interrupt.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) object handle

in *group* The PIE group an interrupt belongs to

in *intSource* The specific interrupt source to disable

11.1.3.8 void PIE_enable (
 PIE_Handle pieHandle)

Enables the peripheral interrupt expansion (PIE)

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) object handle

11.1.3.9 void PIE_enableAdcInt (

PIE_Handle pieHandle,const **ADC_IntNumber_e** intNumber)

Enables the specified ADC interrupt.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) object handlein *intNumber* The interrupt number

11.1.3.10 void PIE_enableCaptureInt (

PIE_Handle pieHandle)

Enables the capture interrupt.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) object handle

11.1.3.11 void PIE_enableExtInt (

PIE_Handle pieHandle,const **CPU_ExtIntNumber_e** intNumber)

Enables the prescribed external interrupt.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) handlein *intNumber* The interrupt number

11.1.3.12 void PIE_enableInt (

PIE_Handle pieHandle,const **PIE_GroupNumber_e** group,const **PIE_InterruptSource_e** intSource)

Enable a specific PIE interrupt.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) object handlein *group* The PIE group an interrupt belongs toin *intSource* The specific interrupt source to enable

11.1.3.13 void PIE_enablePwmInt (

PIE_Handle pieHandle,

const **PWM_Number_e** pwmNumber)

Enables the PWM interrupt.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) handle

in *pwmNumber* The PWM number

11.1.3.14 void PIE_enablePwmTzInt (

PIE_Handle pieHandle,

const **PWM_Number_e** pwmNumber)

Enables the PWM Trip Zone interrupt.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) handle

in *pwmNumber* The PWM number

11.1.3.15 void PIE_enableTimer0Int (

PIE_Handle pieHandle)

Enables the Cpu Timer 0 interrupt.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) handle

11.1.3.16 void PIE_forceInt (

PIE_Handle pieHandle,

const **PIE_GroupNumber_e** group,

const **PIE_InterruptSource_e** intSource)

Force a specific PIE interrupt.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) object handle

in *group* The PIE group an interrupt belongs to

in *intSource* The specific interrupt source to force

11.1.3.17 uint16_t PIE_getExtIntCount (

PIE_Handle pieHandle,const **CPU_ExtIntNumber_e** intNumber)

Gets the external interrupt count value.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) handle

in *intNumber* The external interrupt number

Returns The count value

11.1.3.18 uint16_t PIE_getIntEnables (

PIE_Handle pieHandle,const **PIE_GroupNumber_e** group)

Gets PIE interrupt enable values.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) object handle

in *group* The PIE group the flags belong to

11.1.3.19 uint16_t PIE_getIntFlags (

PIE_Handle pieHandle,const **PIE_GroupNumber_e** group)

Gets PIE interrupt flag values.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) object handle

in *group* The PIE group the flags belong to11.1.3.20 **PIE_Handle** PIE_init (

void * pMemory,

const size_t numBytes)

Initializes the peripheral interrupt expansion (PIE) object handle.

[1]Parameters in *pMemory* A pointer to the memory for the PIE object

in numBytes The number of bytes allocated for the PIE object, bytes

Returns The peripheral interrupt expansion (PIE) object handle

11.1.3.21 void PIE_registerPieIntHandler (

PIE_Handle pieHandle,

const **PIE_GroupNumber_e** groupNumber,

const **PIE_SubGroupNumber_e** subGroupNumber,

const **intVec_t** vector)

Registers a handler for a PIE interrupt.

[1]Parameters *in pieHandle* The peripheral interrupt expansion (PIE) object handle

in groupNumber The PIE group an interrupt belongs to

in subGroupNumber The PIE subgroup an interrupt belongs to

in vector The specific interrupt handler

11.1.3.22 void PIE_registerSystemIntHandler (

PIE_Handle pieHandle,

const **PIE_SystemInterrupts_e** systemInt,

const **intVec_t** vector)

Registers a handler for a PIE interrupt.

[1]Parameters *in pieHandle* The peripheral interrupt expansion (PIE) object handle

in systemInt The system interrupt to register this handler to

in vector The specific interrupt handler

11.1.3.23 void PIE_setDefaultIntVectorTable (

PIE_Handle pieHandle)

Initializes the vector table with illegal ISR handlers.

[1]Parameters *in pieHandle* The peripheral interrupt expansion (PIE) object handle

11.1.3.24 void PIE_setExtIntPolarity (

PIE_Handle pieHandle,
const **CPU_ExtIntNumber_e** intNumber,
const **PIE_ExtIntPolarity_e** polarity)

Sets the external interrupt polarity.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) handle

in *intNumber* The external interrupt number

in *polarity* The signal polarity

11.1.3.25 void PIE_unregisterPieIntHandler (

PIE_Handle pieHandle,
const **PIE_GroupNumber_e** groupNumber,
const **PIE_SubGroupNumber_e** subGroupNumber)

Unregisters a handler for a PIE interrupt.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) object handle

in *groupNumber* The PIE group an interrupt belongs to

in *subGroupNumber* The PIE subgroup an interrupt belongs to

11.1.3.26 void PIE_unregisterSystemIntHandler (

PIE_Handle pieHandle,
const **PIE_SystemInterrupts_e** systemInt)

Unregisters a handler for a PIE interrupt.

[1]Parameters in *pieHandle* The peripheral interrupt expansion (PIE) object handle

in *systemInt* The system interrupt to unregister

12 Phase Locked Loop (PLL)

[Introduction](#)??

[API Functions](#) 103 The Phase Locked Loop (PLL) API provides functions for configuring the device's PLL as well as other miscellaneous clock functions.

This driver is contained in `f2802x/common/source/pll.c`, with `f2802x/common/include/pll.h` containing the API definitions for use by applications.

12.1 PLL

Data Structures

struct `_PLL_Obj`

Macros

`#define PLL_BASE_ADDR`

`#define PLL_PLLCR_DIV_BITS`

`#define PLL_PLLSTS_DIVSEL_BITS`

`#define PLL_PLLSTS_MCLKCLR_BITS`

`#define PLL_PLLSTS_MCLKOFF_BITS`

`#define PLL_PLLSTS_MCLKSTS_BITS`

`#define PLL_PLLSTS_NORMRDYE_BITS`

`#define PLL_PLLSTS_OSCOFF_BITS`

`#define PLL_PLLSTS_PLLOCKES_BITS`

`#define PLL_PLLSTS_PLLOFF_BITS`

Typedefs

`typedef struct _PLL_Obj * PLL_Handle`

`typedef struct _PLL_Obj PLL_Obj`

Enumerations

`enum PLL_ClkStatus_e { PLL_ClkStatus_Normal, PLL_ClkStatus_Missing }`

```
enum PLL_DivideSelect_e { PLL_DivideSelect_ClkIn_by_4, PLL_DivideSelect_ClkIn_by_2,  
PLL_DivideSelect_ClkIn_by_1 }  
  
enum PLL_LockStatus_e { PLL_LockStatus_Locking, PLL_LockStatus_Done }  
  
enum PLL_Multiplier_e {  
PLL_Multiplier_1, PLL_Multiplier_2, PLL_Multiplier_3, PLL_Multiplier_4,  
PLL_Multiplier_5, PLL_Multiplier_6, PLL_Multiplier_7, PLL_Multiplier_8,  
PLL_Multiplier_9, PLL_Multiplier_10, PLL_Multiplier_11, PLL_Multiplier_12 }
```

Functions

```
void PLL_disable (PLL_Handle pllHandle)  
void PLL_disableClkDetect (PLL_Handle pllHandle)  
void PLL_disableNormRdy (PLL_Handle pllHandle)  
void PLL_disableOsc (PLL_Handle pllHandle)  
void PLL_enable (PLL_Handle pllHandle)  
void PLL_enableClkDetect (PLL_Handle pllHandle)  
void PLL_enableNormRdy (PLL_Handle pllHandle)  
void PLL_enableOsc (PLL_Handle pllHandle)  
PLL_ClkStatus_e PLL_getClkStatus (PLL_Handle pllHandle)  
PLL_DivideSelect_e PLL_getDivider (PLL_Handle pllHandle)  
PLL_LockStatus_e PLL_getLockStatus (PLL_Handle pllHandle)  
PLL_Multiplier_e PLL_getMultiplier (PLL_Handle pllHandle)  
PLL_Handle PLL_init (void *pMemory, const size_t numBytes)  
void PLL_resetClkDetect (PLL_Handle pllHandle)  
void PLL_setDivider (PLL_Handle pllHandle, const PLL_DivideSelect_e divSelect)  
void PLL_setLockPeriod (PLL_Handle pllHandle, const uint16_t lockPeriod)  
void PLL_setMultiplier (PLL_Handle pllHandle, const PLL_Multiplier_e freq)  
void PLL_setup (PLL_Handle pllHandle, const PLL_Multiplier_e clkMult, const PLL_DivideSelect_e  
divSelect)
```

12.1.1 Detailed Description

12.1.2 Enumeration Type Documentation

12.1.2.1 enum **PLL_ClkStatus_e**

Enumeration to define the phase lock loop (PLL) clock status.

Enumerator

PLL_ClkStatus_Normal Denotes a normal clock.

PLL_ClkStatus_Missing Denotes a missing clock.

12.1.2.2 enum **PLL_DivideSelect_e**

Enumeration to define the phase lock loop (PLL) divide select.

Enumerator

PLL_DivideSelect_ClkIn_by_4 Denotes a divide select of CLKIN/4.

PLL_DivideSelect_ClkIn_by_2 Denotes a divide select of CLKIN/2.

PLL_DivideSelect_ClkIn_by_1 Denotes a divide select of CLKIN/1.

12.1.2.3 enum **PLL_LockStatus_e**

Enumeration to define the phase lock loop (PLL) clock lock status.

Enumerator

PLL_LockStatus_Locking Denotes that the system is locking to the clock.

PLL_LockStatus_Done Denotes that the system is locked to the clock.

12.1.2.4 enum **PLL_Multiplier_e**

Enumeration to define the phase lock loop (PLL) clock frequency.

Enumerator

PLL_Multiplier_1 Denotes a multiplier of 1.

PLL_Multiplier_2 Denotes a multiplier of 2.

PLL_Multiplier_3 Denotes a multiplier of 3.

PLL_Multiplier_4 Denotes a multiplier of 4.

PLL_Multiplier_5 Denotes a multiplier of 5.

PLL_Multiplier_6 Denotes a multiplier of 6.

PLL_Multiplier_7 Denotes a multiplier of 7.

PLL_Multiplier_8 Denotes a multiplier of 8.

PLL_Multiplier_9 Denotes a multiplier of 9.

PLL_Multiplier_10 Denotes a multiplier of 10.

PLL_Multiplier_11 Denotes a multiplier of 11.

PLL_Multiplier_12 Denotes a multiplier of 12.

12.1.3 Function Documentation

12.1.3.1 void PLL_disable (

PLL_Handle pllHandle)

Disables the phase lock loop (PLL)

[1]Parameters in *pllHandle* The phase lock loop (PLL) object handle

12.1.3.2 void PLL_disableClkDetect (

PLL_Handle pllHandle)

Disables the clock detect logic.

[1]Parameters in *pllHandle* The phase lock loop (PLL) object handle

12.1.3.3 void PLL_disableNormRdy (

PLL_Handle pllHandle)

Disables the NORMRDY signal.

[1]Parameters in *pllHandle* The phase lock loop (PLL) object handle

12.1.3.4 void PLL_disableOsc (

PLL_Handle pllHandle)

Disables the oscillator.

[1]Parameters in *pllHandle* The phase lock loop (PLL) object handle

12.1.3.5 void PLL_enable (

PLL_Handle pllHandle)

Enables the phase lock loop (PLL)

[1]Parameters in *pllHandle* The phase lock loop (PLL) object handle

12.1.3.6 void PLL_enableClkDetect (

722.7

[1]Parameters in *pllHandle* The phase lock loop (PLL) object handle

Returns The lock status

12.1.3.12 **PLL_Multiplier_e** PLL_getMultiplier (**PLL_Handle** pllHandle)

Gets the phase lock loop (PLL) clock frequency.

[1]Parameters in *pllHandle* The phase lock loop (PLL) object handle

Returns The clock frequency

12.1.3.13 **PLL_Handle** PLL_init (void * pMemory, const size_t numBytes)

Initializes the phase lock loop (PLL) object handle.

[1]Parameters in *pMemory* A pointer to the base address of the PLL registers

in *numBytes* The number of bytes allocated for the PLL object, bytes

Returns The phase lock loop (PLL) object handle

12.1.3.14 void PLL_resetClkDetect (**PLL_Handle** pllHandle)

Resets the phase lock loop (PLL) clock detect logic.

[1]Parameters in *pllHandle* The phase lock loop (PLL) object handle

12.1.3.15 void PLL_setDivider (**PLL_Handle** pllHandle, const **PLL_DivideSelect_e** divSelect)

Sets the phase lock loop (PLL) divide select value.

[1]Parameters in *pllHandle* The phase lock loop (PLL) object handle

in *divSelect* The divide select value

12.1.3.16 void PLL_setLockPeriod (

PLL_Handle pllHandle,

const uint16_t lockPeriod)

Sets the phase lock loop (PLL) lock time.

[1]Parameters in *pllHandle* The phase lock loop (PLL) object handle

in lockPeriod The lock period, cycles

12.1.3.17 void PLL_setMultiplier (

PLL_Handle pllHandle,const **PLL_Multiplier_e** freq)

Sets the phase lock loop (PLL) clock frequency.

[1]Parameters in *pllHandle* The phase lock loop (PLL) object handle

in freq The clock frequency

12.1.3.18 void PLL_setup (

PLL_Handle pllHandle,const **PLL_Multiplier_e** clkMult,const **PLL_DivideSelect_e** divSelect)

Sets the phase lock loop (PLL) divider and multiplier.

[1]Parameters in *pllHandle* The phase lock loop (PLL) object handle

in clkMult The clock multiplier value

in divSelect The divide select value

13 Pulse Width Modulator (PWM)

[Introduction](#) [??](#)

[API Functions](#) 111 The pulse width modulation peripheral (PWM) APIs provide functions for configuring and updating the PWM peripherals on this device.

This driver is contained in `f2802x/common/source/pwm.c`, with `f2802x/common/include/pwm.h` containing the API definitions for use by applications.

13.1 PWM

Data Structures

struct [_PWM_Obj](#)

Macros

`#define` [PWM_AQCTL_CAD_BITS](#)

`#define` [PWM_AQCTL_CAU_BITS](#)

`#define` [PWM_AQCTL_CBD_BITS](#)

`#define` [PWM_AQCTL_CBU_BITS](#)

`#define` [PWM_AQCTL_PRD_BITS](#)

`#define` [PWM_AQCTL_ZRO_BITS](#)

`#define` [PWM_CMPCTL_LOADAMODE_BITS](#)

`#define` [PWM_CMPCTL_LOADBMODE_BITS](#)

`#define` [PWM_CMPCTL_SHDWAFULL_BITS](#)

`#define` [PWM_CMPCTL_SHDWAMODE_BITS](#)

`#define` [PWM_CMPCTL_SHDWBFULL_BITS](#)

`#define` [PWM_CMPCTL_SHDWBMODE_BITS](#)

`#define` [PWM_DBCTL_HALFCYCLE_BITS](#)

`#define` [PWM_DBCTL_INMODE_BITS](#)

`#define` [PWM_DBCTL_OUTMODE_BITS](#)

`#define` [PWM_DBCTL_POLSEL_BITS](#)

`#define` [PWM_DCFCTL_BLANKE_BITS](#)

```
#define PWM_DCFCTL_BLANKINV_BITS
#define PWM_DCFCTL_PULSESEL_BITS
#define PWM_DCFCTL_SRCSEL_BITS
#define PWM_DCTRISEL_DCAHCOMPSEL_BITS
#define PWM_DCTRISEL_DCALCOMPSEL_BITS
#define PWM_DCTRISEL_DCBHCOMPSEL_BITS
#define PWM_DCTRISEL_DCBLCOMPSEL_BITS
#define PWM_ePWM1_BASE_ADDR
#define PWM_ePWM2_BASE_ADDR
#define PWM_ePWM3_BASE_ADDR
#define PWM_ePWM4_BASE_ADDR
#define PWM_ETCLR_INT_BITS
#define PWM_ETCLR_SOCA_BITS
#define PWM_ETCLR_SOCB_BITS
#define PWM_ETPS_INTCNT_BITS
#define PWM_ETPS_INTPRD_BITS
#define PWM_ETPS_SOCACNT_BITS
#define PWM_ETPS_SOCAPRD_BITS
#define PWM_ETPS_SOCBCNT_BITS
#define PWM_ETPS_SOCBPRD_BITS
#define PWM_ETSEL_INTEN_BITS
#define PWM_ETSEL_INTSEL_BITS
#define PWM_ETSEL_SOCAEN_BITS
#define PWM_ETSEL_SOCASEL_BITS
#define PWM_ETSEL_SOCBEN_BITS
#define PWM_ETSEL_SOCBSEL_BITS
#define PWM_HRCNFG_AUTOCONV_BITS
#define PWM_HRCNFG_CTLMODE_BITS
#define PWM_HRCNFG_EDGMODE_BITS
```

```
#define PWM_HRCNFG_HRLOAD_BITS
#define PWM_HRCNFG_SELOUTB_BITS
#define PWM_HRCNFG_SWAPAB_BITS
#define PWM_HRPCTL_HRPE_BITS
#define PWM_HRPCTL_PWMSYNCSEL_BITS
#define PWM_HRPCTL_TBPHSHRLOADE_BITS
#define PWM_PCCTL_CHPDUTY_BITS
#define PWM_PCCTL_CHPEN_BITS
#define PWM_PCCTL_CHPFREQ_BITS
#define PWM_PCCTL_OSHTWTH_BITS
#define PWM_TBCTL_CLKDIV_BITS
#define PWM_TBCTL_CTRMODE_BITS
#define PWM_TBCTL_FREESOFT_BITS
#define PWM_TBCTL_HSPCLKDIV_BITS
#define PWM_TBCTL_PHSDIR_BITS
#define PWM_TBCTL_PHSEN_BITS
#define PWM_TBCTL_PRDLD_BITS
#define PWM_TBCTL_SWFSYNC_BITS
#define PWM_TBCTL_SYNCSEL_BITS
#define PWM_TZCLR_CBC_BITS
#define PWM_TZCLR_DCAEVT1_BITS
#define PWM_TZCLR_DCAEVT2_BITS
#define PWM_TZCLR_DCBEVT1_BITS
#define PWM_TZCLR_DCBEVT2_BITS
#define PWM_TZCLR_INT_BITS
#define PWM_TZCLR_OST_BITS
#define PWM_TZCTL_DCAEVT1_BITS
#define PWM_TZCTL_DCAEVT2_BITS
#define PWM_TZCTL_DCBEVT1_BITS
```

```
#define PWM_TZCTL_DCBEVT2_BITS
#define PWM_TZCTL_TZA_BITS
#define PWM_TZCTL_TZB_BITS
#define PWM_TZDCSEL_DCAEVT1_BITS
#define PWM_TZDCSEL_DCAEVT2_BITS
#define PWM_TZDCSEL_DCBEVT1_BITS
#define PWM_TZDCSEL_DCBEVT2_BITS
#define PWM_TZFRC_CBC_BITS
#define PWM_TZFRC_DCAEVT1_BITS
#define PWM_TZFRC_DCAEVT2_BITS
#define PWM_TZFRC_DCBEVT1_BITS
#define PWM_TZFRC_DCBEVT2_BITS
#define PWM_TZFRC_OST_BITS
```

Typedefs

```
typedef struct _PWM_Obj_ * PWM_Handle
typedef struct _PWM_Obj_ PWM_Obj
```

Enumerations

```
enum PWM_ActionQual_e
enum PWM_ChoppingClkFreq_e
enum PWM_ChoppingDutyCycle_e
enum PWM_ChoppingPulseWidth_e
enum PWM_ClkDiv_e
enum PWM_CounterMode_e
enum PWM_DeadBandInputMode_e
enum PWM_DeadBandOutputMode_e
enum PWM_DeadBandPolarity_e
enum PWM_DigitalCompare_FilterSrc_e
```

```
enum PWM_DigitalCompare_Input_e
enum PWM_DigitalCompare_InputSel_e
enum PWM_DigitalCompare_PulseSel_e
enum PWM_HrControlMode_e
enum PWM_HrEdgeMode_e
enum PWM_HrShadowMode_e
enum PWM_HspClkDiv_e
enum PWM_IntMode_e
enum PWM_IntPeriod_e
enum PWM_LoadMode_e
enum PWM_Number_e
enum PWM_PeriodLoad_e
enum PWM_PhaseDir_e
enum PWM_RunMode_e
enum PWM_ShadowMode_e
enum PWM_ShadowStatus_e
enum PWM_SocPeriod_e
enum PWM_SocPulseSrc_e
enum PWM_SyncMode_e
enum PWM_TripZoneDCEventSel_e {
PWM_TripZoneDCEventSel_Disabled,          PWM_TripZoneDCEventSel_DCxHL_DCxLX,
PWM_TripZoneDCEventSel_DCxHH_DCxLX, PWM_TripZoneDCEventSel_DCxHx_DCxLL,
PWM_TripZoneDCEventSel_DCxHx_DCxLH, PWM_TripZoneDCEventSel_DCxHL_DCxLH }
enum PWM_TripZoneFlag_e {
PWM_TripZoneFlag_Global,          PWM_TripZoneFlag_CBC,          PWM_TripZoneFlag_OST,
PWM_TripZoneFlag_DCAEVT1,
PWM_TripZoneFlag_DCAEVT2, PWM_TripZoneFlag_DCBEVT1, PWM_TripZoneFlag_DCBEVT2
}
enum PWM_TripZoneSrc_e
enum PWM_TripZoneState_e
```

Functions

void [PWM_clearIntFlag](#) ([PWM_Handle](#) pwmHandle)

void [PWM_clearOneShotTrip](#) ([PWM_Handle](#) pwmHandle)

void [PWM_clearSocAFlag](#) ([PWM_Handle](#) pwmHandle)

void [PWM_clearSocBFlag](#) ([PWM_Handle](#) pwmHandle)

void [PWM_clearTripZone](#) ([PWM_Handle](#) pwmHandle, const [PWM_TripZoneFlag_e](#) tripZoneFlag)

void [PWM_decrementDeadBandFallingEdgeDelay](#) ([PWM_Handle](#) pwmHandle)

void [PWM_decrementDeadBandRisingEdgeDelay](#) ([PWM_Handle](#) pwmHandle)

void [PWM_disableAutoConvert](#) ([PWM_Handle](#) pwmHandle)

void [PWM_disableChopping](#) ([PWM_Handle](#) pwmHandle)

void [PWM_disableCounterLoad](#) ([PWM_Handle](#) pwmHandle)

void [PWM_disableDeadBand](#) ([PWM_Handle](#) pwmHandle)

void [PWM_disableDeadBandHalfCycle](#) ([PWM_Handle](#) pwmHandle)

void [PWM_disableDigitalCompareBlankingWindow](#) ([PWM_Handle](#) pwmHandle)

void [PWM_disableDigitalCompareBlankingWindowInversion](#) ([PWM_Handle](#) pwmHandle)

void [PWM_disableHrPeriod](#) ([PWM_Handle](#) pwmHandle)

void [PWM_disableHrPhaseSync](#) ([PWM_Handle](#) pwmHandle)

void [PWM_disableInt](#) ([PWM_Handle](#) pwmHandle)

void [PWM_disableSocAPulse](#) ([PWM_Handle](#) pwmHandle)

void [PWM_disableSocBPulse](#) ([PWM_Handle](#) pwmHandle)

void [PWM_disableTripZoneInt](#) ([PWM_Handle](#) pwmHandle, const [PWM_TripZoneFlag_e](#) interrupt-Source)

void [PWM_disableTripZones](#) ([PWM_Handle](#) pwmHandle)

void [PWM_disableTripZoneSrc](#) ([PWM_Handle](#) pwmHandle, const [PWM_TripZoneSrc_e](#) src)

void [PWM_enableAutoConvert](#) ([PWM_Handle](#) pwmHandle)

void [PWM_enableChopping](#) ([PWM_Handle](#) pwmHandle)

void [PWM_enableCounterLoad](#) ([PWM_Handle](#) pwmHandle)

void [PWM_enableDeadBandHalfCycle](#) ([PWM_Handle](#) pwmHandle)

void [PWM_enableDigitalCompareBlankingWindow](#) ([PWM_Handle](#) pwmHandle)

```
void PWM_enableDigitalCompareBlankingWindowInversion (PWM_Handle pwmHandle)
void PWM_enableHrPeriod (PWM_Handle pwmHandle)
void PWM_enableHrPhaseSync (PWM_Handle pwmHandle)
void PWM_enableInt (PWM_Handle pwmHandle)
void PWM_enableSocAPulse (PWM_Handle pwmHandle)
void PWM_enableSocBPulse (PWM_Handle pwmHandle)
void PWM_enableTripZoneInt (PWM_Handle pwmHandle, const PWM_TripZoneFlag_e interrupt-
Source)
void PWM_enableTripZoneSrc (PWM_Handle pwmHandle, const PWM_TripZoneSrc_e src)
void PWM_forceSync (PWM_Handle pwmHandle)
uint16_t PWM_getCmpA (PWM_Handle pwmHandle)
uint16_t PWM_getCmpAHr (PWM_Handle pwmHandle)
uint16_t PWM_getCmpB (PWM_Handle pwmHandle)
uint16_t PWM_getDeadBandFallingEdgeDelay (PWM_Handle pwmHandle)
uint16_t PWM_getDeadBandRisingEdgeDelay (PWM_Handle pwmHandle)
uint16_t PWM_getIntCount (PWM_Handle pwmHandle)
uint16_t PWM_getPeriod (PWM_Handle pwmHandle)
uint16_t PWM_getSocACount (PWM_Handle pwmHandle)
uint16_t PWM_getSocBCount (PWM_Handle pwmHandle)
void PWM_incrementDeadBandFallingEdgeDelay (PWM_Handle pwmHandle)
void PWM_incrementDeadBandRisingEdgeDelay (PWM_Handle pwmHandle)
PWM_Handle PWM_init (void *pMemory, const size_t numBytes)
void PWM_setActionQual_CntDown_CmpA_PwmA (PWM_Handle pwmHandle, const
PWM_ActionQual_e actionQual)
void PWM_setActionQual_CntDown_CmpA_PwmB (PWM_Handle pwmHandle, const
PWM_ActionQual_e actionQual)
void PWM_setActionQual_CntDown_CmpB_PwmA (PWM_Handle pwmHandle, const
PWM_ActionQual_e actionQual)
void PWM_setActionQual_CntDown_CmpB_PwmB (PWM_Handle pwmHandle, const
PWM_ActionQual_e actionQual)
void PWM_setActionQual_CntUp_CmpA_PwmA (PWM_Handle pwmHandle, const
PWM_ActionQual_e actionQual)
```

```
void PWM_setActionQual_CntUp_CmpA_PwmB (PWM_Handle pwmHandle, const
PWM_ActionQual_e actionQual)

void PWM_setActionQual_CntUp_CmpB_PwmA (PWM_Handle pwmHandle, const
PWM_ActionQual_e actionQual)

void PWM_setActionQual_CntUp_CmpB_PwmB (PWM_Handle pwmHandle, const
PWM_ActionQual_e actionQual)

void PWM_setActionQual_Period_PwmA (PWM_Handle pwmHandle, const PWM_ActionQual_e
actionQual)

void PWM_setActionQual_Period_PwmB (PWM_Handle pwmHandle, const PWM_ActionQual_e
actionQual)

void PWM_setActionQual_Zero_PwmA (PWM_Handle pwmHandle, const PWM_ActionQual_e ac-
tionQual)

void PWM_setActionQual_Zero_PwmB (PWM_Handle pwmHandle, const PWM_ActionQual_e ac-
tionQual)

void PWM_setChoppingClkFreq (PWM_Handle pwmHandle, const PWM_ChoppingClkFreq_e clk-
Freq)

void PWM_setChoppingDutyCycle (PWM_Handle pwmHandle, const
PWM_ChoppingDutyCycle_e dutyCycle)

void PWM_setChoppingPulseWidth (PWM_Handle pwmHandle, const
PWM_ChoppingPulseWidth_e pulseWidth)

void PWM_setClkDiv (PWM_Handle pwmHandle, const PWM_ClkDiv_e clkDiv)

void PWM_setCmpA (PWM_Handle pwmHandle, const uint16_t pwmData)

void PWM_setCmpAHr (PWM_Handle pwmHandle, const uint16_t pwmData)

void PWM_setCmpB (PWM_Handle pwmHandle, const uint16_t pwmData)

void PWM_setCount (PWM_Handle pwmHandle, const uint16_t count)

void PWM_setCounterMode (PWM_Handle pwmHandle, const PWM_CounterMode_e counter-
Mode)

void PWM_setDeadBandFallingEdgeDelay (PWM_Handle pwmHandle, const uint16_t delay)

void PWM_setDeadBandInputMode (PWM_Handle pwmHandle, const
PWM_DeadBandInputMode_e inputMode)

void PWM_setDeadBandOutputMode (PWM_Handle pwmHandle, const
PWM_DeadBandOutputMode_e outputMode)

void PWM_setDeadBandPolarity (PWM_Handle pwmHandle, const PWM_DeadBandPolarity_e
polarity)

void PWM_setDeadBandRisingEdgeDelay (PWM_Handle pwmHandle, const uint16_t delay)
```

```
void PWM_setDigitalCompareAEvent1 (PWM_Handle pwmHandle, const bool_t selectFilter, const
bool_t disableSync, const bool_t enableSoc, const bool_t generateSync)

void PWM_setDigitalCompareAEvent2 (PWM_Handle pwmHandle, const bool_t selectFilter, const
bool_t disableSync)

void PWM_setDigitalCompareBEvent1 (PWM_Handle pwmHandle, const bool_t selectFilter, const
bool_t disableSync, const bool_t enableSoc, const bool_t generateSync)

void PWM_setDigitalCompareBEvent2 (PWM_Handle pwmHandle, const bool_t selectFilter, const
bool_t disableSync)

void PWM_setDigitalCompareBlankingPulse (PWM_Handle pwmHandle, const
PWM_DigitalCompare_PulseSel_e pulseSelect)

void PWM_setDigitalCompareFilterOffset (PWM_Handle pwmHandle, const uint16_t offset)

void PWM_setDigitalCompareFilterSource (PWM_Handle pwmHandle, const
PWM_DigitalCompare_FilterSrc_e input)

void PWM_setDigitalCompareFilterWindow (PWM_Handle pwmHandle, const uint16_t window)

void PWM_setDigitalCompareInput (PWM_Handle pwmHandle, const
PWM_DigitalCompare_Input_e input, const PWM_DigitalCompare_InputSel_e inputSel)

void PWM_setHighSpeedClkDiv (PWM_Handle pwmHandle, const PWM_HspClkDiv_e clkDiv)

void PWM_setHrControlMode (PWM_Handle pwmHandle, const PWM_HrControlMode_e controlMode)

void PWM_setHrEdgeMode (PWM_Handle pwmHandle, const PWM_HrEdgeMode_e edgeMode)

void PWM_setHrShadowMode (PWM_Handle pwmHandle, const PWM_HrShadowMode_e shadowMode)

void PWM_setIntMode (PWM_Handle pwmHandle, const PWM_IntMode_e intMode)

void PWM_setIntPeriod (PWM_Handle pwmHandle, const PWM_IntPeriod_e intPeriod)

void PWM_setLoadMode_CmpA (PWM_Handle pwmHandle, const PWM_LoadMode_e loadMode)

void PWM_setLoadMode_CmpB (PWM_Handle pwmHandle, const PWM_LoadMode_e loadMode)

void PWM_setOneShotTrip (PWM_Handle pwmHandle)

void PWM_setPeriod (PWM_Handle pwmHandle, const uint16_t period)

void PWM_setPeriodHr (PWM_Handle pwmHandle, const uint16_t period)

void PWM_setPeriodLoad (PWM_Handle pwmHandle, const PWM_PeriodLoad_e periodLoad)

void PWM_setPhase (PWM_Handle pwmHandle, const uint16_t phase)

void PWM_setPhaseDir (PWM_Handle pwmHandle, const PWM_PhaseDir_e phaseDir)
```

```
void PWM_setRunMode (PWM_Handle pwmHandle, const PWM_RunMode_e runMode)

void PWM_setShadowMode_CmpA (PWM_Handle pwmHandle, const PWM_ShadowMode_e
shadowMode)

void PWM_setShadowMode_CmpB (PWM_Handle pwmHandle, const PWM_ShadowMode_e
shadowMode)

void PWM_setSocAPeriod (PWM_Handle pwmHandle, const PWM_SocPeriod_e intPeriod)

void PWM_setSocAPulseSrc (PWM_Handle pwmHandle, const PWM_SocPulseSrc_e pulseSrc)

void PWM_setSocBPeriod (PWM_Handle pwmHandle, const PWM_SocPeriod_e intPeriod)

void PWM_setSocBPulseSrc (PWM_Handle pwmHandle, const PWM_SocPulseSrc_e pulseSrc)

void PWM_setSwSync (PWM_Handle pwmHandle)

void PWM_setSyncMode (PWM_Handle pwmHandle, const PWM_SyncMode_e syncMode)

void PWM_setTripZoneDCEventSelect_DCAEVT1 (PWM_Handle pwmHandle, const
PWM_TripZoneDCEventSel_e tripZoneEvent)

void PWM_setTripZoneDCEventSelect_DCAEVT2 (PWM_Handle pwmHandle, const
PWM_TripZoneDCEventSel_e tripZoneEvent)

void PWM_setTripZoneDCEventSelect_DCBEVT1 (PWM_Handle pwmHandle, const
PWM_TripZoneDCEventSel_e tripZoneEvent)

void PWM_setTripZoneDCEventSelect_DCBEVT2 (PWM_Handle pwmHandle, const
PWM_TripZoneDCEventSel_e tripZoneEvent)

void PWM_setTripZoneState_DCAEVT1 (PWM_Handle pwmHandle, const
PWM_TripZoneState_e tripZoneState)

void PWM_setTripZoneState_DCAEVT2 (PWM_Handle pwmHandle, const
PWM_TripZoneState_e tripZoneState)

void PWM_setTripZoneState_DCBEVT1 (PWM_Handle pwmHandle, const
PWM_TripZoneState_e tripZoneState)

void PWM_setTripZoneState_DCBEVT2 (PWM_Handle pwmHandle, const
PWM_TripZoneState_e tripZoneState)

void PWM_setTripZoneState_TZA (PWM_Handle pwmHandle, const PWM_TripZoneState_e trip-
ZoneState)

void PWM_setTripZoneState_TZB (PWM_Handle pwmHandle, const PWM_TripZoneState_e trip-
ZoneState)

void PWM_write_CmpA (PWM_Handle pwmHandle, const int16_t pwmData)

void PWM_write_CmpB (PWM_Handle pwmHandle, const int16_t pwmData)
```

13.1.1 Detailed Description

13.1.2 Enumeration Type Documentation

13.1.2.1 enum **PWM_TripZoneDCEventSel_e**

Enumeration to define the pulse width modulation (PWM) trip zone event selections.

Enumerator

PWM_TripZoneDCEventSel_Disabled Event Disabled.

PWM_TripZoneDCEventSel_DCxHL_DCxLX Compare H = Low, Compare L = Don't Care.

PWM_TripZoneDCEventSel_DCxHH_DCxLX Compare H = High, Compare L = Don't Care.

PWM_TripZoneDCEventSel_DCxHx_DCxLL Compare H = Don't Care, Compare L = Low.

PWM_TripZoneDCEventSel_DCxHx_DCxLH Compare H = Don't Care, Compare L = High.

PWM_TripZoneDCEventSel_DCxHL_DCxLH Compare H = Low, Compare L = High.

13.1.2.2 enum **PWM_TripZoneFlag_e**

Enumeration to define the pulse width modulation (PWM) trip zone states.

Enumerator

PWM_TripZoneFlag_Global Global Trip Zone flag.

PWM_TripZoneFlag_CBC Cycle by cycle Trip Zone flag.

PWM_TripZoneFlag_OST One Shot Trip Zone flag.

PWM_TripZoneFlag_DCAEVT1 Digital Compare A Event 1 Trip Zone flag.

PWM_TripZoneFlag_DCAEVT2 Digital Compare A Event 2 Trip Zone flag.

PWM_TripZoneFlag_DCBEVT1 Digital Compare B Event 1 Trip Zone flag.

PWM_TripZoneFlag_DCBEVT2 Digital Compare B Event 2 Trip Zone flag.

13.1.3 Function Documentation

13.1.3.1 void PWM_clearIntFlag (**PWM_Handle** pwmHandle) [inline]

Clears the pulse width modulation (PWM) interrupt flag.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

References *_PWM_Obj_::ETCLR*, and *PWM_ETCLR_INT_BITS*.

13.1.3.2 void PWM_clearOneShotTrip (

PWM_Handle pwmHandle) [inline]

Clears the pulse width modulation (PWM) one shot trip.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

References DISABLE_PROTECTED_REGISTER_WRITE_MODE, EN-
ABLE_PROTECTED_REGISTER_WRITE_MODE, PWM_TZCLR_OST_BITS, and
_PWM_Obj::TZCLR.

13.1.3.3 void PWM_clearSocAFlag (

PWM_Handle pwmHandle) [inline]

Clears the pulse width modulation (PWM) start of conversion (SOC) A flag.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

References _PWM_Obj::ETCLR, and PWM_ETCLR_SOCA_BITS.

13.1.3.4 void PWM_clearSocBFlag (

PWM_Handle pwmHandle) [inline]

Clears the pulse width modulation (PWM) start of conversion (SOC) B flag.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

References _PWM_Obj::ETCLR, and PWM_ETCLR_SOCB_BITS.

13.1.3.5 void PWM_clearTripZone (

PWM_Handle pwmHandle,

const **PWM_TripZoneFlag_e** tripZoneFlag)

Clears the trip zone (TZ) flag specified.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *tripZoneFlag* The trip zone flag to clear

13.1.3.6 void PWM_decrementDeadBandFallingEdgeDelay (

PWM_Handle pwmHandle)

Decrement the dead band falling edge delay.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.7 void PWM_decrementDeadBandRisingEdgeDelay (

PWM_Handle pwmHandle)

Decrement the dead band rising edge delay.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.8 void PWM_disableAutoConvert (

PWM_Handle pwmHandle)

Disables auto conversion of delay line value.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.9 void PWM_disableChopping (

PWM_Handle pwmHandle)

Disables the pulse width modulation (PWM) chopping.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.10 void PWM_disableCounterLoad (

PWM_Handle pwmHandle)

Disables the pulse width modulation (PWM) counter loading from the phase register.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.11 void PWM_disableDeadBand (

PWM_Handle pwmHandle)

Disables the pulse width modulation (PWM) deadband.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.12 void PWM_disableDeadBandHalfCycle (

PWM_Handle pwmHandle)

Disables the pulse width modulation (PWM) deadband half cycle clocking.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.13 void PWM_disableDigitalCompareBlankingWindow (

PWM_Handle pwmHandle)

Disables the pulse width modulation (PWM) digital compare blanking window.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.14 void PWM_disableDigitalCompareBlankingWindowInversion (

PWM_Handle pwmHandle)

Disables the pulse width modulation (PWM) digital compare blanking window inversion.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.15 void PWM_disableHrPeriod (

PWM_Handle pwmHandle)

Disables high resolution period control.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.16 void PWM_disableHrPhaseSync (

PWM_Handle pwmHandle)

Disables high resolution phase synchronization.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.17 void PWM_disableInt (

PWM_Handle pwmHandle)

Disables the pulse width modulation (PWM) interrupt.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.18 void PWM_disableSocAPulse (

PWM_Handle pwmHandle)

Disables the pulse width modulation (PWM) start of conversion (SOC) B pulse generation.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.19 void PWM_disableSocBPulse (

PWM_Handle pwmHandle)

Disables the pulse width modulation (PWM) start of conversion (SOC) B pulse generation.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.20 void PWM_disableTripZoneInt (

PWM_Handle pwmHandle,

const **PWM_TripZoneFlag_e** interruptSource)

Disables the pulse width modulation (PWM) trip zones interrupts.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *interruptSource* The interrupt source to disable

13.1.3.21 void PWM_disableTripZones (

PWM_Handle pwmHandle)

Disables the pulse width modulation (PWM) trip zones.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.22 void PWM_disableTripZoneSrc (

PWM_Handle pwmHandle,

const **PWM_TripZoneSrc_e** src)

Disable the pulse width modulation (PWM) trip zone source.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *src* The pulse width modulation (PWM) trip zone source

13.1.3.23 void PWM_enableAutoConvert (

PWM_Handle pwmHandle)

Enables auto conversion of delay line value.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.24 void PWM_enableChopping (

PWM_Handle pwmHandle)

Enables the pulse width modulation (PWM) chopping.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.25 void PWM_enableCounterLoad (

PWM_Handle pwmHandle)

Enables the pulse width modulation (PWM) counter loading from the phase register.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.26 void PWM_enableDeadBandHalfCycle (

PWM_Handle pwmHandle)

Enables the pulse width modulation (PWM) deadband half cycle clocking.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.27 void PWM_enableDigitalCompareBlankingWindow (

PWM_Handle pwmHandle)

Enables the pulse width modulation (PWM) digital compare blanking window.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.28 void PWM_enableDigitalCompareBlankingWindowInversion (**PWM_Handle** pwmHandle)

Enables the pulse width modulation (PWM) digital compare blanking window inversion.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.29 void PWM_enableHrPeriod (**PWM_Handle** pwmHandle)

Enables high resolution period control.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.30 void PWM_enableHrPhaseSync (**PWM_Handle** pwmHandle)

Enables high resolution phase synchronization.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.31 void PWM_enableInt (**PWM_Handle** pwmHandle)

Enables the pulse width modulation (PWM) interrupt.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.32 void PWM_enableSocAPulse (**PWM_Handle** pwmHandle)

Enables the pulse width modulation (PWM) start of conversion (SOC) A pulse generation.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.33 void PWM_enableSocBPulse (**PWM_Handle** pwmHandle)

Enables the pulse width modulation (PWM) start of conversion (SOC) B pulse generation.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.34 void PWM_enableTripZoneInt (

PWM_Handle pwmHandle,

const **PWM_TripZoneFlag_e** interruptSource)

Enables the pulse width modulation (PWM) trip zones interrupts.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *interruptSource* The interrupt source to enable

13.1.3.35 void PWM_enableTripZoneSrc (

PWM_Handle pwmHandle,

const **PWM_TripZoneSrc_e** src)

Enable the pulse width modulation (PWM) trip zone source.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *src* The pulse width modulation (PWM) trip zone source

13.1.3.36 void PWM_forceSync (

PWM_Handle pwmHandle) [inline]

Force Synchronization.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

References PWM_TBCTL_SWFSYNC_BITS, and _PWM_Obj_::TBCTL.

13.1.3.37 uint16_t PWM_getCmpA (

PWM_Handle pwmHandle) [inline]

Gets the pulse width modulation (PWM) data value from the Counter Compare A hardware.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

Returns The PWM compare data value

References _PWM_Obj_::CMPA.

13.1.3.38 uint16_t PWM_getCmpAHr (

PWM_Handle pwmHandle) [inline]

Gets the pulse width modulation (PWM) data value from the Counter Compare A Hr hardware.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

Returns The PWM compare high resolution data value

References _PWM_Obj_::CMPAHR.

13.1.3.39 uint16_t PWM_getCmpB (

PWM_Handle pwmHandle) [inline]

Gets the pulse width modulation (PWM) data value from the Counter Compare B hardware.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

Returns The PWM compare data value

References _PWM_Obj_::CMPB.

13.1.3.40 uint16_t PWM_getDeadBandFallingEdgeDelay (

PWM_Handle pwmHandle)

Gets the pulse width modulation (PWM) deadband falling edge delay.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

Returns The delay

13.1.3.41 uint16_t PWM_getDeadBandRisingEdgeDelay (

PWM_Handle pwmHandle)

Gets the pulse width modulation (PWM) deadband rising edge delay.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

Returns The delay

13.1.3.42 uint16_t PWM_getIntCount (

PWM_Handle pwmHandle)

Gets the pulse width modulation (PWM) interrupt event count.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

Returns The interrupt event count

13.1.3.43 uint16_t PWM_getPeriod (

PWM_Handle pwmHandle) [inline]

Gets the pulse width modulation (PWM) period value.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

Returns The pwm period value

References _PWM_Obj_::TBPRD.

13.1.3.44 uint16_t PWM_getSocACount (

PWM_Handle pwmHandle)

Gets the pulse width modulation (PWM) start of conversion (SOC) A count.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

Returns The SOC A count

13.1.3.45 uint16_t PWM_getSocBCount (

PWM_Handle pwmHandle)

Gets the pulse width modulation (PWM) start of conversion (SOC) B count.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

Returns The SOC B count

13.1.3.46 void PWM_incrementDeadBandFallingEdgeDelay (

PWM_Handle pwmHandle)

Increment the dead band falling edge delay.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.47 void PWM_incrementDeadBandRisingEdgeDelay (

PWM_Handle pwmHandle)

Increment the dead band rising edge delay.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

13.1.3.48 **PWM_Handle** PWM_init (

void * pMemory,

const size_t numBytes)

Initializes the pulse width modulation (PWM) object handle.

[1]Parameters in *pMemory* A pointer to the base address of the PWM registers

in *numBytes* The number of bytes allocated for the PWM object, bytes

Returns The pulse width modulation (PWM) object handle

13.1.3.49 void PWM_setActionQual_CntDown_CmpA_PwmA (

PWM_Handle pwmHandle,

const **PWM_ActionQual_e** actionQual)

Sets the pulse width modulation (PWM) object action for PWM A when the counter equals CMPA and the counter is decrementing.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *actionQual* The action qualifier

13.1.3.50 void PWM_setActionQual_CntDown_CmpA_PwmB (

PWM_Handle pwmHandle,

const **PWM_ActionQual_e** actionQual)

Sets the pulse width modulation (PWM) object action for PWM B when the counter equals CMPA and the counter is decrementing.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *actionQual* The action qualifier

13.1.3.51 void PWM_setActionQual_CntDown_CmpB_PwmA (

PWM_Handle pwmHandle,

const **PWM_ActionQual_e** actionQual)

Sets the pulse width modulation (PWM) object action for PWM A when the counter equals CMPB and the counter is decrementing.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *actionQual* The action qualifier

13.1.3.52 void PWM_setActionQual_CntDown_CmpB_PwmB (

PWM_Handle pwmHandle,

const **PWM_ActionQual_e** actionQual)

Sets the pulse width modulation (PWM) object action for PWM B when the counter equals CMPB and the counter is decrementing.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *actionQual* The action qualifier

13.1.3.53 void PWM_setActionQual_CntUp_CmpA_PwmA (

PWM_Handle pwmHandle,

const **PWM_ActionQual_e** actionQual)

Sets the pulse width modulation (PWM) object action for PWM A when the counter equals CMPA and the counter is incrementing.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *actionQual* The action qualifier

13.1.3.54 void PWM_setActionQual_CntUp_CmpA_PwmB (

PWM_Handle pwmHandle,

const **PWM_ActionQual_e** actionQual)

Sets the pulse width modulation (PWM) object action for PWM B when the counter equals CMPA and the counter is incrementing.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *actionQual* The action qualifier

13.1.3.55 void PWM_setActionQual_CntUp_CmpB_PwmA (

PWM_Handle pwmHandle,

const **PWM_ActionQual_e** actionQual)

Sets the pulse width modulation (PWM) object action for PWM A when the counter equals CMPB and the counter is incrementing.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *actionQual* The action qualifier

13.1.3.56 void PWM_setActionQual_CntUp_CmpB_PwmB (

PWM_Handle pwmHandle,

const **PWM_ActionQual_e** actionQual)

Sets the pulse width modulation (PWM) object action for PWM B when the counter equals CMPB and the counter is incrementing.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *actionQual* The action qualifier

13.1.3.57 void PWM_setActionQual_Period_PwmA (

PWM_Handle pwmHandle,

const **PWM_ActionQual_e** actionQual)

Sets the pulse width modulation (PWM) object action for PWM A when the counter equals the period.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *actionQual* The action qualifier

13.1.3.58 void PWM_setActionQual_Period_PwmB (

PWM_Handle pwmHandle,

const **PWM_ActionQual_e** actionQual)

Sets the pulse width modulation (PWM) object action for PWM B when the counter equals the period.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *actionQual* The action qualifier

13.1.3.59 void PWM_setActionQual_Zero_PwmA (

PWM_Handle pwmHandle,

const **PWM_ActionQual_e** actionQual)

Sets the pulse width modulation (PWM) object action for PWM A when the counter equals the zero.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *actionQual* The action qualifier

13.1.3.60 void PWM_setActionQual_Zero_PwmB (

PWM_Handle pwmHandle,

const **PWM_ActionQual_e** actionQual)

Sets the pulse width modulation (PWM) object action for PWM B when the counter equals the zero.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *actionQual* The action qualifier

13.1.3.61 void PWM_setChoppingClkFreq (

PWM_Handle pwmHandle,

const **PWM_ChoppingClkFreq_e** clkFreq)

Sets the pulse width modulation (PWM) chopping clock frequency.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *clkFreq* The clock frequency

13.1.3.62 void PWM_setChoppingDutyCycle (

PWM_Handle pwmHandle,const **PWM_ChoppingDutyCycle_e** dutyCycle)

Sets the pulse width modulation (PWM) chopping clock duty cycle.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in dutyCycle The duty cycle

13.1.3.63 void PWM_setChoppingPulseWidth (

PWM_Handle pwmHandle,const **PWM_ChoppingPulseWidth_e** pulseWidth)

Sets the pulse width modulation (PWM) chopping clock pulse width.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in pulseWidth The pulse width

13.1.3.64 void PWM_setClkDiv (

PWM_Handle pwmHandle,const **PWM_ClkDiv_e** clkDiv)

Sets the pulse width modulation (PWM) clock divisor.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in clkDiv The clock divisor

13.1.3.65 void PWM_setCmpA (

PWM_Handle pwmHandle,

const uint16_t pwmData) [inline]

Writes the pulse width modulation (PWM) data value to the Counter Compare A hardware.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in pwmData The PWM data value

References `_PWM_Obj::CMPA`.

722.7

in counterMode The count mode

13.1.3.70 void PWM_setDeadBandFallingEdgeDelay (

PWM_Handle pwmHandle,

const uint16_t delay)

Sets the pulse width modulation (PWM) deadband falling edge delay.

[1]Parameters *in pwmHandle* The pulse width modulation (PWM) object handle

in delay The delay

13.1.3.71 void PWM_setDeadBandInputMode (

PWM_Handle pwmHandle,

const **PWM_DeadBandInputMode_e** inputMode)

Sets the pulse width modulation (PWM) deadband input mode.

[1]Parameters *in pwmHandle* The pulse width modulation (PWM) object handle

in inputMode The input mode

13.1.3.72 void PWM_setDeadBandOutputMode (

PWM_Handle pwmHandle,

const **PWM_DeadBandOutputMode_e** outputMode)

Sets the pulse width modulation (PWM) deadband output mode.

[1]Parameters *in pwmHandle* The pulse width modulation (PWM) object handle

in outputMode The output mode

13.1.3.73 void PWM_setDeadBandPolarity (

PWM_Handle pwmHandle,

const **PWM_DeadBandPolarity_e** polarity)

Sets the pulse width modulation (PWM) deadband polarity.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *polarity* The polarity

13.1.3.74 void PWM_setDeadBandRisingEdgeDelay (

PWM_Handle pwmHandle,

const uint16_t delay)

Sets the pulse width modulation (PWM) deadband rising edge delay.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *delay* The delay

13.1.3.75 void PWM_setDigitalCompareAEvent1 (

PWM_Handle pwmHandle,

const bool_t selectFilter,

const bool_t disableSync,

const bool_t enableSoc,

const bool_t generateSync)

Sets the pulse width modulation (PWM) digital compare A event 1 source parameters.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *selectFilter* Select filter output if true

in *disableSync* Asynchronous if true

in *enableSoc* Enable SOC generation if true

in *generateSync* Generate SYNC if true

13.1.3.76 void PWM_setDigitalCompareAEvent2 (

PWM_Handle pwmHandle,

const bool_t selectFilter,

```
const bool_t disableSync )
```

Sets the pulse width modulation (PWM) digital compare A event 2 source parameters.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *selectFilter* Select filter output if true

in *disableSync* Asynchronous if true

13.1.3.77 void PWM_setDigitalCompareBEvent1 (

PWM_Handle pwmHandle,

const bool_t selectFilter,

const bool_t disableSync,

const bool_t enableSoc,

const bool_t generateSync)

Sets the pulse width modulation (PWM) digital compare B event 1 source parameters.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *selectFilter* Select filter output if true

in *disableSync* Asynchronous if true

in *enableSoc* Enable SOC generation if true

in *generateSync* Generate SYNC if true

13.1.3.78 void PWM_setDigitalCompareBEvent2 (

PWM_Handle pwmHandle,

const bool_t selectFilter,

const bool_t disableSync)

Sets the pulse width modulation (PWM) digital compare B event 2 source parameters.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *selectFilter* Select filter output if true

in *disableSync* Asynchronous if true

13.1.3.79 void PWM_setDigitalCompareBlankingPulse (

PWM_Handle pwmHandle,

const **PWM_DigitalCompare_PulseSel_e** pulseSelect)

Sets the pulse width modulation (PWM) digital compare blanking pulse.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *pulseSelect* The pulse selection

13.1.3.80 void PWM_setDigitalCompareFilterOffset (

PWM_Handle pwmHandle,

const uint16_t offset)

Sets the pulse width modulation (PWM) digital compare filter offset.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *offset* The offset

13.1.3.81 void PWM_setDigitalCompareFilterSource (

PWM_Handle pwmHandle,

const **PWM_DigitalCompare_FilterSrc_e** input)

Sets the pulse width modulation (PWM) digital compare filter source.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *input* The filter's source

13.1.3.82 void PWM_setDigitalCompareFilterWindow (

PWM_Handle pwmHandle,

const uint16_t window)

Sets the pulse width modulation (PWM) digital compare filter offset.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *window* The window

13.1.3.83 void PWM_setDigitalCompareInput (

PWM_Handle pwmHandle,
 const **PWM_DigitalCompare_Input_e** input,
 const **PWM_DigitalCompare_InputSel_e** inputSel)

Sets the pulse width modulation (PWM) digital compare input.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *input* Comparator input to change

in *inputSel* Input selection for designated input

13.1.3.84 void PWM_setHighSpeedClkDiv (

PWM_Handle pwmHandle,
 const **PWM_HspClkDiv_e** clkDiv)

Sets the pulse width modulation (PWM) high speed clock divisor.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *clkDiv* The clock divisor

13.1.3.85 void PWM_setHrControlMode (

PWM_Handle pwmHandle,
 const **PWM_HrControlMode_e** controlMode)

Set the High Resolution Control Mode.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *controlMode* The control mode HRPWM should use

13.1.3.86 void PWM_setHrEdgeMode (

PWM_Handle pwmHandle,
 const **PWM_HrEdgeMode_e** edgeMode)

Set the High Resolution Edge Mode.

722.7

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *loadMode* The load mode

13.1.3.91 void PWM_setLoadMode_CmpB (

PWM_Handle pwmHandle,

const **PWM_LoadMode_e** loadMode)

Sets the pulse width modulation (PWM) load mode for CMPB.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *loadMode* The load mode

13.1.3.92 void PWM_setOneShotTrip (

PWM_Handle pwmHandle) [inline]

Sets the pulse width modulation (PWM) one shot trip.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

References	DISABLE_PROTECTED_REGISTER_WRITE_MODE,	EN-
ABLE_PROTECTED_REGISTER_WRITE_MODE,	PWM_TZFRC_OST_BITS,	and
_PWM_Obj::TZFRC.		

13.1.3.93 void PWM_setPeriod (

PWM_Handle pwmHandle,

const uint16_t period)

Sets the pulse width modulation (PWM) period.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *period* The period

13.1.3.94 void PWM_setPeriodHr (

PWM_Handle pwmHandle,

const uint16_t period)

Sets the pulse width modulation (PWM) high resolution period.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *period* The period

13.1.3.95 void PWM_setPeriodLoad (

PWM_Handle pwmHandle,

const **PWM_PeriodLoad_e** periodLoad)

Sets the pulse width modulation (PWM) period load mode.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *periodLoad* The period load mode

13.1.3.96 void PWM_setPhase (

PWM_Handle pwmHandle,

const uint16_t phase)

Sets the pulse width modulation (PWM) phase.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *phase* The phase

13.1.3.97 void PWM_setPhaseDir (

PWM_Handle pwmHandle,

const **PWM_PhaseDir_e** phaseDir)

Sets the pulse width modulation (PWM) phase direction.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *phaseDir* The phase direction

13.1.3.98 void PWM_setRunMode (

PWM_Handle pwmHandle,

```
const PWM_RunMode_e runMode )
```

Sets the pulse width modulation (PWM) run mode.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *runMode* The run mode

```
13.1.3.99 void PWM_setShadowMode_CmpA (
```

```
    PWM_Handle pwmHandle,
```

```
    const PWM_ShadowMode_e shadowMode )
```

Sets the pulse width modulation (PWM) shadow mode for CMPA.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *shadowMode* The shadow mode

```
13.1.3.100 void PWM_setShadowMode_CmpB (
```

```
    PWM_Handle pwmHandle,
```

```
    const PWM_ShadowMode_e shadowMode )
```

Sets the pulse width modulation (PWM) shadow mode for CMPB.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *shadowMode* The shadow mode

```
13.1.3.101 void PWM_setSocAPeriod (
```

```
    PWM_Handle pwmHandle,
```

```
    const PWM_SocPeriod_e intPeriod )
```

Sets the pulse width modulation (PWM) start of conversion (SOC) A interrupt period.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *intPeriod* The interrupt period

```
13.1.3.102 void PWM_setSocAPulseSrc (
```

```
    PWM_Handle pwmHandle,
```


722.7

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *syncMode* The sync mode

13.1.3.107void PWM_setTripZoneDCEventSelect_DCAEVT1 (

PWM_Handle pwmHandle,

const **PWM_TripZoneDCEventSel_e** tripZoneEvent)

Sets the pulse width modulation (PWM) trip zone digital compare event select for Digital Compare Output A Event 1 (DCAEVT1)

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *tripZoneEvent* The trip zone digital compare event

13.1.3.108void PWM_setTripZoneDCEventSelect_DCAEVT2 (

PWM_Handle pwmHandle,

const **PWM_TripZoneDCEventSel_e** tripZoneEvent)

Sets the pulse width modulation (PWM) trip zone digital compare event select for Digital Compare Output A Event 2 (DCAEVT2)

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *tripZoneEvent* The trip zone digital compare event

13.1.3.109void PWM_setTripZoneDCEventSelect_DCBEVT1 (

PWM_Handle pwmHandle,

const **PWM_TripZoneDCEventSel_e** tripZoneEvent)

Sets the pulse width modulation (PWM) trip zone digital compare event select for Digital Compare Output B Event 1 (DCBEVT1)

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *tripZoneEvent* The trip zone digital compare event

13.1.3.110void PWM_setTripZoneDCEventSelect_DCBEVT2 (

PWM_Handle pwmHandle,

const **PWM_TripZoneDCEventSel_e** tripZoneEvent)

Sets the pulse width modulation (PWM) trip zone digital compare event select for Digital Compare Output B Event 2 (DCBEVT2)

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *tripZoneEvent* The trip zone digital compare event

13.1.3.11 void PWM_setTripZoneState_DCAEVT1 (

PWM_Handle pwmHandle,

const **PWM_TripZoneState_e** tripZoneState)

Sets the pulse width modulation (PWM) trip zone state for Digital Compare Output A Event 1 (DCAEVT1)

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *tripZoneState* The trip zone state

13.1.3.12 void PWM_setTripZoneState_DCAEVT2 (

PWM_Handle pwmHandle,

const **PWM_TripZoneState_e** tripZoneState)

Sets the pulse width modulation (PWM) trip zone state for Digital Compare Output A Event 2 (DCAEVT2)

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *tripZoneState* The trip zone state

13.1.3.13 void PWM_setTripZoneState_DCBEVT1 (

PWM_Handle pwmHandle,

const **PWM_TripZoneState_e** tripZoneState)

Sets the pulse width modulation (PWM) trip zone state for Digital Compare Output B Event 1 (DCBEVT1)

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in tripZoneState The trip zone state

13.1.3.114 void PWM_setTripZoneState_DCBEVT2 (

PWM_Handle pwmHandle,

const **PWM_TripZoneState_e** tripZoneState)

Sets the pulse width modulation (PWM) trip zone state for Digital Compare Output B Event 2 (DCBEVT1)

[1]Parameters *in pwmHandle* The pulse width modulation (PWM) object handle

in tripZoneState The trip zone state

13.1.3.115 void PWM_setTripZoneState_TZA (

PWM_Handle pwmHandle,

const **PWM_TripZoneState_e** tripZoneState)

Sets the pulse width modulation (PWM) trip zone state for Output A (TZA)

[1]Parameters *in pwmHandle* The pulse width modulation (PWM) object handle

in tripZoneState The trip zone state

13.1.3.116 void PWM_setTripZoneState_TZB (

PWM_Handle pwmHandle,

const **PWM_TripZoneState_e** tripZoneState)

Sets the pulse width modulation (PWM) trip zone state for Output B (TZB)

[1]Parameters *in pwmHandle* The pulse width modulation (PWM) object handle

in tripZoneState The trip zone state

13.1.3.117 void PWM_write_CmpA (

PWM_Handle pwmHandle,

const int16_t pwmData) [inline]

Writes the pulse width modulation (PWM) data value to the Counter Compare A hardware.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *pwmData* The PWM data value

References _PWM_Obj_::CMPA, and _PWM_Obj_::TBPRD.

13.1.3.11 void PWM_write_CmpB (

PWM_Handle pwmHandle,

const int16_t pwmData) [inline]

Writes the pulse width modulation (PWM) data value to the Counter Compare B hardware.

[1]Parameters in *pwmHandle* The pulse width modulation (PWM) object handle

in *pwmData* The PWM data value

References _PWM_Obj_::CMPB, and _PWM_Obj_::TBPRD.

14 Power Control (PWR)

[Introduction](#)??

[API Functions](#) .. 151 The power API is a set of functions for configuring low power modes as well as other power related functions such as brown out.

This driver is contained in `f2802x/common/source/pwr.c`, with `f2802x/common/include/pwr.h` containing the API definitions for use by applications.

14.1 PWR

Data Structures

struct [_PWR_Obj_](#)

Macros

`#define` [PWR_BASE_ADDR](#)

`#define` [PWR_BORCFG_BORENZ_BITS](#)

`#define` [PWR_LPMCR0_LPM_BITS](#)

`#define` [PWR_LPMCR0_QUALSTDBY_BITS](#)

`#define` [PWR_LPMCR0_WDINTE_BITS](#)

Typedefs

`typedef struct` [_PWR_Obj_](#) * [PWR_Handle](#)

`typedef struct` [_PWR_Obj_](#) [PWR_Obj](#)

Enumerations

`enum` [PWR_LowPowerMode_e](#) { [PWR_LowPowerMode_Idle](#), [PWR_LowPowerMode_Standby](#), [PWR_LowPowerMode_Halt](#) }

`enum` [PWR_NumStandByClocks_e](#) {
[PWR_NumStandByClocks_2](#), [PWR_NumStandByClocks_3](#), [PWR_NumStandByClocks_4](#),
[PWR_NumStandByClocks_5](#),
[PWR_NumStandByClocks_6](#), [PWR_NumStandByClocks_7](#), [PWR_NumStandByClocks_8](#),
[PWR_NumStandByClocks_9](#),
[PWR_NumStandByClocks_10](#), [PWR_NumStandByClocks_11](#), [PWR_NumStandByClocks_12](#),

PWR_NumStandByClocks_13,
PWR_NumStandByClocks_14, PWR_NumStandByClocks_15, PWR_NumStandByClocks_16,
PWR_NumStandByClocks_17, PWR_NumStandByClocks_19, PWR_NumStandByClocks_20,
PWR_NumStandByClocks_21, PWR_NumStandByClocks_23, PWR_NumStandByClocks_24,
PWR_NumStandByClocks_22, PWR_NumStandByClocks_25, PWR_NumStandByClocks_26, PWR_NumStandByClocks_27, PWR_NumStandByClocks_28,
PWR_NumStandByClocks_29, PWR_NumStandByClocks_30, PWR_NumStandByClocks_31, PWR_NumStandByClocks_32,
PWR_NumStandByClocks_33, PWR_NumStandByClocks_34, PWR_NumStandByClocks_35, PWR_NumStandByClocks_36,
PWR_NumStandByClocks_37, PWR_NumStandByClocks_38, PWR_NumStandByClocks_39, PWR_NumStandByClocks_40,
PWR_NumStandByClocks_41, PWR_NumStandByClocks_42, PWR_NumStandByClocks_43, PWR_NumStandByClocks_44,
PWR_NumStandByClocks_45, PWR_NumStandByClocks_46, PWR_NumStandByClocks_47, PWR_NumStandByClocks_48,
PWR_NumStandByClocks_49, PWR_NumStandByClocks_50, PWR_NumStandByClocks_51, PWR_NumStandByClocks_52,
PWR_NumStandByClocks_53, PWR_NumStandByClocks_54, PWR_NumStandByClocks_55, PWR_NumStandByClocks_56,
PWR_NumStandByClocks_57, PWR_NumStandByClocks_58, PWR_NumStandByClocks_59, PWR_NumStandByClocks_60,
PWR_NumStandByClocks_61, PWR_NumStandByClocks_62, PWR_NumStandByClocks_63, PWR_NumStandByClocks_64,
PWR_NumStandByClocks_65 }

Functions

void PWR_disableBrownOutReset (PWR_Handle pwrHandle)

void PWR_disableWatchDogInt (PWR_Handle pwrHandle)

void PWR_enableBrownOutReset (PWR_Handle pwrHandle)

void PWR_enableWatchDogInt (PWR_Handle pwrHandle)

PWR_Handle PWR_init (void *pMemory, const size_t numBytes)

void PWR_setLowPowerMode (PWR_Handle pwrHandle, const PWR_LowPowerMode_e lowPowerMode)

void PWR_setNumStandByClocks (PWR_Handle pwrHandle, const PWR_NumStandByClocks_e numClkCycles)

14.1.1 Detailed Description

14.1.2 Enumeration Type Documentation

14.1.2.1 enum **PWR_LowPowerMode_e**

Enumeration to define the power (PWR) low power modes.

Enumerator

PWR_LowPowerMode_Idle Denotes the idle mode.

PWR_LowPowerMode_Standby Denotes the standby mode.

PWR_LowPowerMode_Halt Denotes the halt mode.

14.1.2.2 enum **PWR_NumStandByClocks_e**

Enumeration to define the power (PWR) number of standby clock cycles.

Enumerator

PWR_NumStandByClocks_2 Denotes 2 standby clock cycles.

PWR_NumStandByClocks_3 Denotes 3 standby clock cycles.

PWR_NumStandByClocks_4 Denotes 4 standby clock cycles.

PWR_NumStandByClocks_5 Denotes 5 standby clock cycles.

PWR_NumStandByClocks_6 Denotes 6 standby clock cycles.

PWR_NumStandByClocks_7 Denotes 7 standby clock cycles.

PWR_NumStandByClocks_8 Denotes 8 standby clock cycles.

PWR_NumStandByClocks_9 Denotes 9 standby clock cycles.

PWR_NumStandByClocks_10 Denotes 10 standby clock cycles.

PWR_NumStandByClocks_11 Denotes 11 standby clock cycles.

PWR_NumStandByClocks_12 Denotes 12 standby clock cycles.

PWR_NumStandByClocks_13 Denotes 13 standby clock cycles.

PWR_NumStandByClocks_14 Denotes 14 standby clock cycles.

PWR_NumStandByClocks_15 Denotes 15 standby clock cycles.

PWR_NumStandByClocks_16 Denotes 16 standby clock cycles.

PWR_NumStandByClocks_17 Denotes 17 standby clock cycles.

PWR_NumStandByClocks_18 Denotes 18 standby clock cycles.

PWR_NumStandByClocks_19 Denotes 19 standby clock cycles.

PWR_NumStandByClocks_20 Denotes 20 standby clock cycles.

PWR_NumStandByClocks_21 Denotes 21 standby clock cycles.

PWR_NumStandByClocks_22 Denotes 22 standby clock cycles.

PWR_NumStandByClocks_23 Denotes 23 standby clock cycles.

PWR_NumStandByClocks_24 Denotes 24 standby clock cycles.

PWR_NumStandByClocks_25 Denotes 25 standby clock cycles.

PWR_NumStandByClocks_26 Denotes 26 standby clock cycles.

PWR_NumStandByClocks_27 Denotes 27 standby clock cycles.

PWR_NumStandByClocks_28 Denotes 28 standby clock cycles.

<i>PWR_NumStandByClocks_29</i>	Denotes 29 standby clock cycles.
<i>PWR_NumStandByClocks_30</i>	Denotes 30 standby clock cycles.
<i>PWR_NumStandByClocks_31</i>	Denotes 31 standby clock cycles.
<i>PWR_NumStandByClocks_32</i>	Denotes 32 standby clock cycles.
<i>PWR_NumStandByClocks_33</i>	Denotes 33 standby clock cycles.
<i>PWR_NumStandByClocks_34</i>	Denotes 34 standby clock cycles.
<i>PWR_NumStandByClocks_35</i>	Denotes 35 standby clock cycles.
<i>PWR_NumStandByClocks_36</i>	Denotes 36 standby clock cycles.
<i>PWR_NumStandByClocks_37</i>	Denotes 37 standby clock cycles.
<i>PWR_NumStandByClocks_38</i>	Denotes 38 standby clock cycles.
<i>PWR_NumStandByClocks_39</i>	Denotes 39 standby clock cycles.
<i>PWR_NumStandByClocks_40</i>	Denotes 40 standby clock cycles.
<i>PWR_NumStandByClocks_41</i>	Denotes 41 standby clock cycles.
<i>PWR_NumStandByClocks_42</i>	Denotes 42 standby clock cycles.
<i>PWR_NumStandByClocks_43</i>	Denotes 43 standby clock cycles.
<i>PWR_NumStandByClocks_44</i>	Denotes 44 standby clock cycles.
<i>PWR_NumStandByClocks_45</i>	Denotes 45 standby clock cycles.
<i>PWR_NumStandByClocks_46</i>	Denotes 46 standby clock cycles.
<i>PWR_NumStandByClocks_47</i>	Denotes 47 standby clock cycles.
<i>PWR_NumStandByClocks_48</i>	Denotes 48 standby clock cycles.
<i>PWR_NumStandByClocks_49</i>	Denotes 49 standby clock cycles.
<i>PWR_NumStandByClocks_50</i>	Denotes 50 standby clock cycles.
<i>PWR_NumStandByClocks_51</i>	Denotes 51 standby clock cycles.
<i>PWR_NumStandByClocks_52</i>	Denotes 52 standby clock cycles.
<i>PWR_NumStandByClocks_53</i>	Denotes 53 standby clock cycles.
<i>PWR_NumStandByClocks_54</i>	Denotes 54 standby clock cycles.
<i>PWR_NumStandByClocks_55</i>	Denotes 55 standby clock cycles.
<i>PWR_NumStandByClocks_56</i>	Denotes 56 standby clock cycles.
<i>PWR_NumStandByClocks_57</i>	Denotes 57 standby clock cycles.
<i>PWR_NumStandByClocks_58</i>	Denotes 58 standby clock cycles.
<i>PWR_NumStandByClocks_59</i>	Denotes 59 standby clock cycles.
<i>PWR_NumStandByClocks_60</i>	Denotes 60 standby clock cycles.
<i>PWR_NumStandByClocks_61</i>	Denotes 61 standby clock cycles.
<i>PWR_NumStandByClocks_62</i>	Denotes 62 standby clock cycles.
<i>PWR_NumStandByClocks_63</i>	Denotes 63 standby clock cycles.
<i>PWR_NumStandByClocks_64</i>	Denotes 64 standby clock cycles.
<i>PWR_NumStandByClocks_65</i>	Denotes 65 standby clock cycles.

14.1.3 Function Documentation

14.1.3.1 void PWR_disableBrownOutReset (

PWR_Handle pwrHandle)

Disables the brownout reset functions.

[1]Parameters in *pwrHandle* The power (PWR) object handle

722.7

in lowPowerMode The low power mode

14.1.3.7 void PWR_setNumStandByClocks (

PWR_Handle pwrHandle,

const **PWR_NumStandByClocks_e** numClkCycles)

Sets the number of standby clock cycles.

[1]Parameters *in pwrHandle* The power (PWR) object handle

in numClkCycles The number of standby clock cycles

15 Serial Communications Interface (SCI)

[Introduction](#) [??](#)

[API Functions](#) 157 The Serial Communication Interface (SCI) API provides a set of functions for configuring and using the SCI peripheral(s) on this device.

This driver is contained in `f2802x/common/source/sci.c`, with `f2802x/common/include/sci.h` containing the API definitions for use by applications.

15.1 SCI

Data Structures

struct `_SCI_Obj`

Macros

```
#define SCI_SCICCR_CHAR_LENGTH_BITS
#define SCI_SCICCR_LB_ENA_BITS
#define SCI_SCICCR_MODE_BITS
#define SCI_SCICCR_PARITY_BITS
#define SCI_SCICCR_PARITY_ENA_BITS
#define SCI_SCICCR_STOP_BITS
#define SCI_SCICTL1_RESET_BITS
#define SCI_SCICTL1_RX_ERR_INT_ENA_BITS
#define SCI_SCICTL1_RXENA_BITS
#define SCI_SCICTL1_SLEEP_BITS
#define SCI_SCICTL1_TXENA_BITS
#define SCI_SCICTL1_TXWAKE_BITS
#define SCI_SCICTL2_RX_INT_ENA_BITS
#define SCI_SCICTL2_TX_INT_ENA_BITS
#define SCI_SCICTL2_TXEMPTY_BITS
#define SCI_SCICTL2_TXRDY_BITS
#define SCI_SCIFFCT_ABD_BITS
```

```
#define SCI_SCIFFCT_ABDCLR_BITS
#define SCI_SCIFFCT_CDC_BITS
#define SCI_SCIFFCT_DELAY_BITS
#define SCI_SCIFFRX_FIFO_OVF_BITS
#define SCI_SCIFFRX_FIFO_OVFCLR_BITS
#define SCI_SCIFFRX_FIFO_RESET_BITS
#define SCI_SCIFFRX_FIFO_ST_BITS
#define SCI_SCIFFRX_IENA_BITS
#define SCI_SCIFFRX_IL_BITS
#define SCI_SCIFFRX_INT_BITS
#define SCI_SCIFFRX_INTCLR_BITS
#define SCI_SCIFFTX_CHAN_RESET_BITS
#define SCI_SCIFFTX_FIFO_ENA_BITS
#define SCI_SCIFFTX_FIFO_RESET_BITS
#define SCI_SCIFFTX_FIFO_ST_BITS
#define SCI_SCIFFTX_IENA_BITS
#define SCI_SCIFFTX_IL_BITS
#define SCI_SCIFFTX_INT_BITS
#define SCI_SCIFFTX_INTCLR_BITS
#define SCI_SCIRXST_BRKDT_BITS
#define SCI_SCIRXST_FE_BITS
#define SCI_SCIRXST_OE_BITS
#define SCI_SCIRXST_PE_BITS
#define SCI_SCIRXST_RXERROR_BITS
#define SCI_SCIRXST_RXRDY_BITS
#define SCI_SCIRXST_RXWAKE_BITS
#define SCIA_BASE_ADDR
```

Typedefs

```
typedef struct _SCI_Obj * SCI_Handle
```

```
typedef struct _SCI_Obj SCI_Obj
```

Enumerations

```
enum SCI_BaudRate_e { SCI_BaudRate_9_6_kBaud, SCI_BaudRate_19_2_kBaud,  
SCI_BaudRate_57_6_kBaud, SCI_BaudRate_115_2_kBaud }
```

```
enum SCI_CharLength_e {  
SCI_CharLength_1_Bit, SCI_CharLength_2_Bits, SCI_CharLength_3_Bits,  
SCI_CharLength_4_Bits, SCI_CharLength_5_Bits, SCI_CharLength_6_Bits, SCI_CharLength_7_Bits,  
SCI_CharLength_8_Bits }
```

```
enum SCI_FifoLevel_e {  
SCI_FifoLevel_Empty, SCI_FifoLevel_1_Word, SCI_FifoLevel_2_Words, SCI_FifoLevel_3_Words,  
SCI_FifoLevel_4_Words }
```

```
enum SCI_FifoStatus_e {  
SCI_FifoStatus_Empty, SCI_FifoStatus_1_Word, SCI_FifoStatus_2_Words,  
SCI_FifoStatus_3_Words, SCI_FifoStatus_4_Words }
```

```
enum SCI_Mode_e { SCI_Mode_IdleLine, SCI_Mode_AddressBit }
```

```
enum SCI_NumStopBits_e { SCI_NumStopBits_One, SCI_NumStopBits_Two }
```

```
enum SCI_Parity_e { SCI_Parity_Odd, SCI_Parity_Even }
```

```
enum SCI_Priority_e { SCI_Priority_Immediate, SCI_Priority_FreeRun,  
SCI_Priority_AfterRxRxSeq }
```

Functions

```
void SCI_clearAutoBaudDetect (SCI_Handle sciHandle)
```

```
void SCI_clearRxFifoInt (SCI_Handle sciHandle)
```

```
void SCI_clearRxFifoOvf (SCI_Handle sciHandle)
```

```
void SCI_clearTxFifoInt (SCI_Handle sciHandle)
```

```
void SCI_disable (SCI_Handle sciHandle)
```

```
void SCI_disableAutoBaudAlign (SCI_Handle sciHandle)
```

```
void SCI_disableFifoEnh (SCI_Handle sciHandle)
```

```
void SCI_disableLoopBack (SCI_Handle sciHandle)
```

```
void SCI_disableParity (SCI_Handle sciHandle)
void SCI_disableRx (SCI_Handle sciHandle)
void SCI_disableRxErrorInt (SCI_Handle sciHandle)
void SCI_disableRxFifoInt (SCI_Handle sciHandle)
void SCI_disableRxInt (SCI_Handle sciHandle)
void SCI_disableSleep (SCI_Handle sciHandle)
void SCI_disableTx (SCI_Handle sciHandle)
void SCI_disableTxFifoInt (SCI_Handle sciHandle)
void SCI_disableTxInt (SCI_Handle sciHandle)
void SCI_disableTxWake (SCI_Handle sciHandle)
void SCI_enable (SCI_Handle sciHandle)
void SCI_enableAutoBaudAlign (SCI_Handle sciHandle)
void SCI_enableFifoEnh (SCI_Handle sciHandle)
void SCI_enableLoopBack (SCI_Handle sciHandle)
void SCI_enableParity (SCI_Handle sciHandle)
void SCI_enableRx (SCI_Handle sciHandle)
void SCI_enableRxErrorInt (SCI_Handle sciHandle)
void SCI_enableRxFifoInt (SCI_Handle sciHandle)
void SCI_enableRxInt (SCI_Handle sciHandle)
void SCI_enableSleep (SCI_Handle sciHandle)
void SCI_enableTx (SCI_Handle sciHandle)
void SCI_enableTxFifoInt (SCI_Handle sciHandle)
void SCI_enableTxInt (SCI_Handle sciHandle)
void SCI_enableTxWake (SCI_Handle sciHandle)
uint16_t SCI_getData (SCI_Handle sciHandle)
uint16_t SCI_getDataBlocking (SCI_Handle sciHandle)
uint16_t SCI_getDataNonBlocking (SCI_Handle sciHandle, uint16_t *success)
SCI_FifoStatus_e SCI_getRxFifoStatus (SCI_Handle sciHandle)
SCI_FifoStatus_e SCI_getTxFifoStatus (SCI_Handle sciHandle)
```

```
SCI_Handle SCI_init (void *pMemory, const size_t numBytes)

bool_t SCI_isRxDataReady (SCI_Handle sciHandle)

bool_t SCI_isTxReady (SCI_Handle sciHandle)

void SCI_putData (SCI_Handle sciHandle, const uint16_t data)

void SCI_putDataBlocking (SCI_Handle sciHandle, uint16_t data)

uint16_t SCI_putDataNonBlocking (SCI_Handle sciHandle, uint16_t data)

void SCI_reset (SCI_Handle sciHandle)

void SCI_resetChannels (SCI_Handle sciHandle)

void SCI_resetRxFifo (SCI_Handle sciHandle)

void SCI_resetTxFifo (SCI_Handle sciHandle)

void SCI_setBaudRate (SCI_Handle sciHandle, const SCI_BaudRate_e baudRate)

void SCI_setCharLength (SCI_Handle sciHandle, const SCI_CharLength_e charLength)

void SCI_setMode (SCI_Handle sciHandle, const SCI_Mode_e mode)

void SCI_setNumStopBits (SCI_Handle sciHandle, const SCI_NumStopBits_e numBits)

void SCI_setParity (SCI_Handle sciHandle, const SCI_Parity_e parity)

void SCI_setPriority (SCI_Handle sciHandle, const SCI_Priority_e priority)

void SCI_setRxFifoIntLevel (SCI_Handle sciHandle, const SCI_FifoLevel_e fifoLevel)

void SCI_setTxDelay (SCI_Handle sciHandle, const uint8_t delay)

void SCI_setTxFifoIntLevel (SCI_Handle sciHandle, const SCI_FifoLevel_e fifoLevel)
```

15.1.1 Detailed Description

15.1.2 Enumeration Type Documentation

15.1.2.1 enum **SCI_BaudRate_e**

Enumeration to define the serial communications interface (SCI) baud rates. This enumeration assume a device clock of 60Mhz and a LSPCLK of 15MHz.

Enumerator

SCI_BaudRate_9_6_kBaud Denotes 9.6 kBaud.
SCI_BaudRate_19_2_kBaud Denotes 19.2 kBaud.
SCI_BaudRate_57_6_kBaud Denotes 57.6 kBaud.
SCI_BaudRate_115_2_kBaud Denotes 115.2 kBaud.

15.1.2.2 enum **SCI_CharLength_e**

Enumeration to define the serial communications interface (SCI) character lengths.

Enumerator

- SCI_CharLength_1_Bit** Denotes a character length of 1 bit.
- SCI_CharLength_2_Bits** Denotes a character length of 2 bits.
- SCI_CharLength_3_Bits** Denotes a character length of 3 bits.
- SCI_CharLength_4_Bits** Denotes a character length of 4 bits.
- SCI_CharLength_5_Bits** Denotes a character length of 5 bits.
- SCI_CharLength_6_Bits** Denotes a character length of 6 bits.
- SCI_CharLength_7_Bits** Denotes a character length of 7 bits.
- SCI_CharLength_8_Bits** Denotes a character length of 8 bits.

15.1.2.3 enum **SCI_FifoLevel_e**

Enumeration to define the serial communications interface (SCI) FIFO level.

Enumerator

- SCI_FifoLevel_Empty** Denotes the fifo is empty.
- SCI_FifoLevel_1_Word** Denotes the fifo contains 1 word.
- SCI_FifoLevel_2_Words** Denotes the fifo contains 2 words.
- SCI_FifoLevel_3_Words** Denotes the fifo contains 3 words.
- SCI_FifoLevel_4_Words** Denotes the fifo contains 4 words.

15.1.2.4 enum **SCI_FifoStatus_e**

Enumeration to define the serial communications interface (SCI) FIFO status.

Enumerator

- SCI_FifoStatus_Empty** Denotes the fifo is empty.
- SCI_FifoStatus_1_Word** Denotes the fifo contains 1 word.
- SCI_FifoStatus_2_Words** Denotes the fifo contains 2 words.
- SCI_FifoStatus_3_Words** Denotes the fifo contains 3 words.
- SCI_FifoStatus_4_Words** Denotes the fifo contains 4 words.

15.1.2.5 enum **SCI_Mode_e**

Enumeration to define the serial communications interface (SCI) multiprocessor protocol mode.

Enumerator

- SCI_Mode_IdleLine** Denotes idle-line mode protocol.
- SCI_Mode_AddressBit** Denotes address-bit mode protocol.

15.1.2.6 enum **SCI_NumStopBits_e**

Enumeration to define the serial communications interface (SCI) number of stop bits.

Enumerator

SCI_NumStopBits_One Denotes 1 stop bit.

SCI_NumStopBits_Two Denotes 2 stop bits.

15.1.2.7 enum **SCI_Parity_e**

Enumeration to define the serial communications interface (SCI) parity.

Enumerator

SCI_Parity_Odd Denotes odd parity.

SCI_Parity_Even Denotes even parity.

15.1.2.8 enum **SCI_Priority_e**

Enumeration to define the serial communications interface (SCI) emulation suspend priority.

Enumerator

SCI_Priority_Immediate Denotes an immediate stop.

SCI_Priority_FreeRun Denotes free running.

SCI_Priority_AfterRxRxSeq Denotes that a stop after the current receive/transmit sequence.

15.1.3 Function Documentation

15.1.3.1 void SCI_clearAutoBaudDetect (

SCI_Handle sciHandle)

Clears the auto baud detect mode.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.2 void SCI_clearRxFifoInt (

SCI_Handle sciHandle)

Clears the Rx FIFO interrupt flag.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.3 void SCI_clearRxFifoOvf (

SCI_Handle sciHandle)

Clears the Rx FIFO overflow flag.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.4 void SCI_clearTxFifoInt (

SCI_Handle sciHandle)

Clears the Tx FIFO interrupt flag.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.5 void SCI_disable (

SCI_Handle sciHandle)

Disables the serial communications interface (SCI) interrupt.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.6 void SCI_disableAutoBaudAlign (

SCI_Handle sciHandle)

Disable the serial communications interface (SCI) auto baud alignment.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.7 void SCI_disableFifoEnh (

SCI_Handle sciHandle)

Disables the serial communications interface (SCI) FIFO enhancements.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.8 void SCI_disableLoopBack (

SCI_Handle sciHandle)

Disables the serial peripheral interface (SCI) loop back mode.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.9 void SCI_disableParity (

SCI_Handle sciHandle)

Disable the parity.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.10 void SCI_disableRx (

SCI_Handle sciHandle)

Disables the serial communications interface (SCI) master/slave receive mode.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.11 void SCI_disableRxErrorInt (

SCI_Handle sciHandle)

Disables the serial communications interface (SCI) receive error interrupt.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.12 void SCI_disableRxFifoInt (

SCI_Handle sciHandle)

Disables the serial communications interface (SCI) receive FIFO interrupt.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.13 void SCI_disableRxInt (

SCI_Handle sciHandle)

Disables the serial communications interface (SCI) receive interrupt.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.14 void SCI_disableSleep (

SCI_Handle sciHandle)

Disables the serial communications interface (SCI) sleep mode.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.15 void SCI_disableTx (

SCI_Handle sciHandle)

Disables the serial communications interface (SCI) master/slave transmit mode.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.16 void SCI_disableTxFifoInt (

SCI_Handle sciHandle)

Disables the serial communications interface (SCI) transmit FIFO interrupt.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.17 void SCI_disableTxInt (

SCI_Handle sciHandle)

Disables the serial communications interface (SCI) transmit interrupt.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.18 void SCI_disableTxWake (

SCI_Handle sciHandle)

Disables the serial communications interface (SCI) wakeup method.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.19 void SCI_enable (

SCI_Handle sciHandle)

Enables the serial communications interface (SCI)

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.20 void SCI_enableAutoBaudAlign (

SCI_Handle sciHandle)

Enable the serial communications interface (SCI) auto baud alignment.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.21 void SCI_enableFifoEnh (

SCI_Handle sciHandle)

Enables the serial communications interface (SCI) FIFO enhancements.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.22 void SCI_enableLoopBack (

SCI_Handle sciHandle)

Enables the serial peripheral interface (SCI) loop back mode.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.23 void SCI_enableParity (

SCI_Handle sciHandle)

Enables the serial peripheral interface (SCI) parity.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.24 void SCI_enableRx (

SCI_Handle sciHandle)

Enables the serial peripheral interface (SCI) receiver.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.25 void SCI_enableRxErrorInt (

SCI_Handle sciHandle)

Enables the serial communications interface (SCI) receive error interrupt.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.26 void SCI_enableRxFifoInt (

SCI_Handle sciHandle)

Enables the serial communications interface (SCI) receive FIFO interrupt.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.27 void SCI_enableRxInt (

SCI_Handle sciHandle)

Enables the serial communications interface (SCI) receive interrupt.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.28 void SCI_enableSleep (

SCI_Handle sciHandle)

Enables the serial communications interface (SCI) sleep mode.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.29 void SCI_enableTx (

SCI_Handle sciHandle)

Enables the serial communications interface (SCI) master/slave transmit mode.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.30 void SCI_enableTxFifoInt (

SCI_Handle sciHandle)

Enables the serial communications interface (SCI) transmit FIFO interrupt.

722.7

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

out *success* Pointer to a variable which will house whether the read was successful or not (true on success)

Returns Data if successful, or NULL if no characters

15.1.3.36 **SCI_FifoStatus_e** SCI_getRxFifoStatus (

SCI_Handle *sciHandle*)

Gets the serial communications interface (SCI) receive FIFO status.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

Returns The receive FIFO status

15.1.3.37 **SCI_FifoStatus_e** SCI_getTxFifoStatus (

SCI_Handle *sciHandle*)

Gets the serial communications interface (SCI) transmit FIFO status.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

Returns The transmit FIFO status

15.1.3.38 **SCI_Handle** SCI_init (

void * *pMemory*,

const size_t *numBytes*)

Initializes the serial communications interface (SCI) object handle.

[1]Parameters in *pMemory* A pointer to the base address of the SCI registers

in *numBytes* The number of bytes allocated for the SCI object, bytes

Returns The serial communications interface (SCI) object handle

15.1.3.39 bool_t SCI_isRxDataReady (

SCI_Handle *sciHandle*) [inline]

Determines if the serial communications interface (SCI) has receive data ready.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

Returns The receive data status

References SCI_SCIRXST_RXRDY_BITS, and _SCI_Obj_::SCIRXST.

15.1.3.40 `bool_t SCI_isTxReady (`
SCI_Handle *sciHandle*) `[inline]`

Determines if the serial communications interface (SCI) is ready to transmit.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

Returns The transmit status

References SCI_SCICTL2_TXRDY_BITS, and _SCI_Obj_::SCICTL2.

15.1.3.41 `void SCI_putData (`
SCI_Handle *sciHandle*,
`const uint16_t data ) [inline]`

Writes data to the serial communications interface (SCI)

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

in *data* The data value

References _SCI_Obj_::SCITXBUF.

15.1.3.42 `void SCI_putDataBlocking (`
SCI_Handle *sciHandle*,
`uint16_t data )`

Writes data to the serial communications interface (Blocking)

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

in *data* The data value

15.1.3.43 `uint16_t SCI_putDataNonBlocking (`
SCI_Handle *sciHandle*,

uint16_t data)

Writes data to the serial communications interface (Non-Blocking)

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

in *data* The data value

Returns True on successful write, false if no space is available in the transmit buffer

15.1.3.44 void SCI_reset (

SCI_Handle sciHandle)

Resets the serial communications interface (SCI)

[1]Parameters in *sciHandle* The serial communication interface (SCI) object handle

15.1.3.45 void SCI_resetChannels (

SCI_Handle sciHandle)

Resets the serial communications interface (SCI) transmit and receive channels.

[1]Parameters in *sciHandle* The serial communication interface (SCI) object handle

15.1.3.46 void SCI_resetRxFifo (

SCI_Handle sciHandle)

Resets the serial communications interface (SCI) receive FIFO.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.47 void SCI_resetTxFifo (

SCI_Handle sciHandle)

Resets the serial communications interface (SCI) transmit FIFO.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

15.1.3.48 void SCI_setBaudRate (

SCI_Handle sciHandle,

const **SCI_BaudRate_e** baudRate)

Sets the serial communications interface (SCI) baud rate.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

in *baudRate* The baud rate

15.1.3.49 void SCI_setCharLength (

SCI_Handle sciHandle,

const **SCI_CharLength_e** charLength)

Sets the serial communications interface (SCI) character length.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

in *charLength* The character length

15.1.3.50 void SCI_setMode (

SCI_Handle sciHandle,

const **SCI_Mode_e** mode)

Sets the serial communications interface (SCI) multiprocessor mode.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

in *mode* The multiprocessor mode

15.1.3.51 void SCI_setNumStopBits (

SCI_Handle sciHandle,

const **SCI_NumStopBits_e** numBits)

Sets the serial communications interface (SCI) number of stop bits.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

in *numBits* The number of bits

15.1.3.52 void SCI_setParity (

SCI_Handle sciHandle,

const **SCI_Parity_e** parity)

Sets the serial communications interface (SCI) parity.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

in *parity* The parity

15.1.3.53 void SCI_setPriority (

SCI_Handle sciHandle,

const **SCI_Priority_e** priority)

Sets the serial communications interface (SCI) priority.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

in *priority* The priority

15.1.3.54 void SCI_setRxFifoIntLevel (

SCI_Handle sciHandle,

const **SCI_FifoLevel_e** fifoLevel)

Sets the serial communications interface (SCI) receive FIFO level for generating an interrupt.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

in *fifoLevel* The FIFO level

15.1.3.55 void SCI_setTxDelay (

SCI_Handle sciHandle,

const uint8_t delay)

Sets the serial communications interface (SCI) transmit delay.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

in *delay* The transmit delay

15.1.3.56 void SCI_setTxFifoIntLevel (

SCI_Handle sciHandle,

const **SCI_FifoLevel_e** fifoLevel)

Sets the serial communications interface (SCI) transmit FIFO level for generating an interrupt.

[1]Parameters in *sciHandle* The serial communications interface (SCI) object handle

in *fifoLevel* The FIFO level

16 Serial Peripheral Interface (SPI)

Introduction ??
 API Functions 177 The Serial Peripheral Interface (SPI) API provides a set of functions for configuring and using the device's SPI peripheral(s).

This driver is contained in `f2802x/common/source/spi.c`, with `f2802x/common/include/spi.h` containing the API definitions for use by applications.

16.1 SPI

Macros

```
#define SPI_SPICCR_CHAR_LENGTH_BITS
#define SPI_SPICCR_CLKPOL_BITS
#define SPI_SPICCR_RESET_BITS
#define SPI_SPICCR_SPILBK_BITS
#define SPI_SPICTL_CLK_PHASE_BITS
#define SPI_SPICTL_INT_ENA_BITS
#define SPI_SPICTL_MODE_BITS
#define SPI_SPICTL_OVRRUN_INT_ENA_BITS
#define SPI_SPICTL_TALK_BITS
#define SPI_SPIFFRX_FIFO_OVF_BITS
#define SPI_SPIFFRX_FIFO_OVFCLR_BITS
#define SPI_SPIFFRX_FIFO_RESET_BITS
#define SPI_SPIFFRX_FIFO_ST_BITS
#define SPI_SPIFFRX_IENA_BITS
#define SPI_SPIFFRX_IL_BITS
#define SPI_SPIFFRX_INT_BITS
#define SPI_SPIFFRX_INTCLR_BITS
#define SPI_SPIFFTX_CHAN_RESET_BITS
#define SPI_SPIFFTX_FIFO_ENA_BITS
#define SPI_SPIFFTX_FIFO_RESET_BITS
```

```
#define SPI_SPIFFTX_FIFO_ST_BITS  
  
#define SPI_SPIFFTX_IENA_BITS  
  
#define SPI_SPIFFTX_IL_BITS  
  
#define SPI_SPIFFTX_INT_BITS  
  
#define SPI_SPIFFTX_INTCLR_BITS  
  
#define SPIA_BASE_ADDR
```

Typedefs

```
typedef struct _SPI_Obj * SPI_Handle
```

Enumerations

```
enum SPI_BaudRate_e { SPI_BaudRate_500_KBaud, SPI_BaudRate_1_MBaud }  
  
enum SPI_CharLength_e {  
    SPI_CharLength_1_Bit,          SPI_CharLength_2_Bits,          SPI_CharLength_3_Bits,  
    SPI_CharLength_4_Bits,          SPI_CharLength_6_Bits,          SPI_CharLength_7_Bits,  
    SPI_CharLength_5_Bits,          SPI_CharLength_8_Bits,          SPI_CharLength_9_Bits,  
    SPI_CharLength_10_Bits,         SPI_CharLength_11_Bits,         SPI_CharLength_12_Bits,  
    SPI_CharLength_13_Bits,         SPI_CharLength_14_Bits,         SPI_CharLength_15_Bits,  
    SPI_CharLength_16_Bits }  
  
enum SPI_ClkPhase_e { SPI_ClkPhase_Normal, SPI_ClkPhase_Delayed }  
  
enum SPI_ClkPolarity_e { SPI_ClkPolarity_OutputRisingEdge_InputFallingEdge,  
    SPI_ClkPolarity_OutputFallingEdge_InputRisingEdge }  
  
enum SPI_FifoLevel_e {  
    SPI_FifoLevel_Empty, SPI_FifoLevel_1_Word, SPI_FifoLevel_2_Words, SPI_FifoLevel_3_Words,  
    SPI_FifoLevel_4_Words }  
  
enum SPI_FifoStatus_e {  
    SPI_FifoStatus_Empty, SPI_FifoStatus_1_Word, SPI_FifoStatus_2_Words,  
    SPI_FifoStatus_3_Words, SPI_FifoStatus_4_Words }  
  
enum SPI_Mode_e { SPI_Mode_Slave, SPI_Mode_Master }  
  
enum SPI_Priority_e { SPI_Priority_Immediate, SPI_Priority_FreeRun, SPI_Priority_AfterRxRxSeq  
    }  
  
enum SPI_StelInv_e { SPI_StelInv_ActiveLow, SPI_StelInv_ActiveHigh }  
  
enum SPI_TriWire_e { SPI_TriWire_NormalFourWire, SPI_TriWire_ThreeWire }
```

Functions

`void SPI_clearRxFifoInt (SPI_Handle spiHandle)`
`void SPI_clearRxFifoOvf (SPI_Handle spiHandle)`
`void SPI_clearTxFifoInt (SPI_Handle spiHandle)`
`void SPI_disable (SPI_Handle spiHandle)`
`void SPI_disableChannels (SPI_Handle spiHandle)`
`void SPI_disableInt (SPI_Handle spiHandle)`
`void SPI_disableLoopBack (SPI_Handle spiHandle)`
`void SPI_disableOverRunInt (SPI_Handle spiHandle)`
`void SPI_disableRxFifo (SPI_Handle spiHandle)`
`void SPI_disableRxFifoInt (SPI_Handle spiHandle)`
`void SPI_disableTx (SPI_Handle spiHandle)`
`void SPI_disableTxFifo (SPI_Handle spiHandle)`
`void SPI_disableTxFifoEnh (SPI_Handle spiHandle)`
`void SPI_disableTxFifoInt (SPI_Handle spiHandle)`
`void SPI_enable (SPI_Handle spiHandle)`
`void SPI_enableChannels (SPI_Handle spiHandle)`
`void SPI_enableFifoEnh (SPI_Handle spiHandle)`
`void SPI_enableInt (SPI_Handle spiHandle)`
`void SPI_enableLoopBack (SPI_Handle spiHandle)`
`void SPI_enableOverRunInt (SPI_Handle spiHandle)`
`void SPI_enableRxFifo (SPI_Handle spiHandle)`
`void SPI_enableRxFifoInt (SPI_Handle spiHandle)`
`void SPI_enableTx (SPI_Handle spiHandle)`
`void SPI_enableTxFifo (SPI_Handle spiHandle)`
`void SPI_enableTxFifoInt (SPI_Handle spiHandle)`
`SPI_FifoStatus_e SPI_getRxFifoStatus (SPI_Handle spiHandle)`
`SPI_FifoStatus_e SPI_getTxFifoStatus (SPI_Handle spiHandle)`
`SPI_Handle SPI_init (void *pMemory, const size_t numBytes)`

```
uint16_t SPI_read (SPI_Handle spiHandle)
void SPI_reset (SPI_Handle spiHandle)
void SPI_resetChannels (SPI_Handle spiHandle)
void SPI_resetRxFifo (SPI_Handle spiHandle)
void SPI_resetTxFifo (SPI_Handle spiHandle)
void SPI_setBaudRate (SPI_Handle spiHandle, const SPI_BaudRate_e baudRate)
void SPI_setCharLength (SPI_Handle spiHandle, const SPI_CharLength_e length)
void SPI_setClkPhase (SPI_Handle spiHandle, const SPI_ClkPhase_e clkPhase)
void SPI_setClkPolarity (SPI_Handle spiHandle, const SPI_ClkPolarity_e polarity)
void SPI_setMode (SPI_Handle spiHandle, const SPI_Mode_e mode)
void SPI_setPriority (SPI_Handle spiHandle, const SPI_Priority_e priority)
void SPI_setRxFifoIntLevel (SPI_Handle spiHandle, const SPI_FifoLevel_e fifoLevel)
void SPI_setStelInv (SPI_Handle spiHandle, const SPI_StelInv_e stelInv)
void SPI_setTriWire (SPI_Handle spiHandle, const SPI_TriWire_e triWire)
void SPI_setTxDelay (SPI_Handle spiHandle, const uint8_t delay)
void SPI_setTxFifoIntLevel (SPI_Handle spiHandle, const SPI_FifoLevel_e fifoLevel)
void SPI_write (SPI_Handle spiHandle, const uint16_t data)
void SPI_write8 (SPI_Handle spiHandle, const uint16_t data)
```

16.1.1 Detailed Description

16.1.2 Enumeration Type Documentation

16.1.2.1 enum **SPI_BaudRate_e**

Enumeration to define the serial peripheral interface (SPI) baud rates. These assume a LSCLK of 12.5MHz.

Enumerator

SPI_BaudRate_500_KBaud Denotes 500 KBaud.

SPI_BaudRate_1_MBaud Denotes 1 MBaud.

16.1.2.2 enum **SPI_CharLength_e**

Enumeration to define the serial peripheral interface (SPI) character lengths.

Enumerator

- SPI_CharLength_1_Bit** Denotes a character length of 1 bit.
- SPI_CharLength_2_Bits** Denotes a character length of 2 bits.
- SPI_CharLength_3_Bits** Denotes a character length of 3 bits.
- SPI_CharLength_4_Bits** Denotes a character length of 4 bits.
- SPI_CharLength_5_Bits** Denotes a character length of 5 bits.
- SPI_CharLength_6_Bits** Denotes a character length of 6 bits.
- SPI_CharLength_7_Bits** Denotes a character length of 7 bits.
- SPI_CharLength_8_Bits** Denotes a character length of 8 bits.
- SPI_CharLength_9_Bits** Denotes a character length of 9 bits.
- SPI_CharLength_10_Bits** Denotes a character length of 10 bits.
- SPI_CharLength_11_Bits** Denotes a character length of 11 bits.
- SPI_CharLength_12_Bits** Denotes a character length of 12 bits.
- SPI_CharLength_13_Bits** Denotes a character length of 13 bits.
- SPI_CharLength_14_Bits** Denotes a character length of 14 bits.
- SPI_CharLength_15_Bits** Denotes a character length of 15 bits.
- SPI_CharLength_16_Bits** Denotes a character length of 16 bits.

16.1.2.3 enum **SPI_ClkPhase_e**

Enumeration to define the serial peripheral interface (SPI) clock phase.

Enumerator

- SPI_ClkPhase_Normal** Denotes a normal clock scheme.
- SPI_ClkPhase_Delayed** Denotes that the SPICLK signal is delayed by one half-cycle.

16.1.2.4 enum **SPI_ClkPolarity_e**

Enumeration to define the serial peripheral interface (SPI) clock polarity for the input and output data.

Enumerator

- SPI_ClkPolarity_OutputRisingEdge_InputFallingEdge** Denotes that the tx data is output on the rising edge, the rx data is latched on the falling edge.
- SPI_ClkPolarity_OutputFallingEdge_InputRisingEdge** Denotes that the tx data is output on the falling edge, the rx data is latched on the rising edge.

16.1.2.5 enum **SPI_FifoLevel_e**

Enumeration to define the serial peripheral interface (SPI) FIFO level.

Enumerator

- SPI_FifoLevel_Empty** Denotes the fifo is empty.

SPI_FifoLevel_1_Word Denotes the fifo contains 1 word.
SPI_FifoLevel_2_Words Denotes the fifo contains 2 words.
SPI_FifoLevel_3_Words Denotes the fifo contains 3 words.
SPI_FifoLevel_4_Words Denotes the fifo contains 4 words.

16.1.2.6 enum **SPI_FifoStatus_e**

Enumeration to define the serial peripheral interface (SPI) FIFO status.

Enumerator

SPI_FifoStatus_Empty Denotes the fifo is empty.
SPI_FifoStatus_1_Word Denotes the fifo contains 1 word.
SPI_FifoStatus_2_Words Denotes the fifo contains 2 words.
SPI_FifoStatus_3_Words Denotes the fifo contains 3 words.
SPI_FifoStatus_4_Words Denotes the fifo contains 4 words.

16.1.2.7 enum **SPI_Mode_e**

Enumeration to define the serial peripheral interface (SPI) network mode control.

Enumerator

SPI_Mode_Slave Denotes slave mode.
SPI_Mode_Master Denotes master mode.

16.1.2.8 enum **SPI_Priority_e**

Enumerator

SPI_Priority_Immediate Stops immediately after EMU halt.
SPI_Priority_FreeRun Doesn't stop after EMU halt.
SPI_Priority_AfterRxRxSeq Stops after EMU halt and next rx/rx sequence

16.1.2.9 enum **SPI_Stelnv_e**

Enumerator

SPI_Stelnv_ActiveLow Denotes active low STE pin.
SPI_Stelnv_ActiveHigh Denotes active high STE pin.

16.1.2.10 enum **SPI_TriWire_e**

Enumerator

SPI_TriWire_NormalFourWire Denotes 4 wire SPI mode.
SPI_TriWire_ThreeWire Denotes 3 wire SPI mode.

16.1.3 Function Documentation

16.1.3.1 void SPI_clearRxFifoInt (

SPI_Handle spiHandle)

Clears the Rx FIFO interrupt flag.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.2 void SPI_clearRxFifoOvf (

SPI_Handle spiHandle)

Clears the Rx FIFO overflow flag.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.3 void SPI_clearTxFifoInt (

SPI_Handle spiHandle)

Clears the Tx FIFO interrupt flag.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.4 void SPI_disable (

SPI_Handle spiHandle)

Disables the serial peripheral interface (SPI)

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.5 void SPI_disableChannels (

SPI_Handle spiHandle)

Disables the serial peripheral interface (SPI) transmit and receive channels.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.6 void SPI_disableInt (

SPI_Handle spiHandle)

Disables the serial peripheral interface (SPI) interrupt.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.7 void SPI_disableLoopBack (

SPI_Handle spiHandle)

Disables the serial peripheral interface (SPI) loop back mode.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.8 void SPI_disableOverRunInt (

SPI_Handle spiHandle)

Disables the serial peripheral interface (SPI) over-run interrupt.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.9 void SPI_disableRxFifo (

SPI_Handle spiHandle)

Disables the serial peripheral interface (SPI) receive FIFO.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.10 void SPI_disableRxFifoInt (

SPI_Handle spiHandle)

Disables the serial peripheral interface (SPI) receive FIFO interrupt.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.11 void SPI_disableTx (

SPI_Handle spiHandle)

Disables the serial peripheral interface (SPI) master/slave transmit mode.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.12 void SPI_disableTxFifo (

SPI_Handle spiHandle)

Disables the serial peripheral interface (SPI) transmit FIFO.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.13 void SPI_disableTxFifoEnh (

SPI_Handle spiHandle)

Disables the serial peripheral interface (SPI) transmit FIFO enhancements.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.14 void SPI_disableTxFifoInt (

SPI_Handle spiHandle)

Disables the serial peripheral interface (SPI) transmit FIFO interrupt.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.15 void SPI_enable (

SPI_Handle spiHandle)

Enables the serial peripheral interface (SPI)

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.16 void SPI_enableChannels (

SPI_Handle spiHandle)

Enables the serial peripheral interface (SPI) transmit and receive channels.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.17 void SPI_enableFifoEnh (

SPI_Handle spiHandle)

Enables the serial peripheral interface (SPI) transmit FIFO enhancements.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.18 void SPI_enableInt (
SPI_Handle spiHandle)

Enables the serial peripheral interface (SPI) interrupt.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.19 void SPI_enableLoopBack (
SPI_Handle spiHandle)

Enables the serial peripheral interface (SPI) loop back mode.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.20 void SPI_enableOverRunInt (
SPI_Handle spiHandle)

Enables the serial peripheral interface (SPI) over-run interrupt.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.21 void SPI_enableRxFifo (
SPI_Handle spiHandle)

Enables the serial peripheral interface (SPI) receive FIFO.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.22 void SPI_enableRxFifoInt (
SPI_Handle spiHandle)

Enables the serial peripheral interface (SPI) receive FIFO interrupt.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.23 void SPI_enableTx (
 SPI_Handle spiHandle)

Enables the serial peripheral interface (SPI) master/slave transmit mode.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.24 void SPI_enableTxFifo (
 SPI_Handle spiHandle)

Enables the serial peripheral interface (SPI) transmit FIFO.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.25 void SPI_enableTxFifoInt (
 SPI_Handle spiHandle)

Enables the serial peripheral interface (SPI) transmit FIFO interrupt.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.26 **SPI_FifoStatus_e** SPI_getRxFifoStatus (
 SPI_Handle spiHandle)

Gets the serial peripheral interface (SPI) receive FIFO status.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

Returns The receive FIFO status

16.1.3.27 **SPI_FifoStatus_e** SPI_getTxFifoStatus (
 SPI_Handle spiHandle)

Gets the serial peripheral interface (SPI) transmit FIFO status.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

Returns The transmit FIFO status

16.1.3.28 **SPI_Handle** SPI_init (
 void * pMemory,

const size_t numBytes)

Initializes the serial peripheral interface (SPI) object handle.

[1]Parameters in *pMemory* A pointer to the base address of the SPI registers

in *numBytes* The number of bytes allocated for the SPI object, bytes

Returns The serial peripheral interface (SPI) object handle

16.1.3.29 **uint16_t SPI_read (**

SPI_Handle spiHandle) [inline]

Reads data from the serial peripheral interface (SPI)

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

Returns The received data value

16.1.3.30 **void SPI_reset (**

SPI_Handle spiHandle)

Resets the serial peripheral interface (SPI)

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.31 **void SPI_resetChannels (**

SPI_Handle spiHandle)

Resets the serial peripheral interface (SPI) transmit and receive channels.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.32 **void SPI_resetRxFifo (**

SPI_Handle spiHandle)

Resets the serial peripheral interface (SPI) receive FIFO.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.33 **void SPI_resetTxFifo (**

SPI_Handle spiHandle)

Resets the serial peripheral interface (SPI) transmit FIFO.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

16.1.3.34 void SPI_setBaudRate (

SPI_Handle spiHandle,

const **SPI_BaudRate_e** baudRate)

Sets the serial peripheral interface (SPI) baud rate.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

in *baudRate* The baud rate

16.1.3.35 void SPI_setCharLength (

SPI_Handle spiHandle,

const **SPI_CharLength_e** length)

Sets the serial peripheral interface (SPI) character length.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

in *length* The character length

16.1.3.36 void SPI_setClkPhase (

SPI_Handle spiHandle,

const **SPI_ClkPhase_e** clkPhase)

Sets the serial peripheral interface (SPI) clock phase.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

in *clkPhase* The clock phase

16.1.3.37 void SPI_setClkPolarity (

SPI_Handle spiHandle,

const **SPI_ClkPolarity_e** polarity)

Sets the serial peripheral interface (SPI) clock polarity.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

in *polarity* The clock polarity

16.1.3.38 void SPI_setMode (

SPI_Handle spiHandle,

const **SPI_Mode_e** mode)

Sets the serial peripheral interface (SPI) network mode.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

in *mode* The network mode

16.1.3.39 void SPI_setPriority (

SPI_Handle spiHandle,

const **SPI_Priority_e** priority)

Sets the priority of the SPI port vis-a-vis the EMU.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

in *priority* The priority of the SPI port vis-a-vis the EMU

16.1.3.40 void SPI_setRxFifoIntLevel (

SPI_Handle spiHandle,

const **SPI_FifoLevel_e** fifoLevel)

Sets the serial peripheral interface (SPI) receive FIFO level for generating an interrupt.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

in *fifoLevel* The FIFO level

16.1.3.41 void SPI_setStelInv (

SPI_Handle spiHandle,

const **SPI_SteInv_e** steinv)

Controls pin inversion of STE pin.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

in *steinv* Polarity of STE pin

16.1.3.42 void SPI_setTriWire (

SPI_Handle spiHandle,

const **SPI_TriWire_e** triwire)

Sets SPI port operating mode.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

in *triwire* 3 or 4 wire mode

16.1.3.43 void SPI_setTxDelay (

SPI_Handle spiHandle,

const uint8_t delay)

Sets the serial peripheral interface (SPI) transmit delay.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

in *delay* The delay value

16.1.3.44 void SPI_setTxFifoIntLevel (

SPI_Handle spiHandle,

const **SPI_FifoLevel_e** fifoLevel)

Sets the serial peripheral interface (SPI) transmit FIFO level for generating an interrupt.

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

in *fifoLevel* The FIFO level

16.1.3.45 void SPI_write (

SPI_Handle spiHandle,

`const uint16_t data ) [inline]`

Writes data to the serial peripheral interface (SPI)

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

in *data* The data value

16.1.3.46 void SPI_write8 (

SPI_Handle spiHandle,

`const uint16_t data ) [inline]`

Writes a byte of data to the serial peripheral interface (SPI)

[1]Parameters in *spiHandle* The serial peripheral interface (SPI) object handle

in *data* The data value

17 Timer

[Introduction](#) [??](#)

[API Functions](#) . 193 The timer API provides a set of functions for configuring, starting, stopping, and reading the CPU timers.

This driver is contained in `f2802x/common/source/timer.c`, with `f2802x/common/include/timer.h` containing the API definitions for use by applications.

17.1 TIMER

Data Structures

struct `_TIMER_Obj_`

Macros

```
#define TIMER0_BASE_ADDR
#define TIMER1_BASE_ADDR
#define TIMER2_BASE_ADDR
#define TIMER_TCR_FREESOFT_BITS
#define TIMER_TCR_TIE_BITS
#define TIMER_TCR_TIF_BITS
#define TIMER_TCR_TRB_BITS
#define TIMER_TCR_TSS_BITS
```

Typedefs

```
typedef struct _TIMER_Obj_ * TIMER_Handle
typedef struct _TIMER_Obj_ TIMER_Obj
```

Enumerations

```
enum TIMER_EmulationMode_e { TIMER_EmulationMode_StopAfterNextDecrement,
TIMER_EmulationMode_StopAtZero, TIMER_EmulationMode_RunFree }

enum TIMER_Status_e { TIMER_Status_CntIsNotZero, TIMER_Status_CntIsZero }
```

Functions

```
void TIMER_clearFlag (TIMER_Handle timerHandle)
void TIMER_disableInt (TIMER_Handle timerHandle)
void TIMER_enableInt (TIMER_Handle timerHandle)
uint32_t TIMER_getCount (TIMER_Handle timerHandle)
TIMER_Status_e TIMER_getStatus (TIMER_Handle timerHandle)
TIMER_Handle TIMER_init (void *pMemory, const size_t numBytes)
void TIMER_reload (TIMER_Handle timerHandle)
void TIMER_setDecimationFactor (TIMER_Handle timerHandle, const uint16_t decFactor)
void TIMER_setEmulationMode (TIMER_Handle timerHandle, const TIMER_EmulationMode_e mode)
void TIMER_setPeriod (TIMER_Handle timerHandle, const uint32_t period)
void TIMER_setPreScaler (TIMER_Handle timerHandle, const uint16_t preScaler)
void TIMER_start (TIMER_Handle timerHandle)
void TIMER_stop (TIMER_Handle timerHandle)
```

17.1.1 Detailed Description

17.1.2 Enumeration Type Documentation

17.1.2.1 enum **TIMER_EmulationMode_e**

Enumeration to define the timer (TIMER) emulation mode.

Enumerator

TIMER_EmulationMode_StopAfterNextDecrement Denotes that the timer will stop after the next decrement.

TIMER_EmulationMode_StopAtZero Denotes that the timer will stop when it reaches zero.

TIMER_EmulationMode_RunFree Denotes that the timer will run free.

17.1.2.2 enum **TIMER_Status_e**

Enumeration to define the timer (TIMER) status.

Enumerator

TIMER_Status_CntIsNotZero Denotes that the counter is non-zero.

TIMER_Status_CntIsZero Denotes that the counter is zero.

17.1.3 Function Documentation

17.1.3.1 void `TIMER_clearFlag` (
 TIMER_Handle timerHandle)

Clears the timer (TIMER) flag.

[1]Parameters in *timerHandle* The timer (TIMER) object handle

17.1.3.2 void `TIMER_disableInt` (
 TIMER_Handle timerHandle)

Disables the timer (TIMER) interrupt.

[1]Parameters in *timerHandle* The timer (TIMER) object handle

17.1.3.3 void `TIMER_enableInt` (
 TIMER_Handle timerHandle)

Enables the timer (TIMER) interrupt.

[1]Parameters in *timerHandle* The timer (TIMER) object handle

17.1.3.4 uint32_t `TIMER_getCount` (
 TIMER_Handle timerHandle) [inline]

Gets the timer (TIMER) count.

[1]Parameters in *timerHandle* The timer (TIMER) object handle

Returns The timer (TIMER) count

References `_TIMER_Obj_::TIM`.

17.1.3.5 **TIMER_Status_e** `TIMER_getStatus` (
 TIMER_Handle timerHandle)

Gets the timer (TIMER) status.

[1]Parameters in *timerHandle* The timer (TIMER) object handle

Returns The timer (TIMER) status

17.1.3.6 TIMER_Handle TIMER_init (

void * pMemory,
const size_t numBytes)

Initializes the timer (TIMER) object handle.

[1]Parameters in *pMemory* A pointer to the base address of the TIMER registers

in *numBytes* The number of bytes allocated for the TIMER object, bytes

Returns The timer (CLK) object handle

17.1.3.7 void TIMER_reload (

TIMER_Handle timerHandle)

Reloads the timer (TIMER) value.

[1]Parameters in *timerHandle* The timer (TIMER) object handle

17.1.3.8 void TIMER_setDecimationFactor (

TIMER_Handle timerHandle,

const uint16_t decFactor)

Sets the timer (TIMER) decimation factor.

[1]Parameters in *timerHandle* The timer (TIMER) object handle

in *decFactor* The timer decimation factor

17.1.3.9 void TIMER_setEmulationMode (

TIMER_Handle timerHandle,

const **TIMER_EmulationMode_e** mode)

Sets the timer (TIMER) emulation mode.

[1]Parameters in *timerHandle* The timer (TIMER) object handle

in *mode* The emulation mode

17.1.3.10 void TIMER_setPeriod (

TIMER_Handle timerHandle,

const uint32_t period)

Sets the timer (TIMER) period.

[1]Parameters in *timerHandle* The timer (TIMER) object handle

in *period* The period

17.1.3.11 void TIMER_setPreScaler (

TIMER_Handle timerHandle,

const uint16_t preScaler)

Sets the timer (TIMER) prescaler.

[1]Parameters in *timerHandle* The timer (TIMER) object handle

in *preScaler* The preScaler value

17.1.3.12 void TIMER_start (

TIMER_Handle timerHandle)

Starts the timer (TIMER)

[1]Parameters in *timerHandle* The timer (TIMER) object handle

17.1.3.13 void TIMER_stop (

TIMER_Handle timerHandle)

Stops the timer (TIMER)

[1]Parameters in *timerHandle* The timer (TIMER) object handle

18 Watchdog Timer

[Introduction](#) [??](#)
[API Functions](#) 199 The Watchdog Timer API provides a set of functions for using the watchdog module.

This driver is contained in `f2802x/common/source/watchdog.c`, with `f2802x/common/include/watchdog.h` containing the API definitions for use by applications.

18.1 WDOG

Data Structures

struct [_WD OG_Obj_](#)

Macros

```
#define WDOG\_BASE\_ADDR
#define WDOG\_SCSR\_WDENINT\_BITS
#define WDOG\_SCSR\_WDINTS\_BITS
#define WDOG\_SCSR\_WDOVERRIDE\_BITS
#define WDOG\_WDCNTR\_BITS
#define WDOG\_WDCR\_WDCHK\_BITS
#define WDOG\_WDCR\_WDDIS\_BITS
#define WDOG\_WDCR\_WDFLAG\_BITS
#define WDOG\_WDCR\_WDPS\_BITS
#define WDOG\_WDCR\_WRITE\_ENABLE
#define WDOG\_WDKEY\_BITS
```

Typedefs

```
typedef struct \_WD OG\_Obj\_ * WDOG\_Handle
typedef struct \_WD OG\_Obj\_ WDOG\_Obj
```

Enumerations

```
enum WDOG_IntStatus_e { WDOG_IntStatus_Active, WDOG_IntStatus_InActive }

enum WDOG_PreScaler_e {
    WDOG_PreScaler_OscClk_by_512_by_1,          WDOG_PreScaler_OscClk_by_512_by_2,
    WDOG_PreScaler_OscClk_by_512_by_4, WDOG_PreScaler_OscClk_by_512_by_8,
    WDOG_PreScaler_OscClk_by_512_by_16,         WDOG_PreScaler_OscClk_by_512_by_32,
    WDOG_PreScaler_OscClk_by_512_by_64 }
```

Functions

```
void WDOG_clearCounter (WDOG_Handle wdogHandle)

void WDOG_disable (WDOG_Handle wdogHandle)

void WDOG_disableInt (WDOG_Handle wdogHandle)

void WDOG_enable (WDOG_Handle wdogHandle)

void WDOG_enableInt (WDOG_Handle wdogHandle)

void WDOG_enableOverRide (WDOG_Handle wdogHandle)

WDOG_IntStatus_e WDOG_getIntStatus (WDOG_Handle wdogHandle)

WDOG_Handle WDOG_init (void *pMemory, const size_t numBytes)

void WDOG_setCount (WDOG_Handle wdogHandle, const uint8_t count)

void WDOG_setPreScaler (WDOG_Handle wdogHandle, const WDOG_PreScaler_e preScaler)
```

18.1.1 Detailed Description

18.1.2 Enumeration Type Documentation

18.1.2.1 enum **WDOG_IntStatus_e**

Enumeration to define the watchdog (WDOG) interrupt status.

Enumerator

WDOG_IntStatus_Active Denotes an active interrupt status.

WDOG_IntStatus_InActive Denotes an in-active interrupt status.

18.1.2.2 enum **WDOG_PreScaler_e**

Enumeration to define the watchdog (WDOG) timer clock prescaler, which sets the clock frequency.

Enumerator

WDOG_PreScaler_OscClk_by_512_by_1 Denotes WDCLK = OSCCLK/512/1.
WDOG_PreScaler_OscClk_by_512_by_2 Denotes WDCLK = OSCCLK/512/2.
WDOG_PreScaler_OscClk_by_512_by_4 Denotes WDCLK = OSCCLK/512/4.
WDOG_PreScaler_OscClk_by_512_by_8 Denotes WDCLK = OSCCLK/512/8.
WDOG_PreScaler_OscClk_by_512_by_16 Denotes WDCLK = OSCCLK/512/16.
WDOG_PreScaler_OscClk_by_512_by_32 Denotes WDCLK = OSCCLK/512/32.
WDOG_PreScaler_OscClk_by_512_by_64 Denotes WDCLK = OSCCLK/512/64.

18.1.3 Function Documentation

18.1.3.1 void WDOG_clearCounter (

WDOG_Handle wdogHandle)

Clears the watchdog (WDOG) counter.

[1]Parameters in *wdogHandle* The watchdog (WDOG) timer object handle

18.1.3.2 void WDOG_disable (

WDOG_Handle wdogHandle)

Disables the watchdog (WDOG) timer.

[1]Parameters in *wdogHandle* The watchdog (WDOG) timer object handle

18.1.3.3 void WDOG_disableInt (

WDOG_Handle wdogHandle)

Disables the watchdog (WDOG) timer interrupt.

[1]Parameters in *wdogHandle* The watchdog (WDOG) timer object handle

18.1.3.4 void WDOG_enable (

WDOG_Handle wdogHandle)

Enables the watchdog (WDOG) timer.

[1]Parameters in *wdogHandle* The watchdog (WDOG) timer object handle

18.1.3.5 void WDOG_enableInt (
 WDOG_Handle wdogHandle)

Enables the watchdog (WDOG) timer interrupt.

[1]Parameters in *wdogHandle* The watchdog (WDOG) timer object handle

18.1.3.6 void WDOG_enableOverRide (
 WDOG_Handle wdogHandle)

Enables the watchdog (WDOG) timer override.

[1]Parameters in *wdogHandle* The watchdog (WDOG) timer object handle

18.1.3.7 **WDOG_IntStatus_e** WDOG_getIntStatus (
 WDOG_Handle wdogHandle)

Gets the watchdog (WDOG) interrupt status.

[1]Parameters in *wdogHandle* The watchdog (WDOG) timer object handle

Returns The interrupt status

18.1.3.8 **WDOG_Handle** WDOG_init (
 void * pMemory,
 const size_t numBytes)

Initializes the watchdog (WDOG) object handle.

[1]Parameters in *pMemory* A pointer to the base address of the WDOG registers

in *numBytes* The number of bytes allocated for the WDOG object, bytes

Returns The watchdog (WDOG) object handle

18.1.3.9 void WDOG_setCount (
 WDOG_Handle wdogHandle,
 const uint8_t count)

Sets the watchdog (WDOG) counter.

[1]Parameters *in wdogHandle* The watchdog (WDOG) timer object handle

in count The count

18.1.3.10 void WDOG_setPreScaler (

WDOG_Handle wdogHandle,

const **WDOG_PreScaler_e** preScaler)

Sets the watchdog (WDOG) timer clock prescaler.

[1]Parameters *in wdogHandle* The watchdog (WDOG) timer object handle

in preScaler The prescaler

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